

*Doux Commerce**

Max Posch[†]
University of Exeter

Itzhak Tzachi Raz[‡]
Hebrew University and CEPR

October, 2026

[\[Please read the latest version, available here\]](#)

Abstract

Can markets help create the cultural foundations for cooperation beyond the local community? We study U.S. market integration during railroad expansion, 1850-1920. Following domestic migrant families across censuses, we find that gaining rather than losing market access led to a rapid and persistent decline in parochial orientation, reflected in less locally distinctive names for their children. County-level analysis shows that market integration results in higher voter turnout, more trusting newspaper language, and greater diversity in family-related behaviors and religious denominations, suggesting greater impersonal prosociality and tolerance, while reliance on relatives for economic support declines. The decline in parochial orientation is stronger for migrants working in market-oriented industries. Together with newspaper evidence, this pattern suggests that exchange with strangers and interdependence with distant others contribute to these changes. Markets may therefore encourage a broader social orientation alongside reduced dependence on family and local ties.

Keywords: Market integration, Market access, Cultural adaptation, Cooperation, Universalism, Tolerance, Trust

JEL codes: Z10, Z13, N71, N72, R49

*A version of this paper was previously circulated under the title “Markets Make Humans WEIRD: Evidence from 1850-1920 United States”. We thank Anne Sofie Beck Knudsen, Netanel Ben-Porath, Benjamin Enke, Martin Fiszbein, Ori Heffetz, Richard Hornbeck, Elias Papaioannou, Devesh Rustagi, and Moshe Shayo for helpful feedback and conversations. We thank seminar participants at Ben-Gurion, Brown, Cambridge, Hebrew University, King’s College London, Koç, Lugano, Lund, Oxford, Queen Mary University of London, the University of Southern Denmark, Tel Aviv, and Yale, and participants at ASREC Europe and GMU, the BSE Summer Forum (Economics of Institutions), the CEPR Economic History Symposium, the Growth, History and Development Workshop at SDU, the Hayek Workshop at LSE, the NBER Summer Institute (DAE), NICEP, the Workshop on Persistence and Cultural Transmission at Queen’s University Belfast, SCHEME at Birmingham, SIOE, and the Swiss Economists Abroad End-of-Year Conference for helpful comments. Roee Diler provided valuable research assistance. We thank Matt Nelson for sharing kin propinquity data. All remaining errors are our own. This research was supported by The Israel Science Foundation (grant No. 2798/25). For the purpose of open access, the authors have applied a Creative Commons Attribution (CC BY) licence to any Author Accepted Manuscript version arising from this submission.

[†]Department of Economics, University of Exeter. Email: m.posch@exeter.ac.uk.

[‡]Department of Economics and PPE, The Hebrew University of Jerusalem, and CEPR. Email: iraz@mail.huji.ac.il.

“Commerce is a cure for the most destructive prejudices; for it is almost a general rule that wherever manners are gentle there is commerce; and wherever there is commerce, manners are gentle.”

— Montesquieu, *The Spirit of the Laws*, Book 20, Chapter I

1 Introduction

Social scientists and philosophers have long debated how the spread of markets changes the way people relate to one another (Hirschman, 1982). Enlightenment thinkers such as Montesquieu (1748), Smith (1776) and Hume (1758) argued that market exchange broadens the social scope of cooperation, a view captured by the *doux commerce* hypothesis. A contrasting view, most influentially expressed by Polanyi (1944), emphasizes that the expansion of market relations can erode the social bonds on which people depend. Whether market integration broadens people’s social orientation, reduces their reliance on local ties, or does both remains an open empirical question.

In this paper, we provide new evidence on this question by studying the United States between 1850 and 1920, one of history’s most significant episodes of market integration. Railroad expansion connected previously localized economies with national markets, and people’s livelihoods became increasingly dependent on distant buyers, suppliers, and creditors. We find that this market integration fostered more universal and less parochial social orientations, while reducing reliance on relatives for economic support.

Two related mechanisms motivate this argument. First, exchange with unfamiliar people through markets can make shared norms of fairness and trust beyond the local community more valuable (Henrich et al., 2010; Tabellini, 2008). Second, market integration can make people’s livelihoods depend on distant others whom they never meet: a farmer selling through a local merchant, for example, may depend on buyers and creditors far beyond the community. Such relationships can provide experience of benefiting from the actions of anonymous others, encouraging trust beyond the immediate community (Jha et al., 2025). At the same time, wider markets can offer alternatives to local trade and support, making it less costly to depart from local expectations (Henrich, 2020). Market integration may therefore help create the cultural foundations of impersonal cooperation while reducing dependence on family and local approval.

We measure market integration using county-level market access (Donaldson and Hornbeck, 2016), which combines transportation costs to other counties with their populations to capture the potential for trade. Market access increased unevenly across counties as the transportation network expanded and populations grew.

We use two complementary research designs, combining full-count census data with historical

newspapers, religious censuses, and election returns. The first follows domestic migrant families, comparing changes in their parochial orientation, measured by the local distinctiveness of names they gave their children, between families whose moves increased their market access and those whose moves reduced it. This allows us to study cultural adjustment within families as their market access environment changes. The second follows counties over time, asking how market access changed parochial orientation, impersonal prosociality, tolerance of different norms and religious identities, and reliance on relatives for support in the wider population.

Our migrant design uses the Local Name Index (LNI), which measures how distinctively local a child's first name is by comparing its frequency in the child's county with its frequency elsewhere in the country (Raz, 2025). Such names can reflect parochial orientation: the extent to which parents are attached to the community in which they live or conform to its norms. We identify domestic migrants and infer migration timing using linked census records and children's birthplaces. Leveraging a difference-in-differences design, we study the change in siblings' names before and after migration, comparing families whose moves increased market access relative to those whose moves reduced it. Family fixed effects absorb permanent differences across families, including stable naming preferences.

We find that children's LNI scores fall after migration for families gaining market access relative to families losing it. The difference appears immediately after the inferred move and persists for at least a decade, averaging 2.18 LNI points (0.132 of a pre-migration standard deviation). This pattern is consistent with a negative impact of market integration on parochial orientation.

Two alternative interpretations could produce this pattern without any decline in parochial orientation. First, parents might remain oriented toward the community they left, choosing names characteristic of it that are less local to their new county. Second, their naming choices might not change at all, but the same kinds of names score as less local in the destination. Two variants of the LNI weigh against these interpretations. Gaining rather than losing market access also lowers the LNI of post-move names when we score them in the origin county, which suggests that parochial orientation toward the origin also declines. It also lowers their LNI relative to earlier siblings' names scored in the same destination, suggesting that changes in parents' naming behavior contribute to the relative decline in LNI, rather than the decline arising solely from the changes in the local distribution of names.

The naming response is stronger among families whose livelihoods depend more on exchange with strangers and distant markets. We compare market-oriented industries, such as agriculture, manufacturing, and trade, with industries serving mainly local needs, such as construction and public administration. Gaining rather than losing market access reduces LNI by an additional 2.28 points among market-oriented families. This difference does not appear to be explained by observed pre-existing characteristics, such as rural versus urban origin, family size, or prior cultural traits.

The stronger response is also evident among families that moved from the same origin to the same destination in the same decade, making differences in migration routes a less likely explanation. These comparisons support our account of adaptation to wider economic relationships.

Market integration could also broaden social orientation by raising incomes or drawing workers out of agriculture (e.g., [Inglehart, 1997](#); [Inglehart and Welzel, 2005](#); [Fouka and Serlin, 2024](#)). We therefore ask whether the naming response is stronger among families who enter better-paying occupations or leave farming. Instead, we find similar responses among migrants who moved down the occupational ladder and those who moved up or stayed at the same level. Among families initially in agriculture, those who remain farmers also respond much like those who leave it.

Another possibility is that market integration changes attitudes by drawing people away from relatives and exposing them to a more diverse population (e.g., [Schulz et al., 2019](#); [Enke, 2019](#); [Allport, 1954](#); [Bursztyrn et al., 2024](#)). If separation from kin were central, we would expect a weaker naming response among migrants who live near relatives after moving. Instead, we find that their response is similar to that of migrants who do not live near kin. Nor is the response stronger among families moving to neighborhoods with a smaller share of people of their own ancestry than in the places they left. We also account for differences in prosperity, urbanization, population diversity, and local institutions that could accompany changes in market access. The naming response remains negative and statistically significant, although the estimate is about a third smaller when compared in the same sample.

Our county-level analysis follows places as their market access changes, covering both residents who stay and those who move in. It extends the analysis beyond migrant families and beyond naming, to impersonal prosociality, tolerance, and reliance on kin for economic support.

Alongside the Local Name Index, we measure impersonal prosociality using presidential-election turnout, which captures civic participation (e.g., [Putnam, 1995](#); [Alesina and La Ferrara, 2000](#); [Rustagi, 2024](#)), and newspaper language about the trustworthiness of people in general. We measure tolerance using the Norms Tolerance Index, which captures dispersion in family-related behaviors such as mothers' age at the birth of their oldest child, and the Religious Diversity Index, which captures the diversity of religious denominations ([Raz, 2025](#)). These validated measures reflect the idea that communities that tolerate departures from local expectations permit greater diversity in how people live and worship. Finally, we measure reliance on kin by the share of vulnerable people—orphans, people with disabilities, and the elderly—living with relatives at home ([Ghosh et al., 2023](#)). Our preferred summary is a Composite Universalism Index that increases with turnout and norm tolerance and decreases with locally distinctive naming and family care.¹ We report all six individual measures alongside it.

¹We exclude generalized trust and religious diversity from the composite to maximize its coverage.

We compare changes in market access and these outcomes within counties, accounting for permanent county differences, state-by-year shocks, and broad geographic trends. Railroad construction may have responded to the same local economic and cultural changes that shaped our outcomes. A county's market access, however, depends on the entire transportation network: a railroad built far away can improve access to distant markets even when local infrastructure remains unchanged. Following [Donaldson and Hornbeck \(2016\)](#) and [Hornbeck and Rotemberg \(2024\)](#), we flexibly control for railroad construction and population in and around each county, focusing on changes in the wider network that are less likely to reflect local conditions. We also account for the possibility that some places may have been better positioned to gain access to markets regardless of railroad construction. We therefore recenter actual market access by controlling for the expected market access under a counterfactual expansion of the canal network without railroads ([Borusyak and Hull, 2023](#); [Hornbeck and Rotemberg, 2024](#); [Fogel, 1964](#)).

We find that counties that gained greater market access scored higher on our composite Universalism Index. A 10 percent increase in market access raises the index by about 0.025 standard deviations in the baseline specification and 0.015 standard deviations when we combine recentering with the controls for local railroad and population. Both estimates are highly statistically significant. We also obtain a positive and statistically significant effect on the composite index when we instrument for market access using counties' access to water transportation in 1850, following [Donaldson and Hornbeck \(2016\)](#) and [Hornbeck and Rotemberg \(2024\)](#). The individual measures show that this result spans distinct aspects of social life: locally distinctive naming declines, family-related behaviors become more diverse, and voter turnout rises. At the same time, fewer vulnerable people live with relatives at home, consistent with reduced reliance on kin for economic support. The county relationships could also reflect changes in prosperity, urbanization, population diversity, or local institutions. Accounting for these factors leaves the main pattern intact. We also find greater religious diversity and, in the main specifications, higher newspaper-based social trust, providing further evidence beyond the components of the index.

The last part of the paper provides additional evidence on mechanisms. First, families who are already less locally oriented might be more willing to move to places gaining market access. We therefore examine selective in-migration among incoming domestic migrants. We find no evidence that counties with greater market access growth attracted families who had previously chosen less locally distinctive names, and little systematic selection on other observed demographic, social, and economic characteristics. Together with the within-family naming results, this highlights the role of cultural adjustment.

Next, we examine whether market integration expanded commerce and the wider economic relationships emphasized in our account. We measure commerce using the share of employed residents working in wholesale and retail trade. Two newspaper indices capture discussion of

exchange with strangers, such as references to “traveling salesmen” or “commodity buyer”, and interdependence with distant others, such as “loan agent,” the “Chicago market,” and “grain inspection.” We find that all three measures increase with market access, including when we account for counterfactual market access and local changes in railroads and population. Together with the stronger naming response among market-oriented families, these patterns are consistent with cultural adjustment to greater exchange with strangers and dependence on distant others.

Finally, we ask whether becoming less parochial was associated with better outcomes in a more market-integrated environment. Comparing families who moved from the same origin to the same destination in the same decade, we find that becoming less parochial after migration is more positively associated with children’s survival and real-property holdings when the move increased market access. These comparisons provide suggestive evidence of benefits from adaptation, but do not identify causal returns.

Related Literature. Our main contribution is to show how the expansion of markets can broaden social orientation while reducing reliance on relatives for economic support. Influential work by [Henrich et al., eds \(2004\)](#), [Henrich et al. \(2010\)](#), and [Enke \(2023\)](#) links market exposure to fairness, trust, and a broader scope of moral concern. More recently, [Rustagi \(2026\)](#) provides evidence that historical proximity to a marketplace fostered civic values and cooperation in Ethiopia.² We bring together evidence on adjustment within existing families and on changes in the broader population during the emergence of a nationally integrated economy. Together, the two analyses provide evidence on how the cultural foundations of cooperation beyond the community can develop as livelihoods become more closely tied to distant markets.

We also contribute to understanding why market integration can bring about these changes. The incentives to cooperate with unfamiliar trading partners are central to the account in [Henrich et al. \(2010\)](#) and to the evidence in [Rustagi \(2026\)](#). A related possibility is that markets give people a stake in the welfare of distant others whom they never meet. [Jha et al. \(2025\)](#) provide experimental evidence that financial-market exposure increases generalized trust, consistent with the experience of relying on unfamiliar others whose interests are aligned with one’s own.³ We examine these two features together in the expansion of markets for goods and services. Differences in families’ responses across livelihoods, together with newspaper evidence on unfamiliar exchange and distant-

²Related evidence links market exposure to behavior measured in a nationwide lost-letter experiment in Italy ([Baldassarri, 2020](#)) and an experimental study of moral decision-making in Greenland ([Agneman and Chevrot-Bianco, 2023](#)). In these studies, market exposure itself is not randomly assigned. [Ngoma et al. \(2025\)](#) experimentally increase access to urban markets in the Democratic Republic of Congo. They find lasting changes in economic activity and values, but no average effect on prosociality or moral universalism.

³Related financial-market experiments study effects on support for peace ([Jha and Shayo, 2019](#)) and on economic values and political preferences ([Margalit and Shayo, 2021](#)). Outside financial-market experiments, related work examines how gains from inter-ethnic exchange can support tolerance ([Jha, 2013](#)), and how disruptions to credit and trading relationships during economic and political crises can lead to violence ([Grosfeld et al., 2020](#)).

market dependence, allow us to assess this account in a setting where markets increasingly organized how people earned their living. Taken together, these patterns are consistent with cultural adjustment to wider economic relationships involving both exchange with strangers and interdependence with distant others.

Our findings also contribute to research on how economic relationships shape the scope of cooperation (Platteau, 2000; Tabellini, 2008; Greif and Tabellini, 2017; Enke, 2024; Greif et al., 2025). Our evidence on social orientation complements Ghosh et al. (2026b), who show that access to manufacturing employment reduced marriages within closely connected social networks in the historical United States. We examine changes in people’s orientation towards those outside their community alongside changes in their reliance on relatives for support. These describe features of social networks; our outcomes concern social orientation and family support arrangements. These findings speak to the long-standing debate about markets and social life (Hirschman, 1982). They suggest that broader cooperation can develop alongside reduced reliance on family-based support.⁴

More broadly, our study adds to the literature on the origins and determinants of prosocial culture and cooperation. A large literature links historical experiences, institutions and kinship organization to differences in culture and relations within and across groups (Nunn and Wantchekon, 2011; Lowes et al., 2017; Enke, 2019; Schulz et al., 2019; Rustagi, 2024). Changes in economic incentives and social experience can also alter cultural practices and prosocial behavior, sometimes even rapidly (Francois et al., 2018; Bau, 2021; Rao, 2019; Lowe, 2021; Ghosh et al., 2026a; Raz, 2025). We show that families adjust their naming choices in response to market integration within a single parental generation. This provides evidence on how changes in adult choices contribute to cultural change during a wider economic transformation. Our account also complements research on the cultural legacy of the American frontier (Bazzi et al., 2020, 2024), by examining how integration into wider markets shapes orientation towards the local community.

Finally, our paper complements the literature on the economic impact of railroads in the United States (Fogel, 1964; Atack et al., 2010, 2011; Donaldson and Hornbeck, 2016; Chan, 2022; Hornbeck and Rotemberg, 2024). We study the cultural consequences of the market integration that these networks made possible. In the same historical setting, Anelli et al. (2026) show that exposure to increased trade competition encouraged support for the People’s Party, distinguishing this competitive pressure from gains in market access. Their findings concern the political response to the distributional consequences of integration; ours concern changes in social orientation. Our findings also connect this literature to research on the role of culture in economic development and innovation (Gorodnichenko and Roland, 2017; Posch et al., 2026). They raise the possibility that transport infrastructure influences development partly through the cultural changes that accompany

⁴A separate experimental literature asks whether trading in markets changes willingness to avoid harm to third parties (Falk and Szech, 2013; Bartling et al., 2023).

wider exchange.

Outline. The rest of this paper is organized as follows. Section 2 provides historical background, and Section 3 lays out the conceptual framework. Section 4 describes the data and measures. Section 5 examines cultural adaptation using domestic migrants, and Section 6 presents the county-level analysis. Section 7 provides additional evidence on mechanisms. Section 8 concludes.

2 Historical Background

By the early twentieth century, the United States had undergone a profound transformation into what Polanyi (1944) termed a “market society.” Market integration fundamentally reshaped the country’s social and economic organization. Farmers and other producers shifted from local subsistence to market-oriented production, interregional and long-distance trade expanded rapidly, and households grew increasingly dependent on markets to meet their consumption needs (e.g., Clark, 1990; Cronon, 1991; Bruegel, 2002).

Two key developments drove this integration. First, the period from 1850 to 1920 saw the explosive expansion of the U.S. railroad network: total track mileage grew from just under 9,000 miles in 1850 to nearly 238,000 by 1920 (Appendix Figure A.1), which dramatically reduced the cost of trade. Second, during the “Age of Mass Migration”, roughly 30 million immigrants arrived in the United States (Abramitzky et al., 2014), and the country’s population rose from about 23 million in 1850 to 106 million by 1920. The result was a steep and uneven rise in counties’ *market access*—their potential to trade with the rest of the country. This facilitated economic development, the settlement and growth of rural areas (Donaldson and Hornbeck, 2016; Chan, 2022), a surge in manufacturing activity and aggregate productivity (Atack et al., 2011; Hornbeck and Rotemberg, 2024), and urbanization (Atack et al., 2010).

Consider how market integration changed a typical farmer’s economic relationships. Before market integration, rural life was rooted in localized, personal exchange. Daily survival and farm operations relied heavily on non-market networks of reciprocal mutual aid—such as labor swapping, tool sharing, and communal works—conducted entirely outside the cash economy (Clark, 1990; Bruegel, 2002). A farm family consumed a substantial share of its harvest and brought only the remaining surplus to market. Transactions were conducted face-to-face, including dealings with local millers and in-kind barter at the general store. The merchant or general storekeeper served as an important financial intermediary in the local community, extending informal “book credit” based on long-term, intimate, personal knowledge of the farmer’s character and family circumstances (Atherton, 1971; Clark, 1990). If they sold directly to a more distant market, farmers transported their grain to the city themselves, mostly by waterway, and sold it there, often through kinship or

personal ties (Cronon, 1991; Bruegel, 2002). In this localized economy, the physical handling of the crop preserved the farmer's personal identity: wheat was packed, stored, and transported into discrete burlap sacks, each stenciled with the grower's name. A farmer's name was their brand, and their personal reputation stood as the primary guarantee of quality (Cronon, 1991).

This personalized system of grain handling and exchange changed in the second half of the nineteenth century by market integration, the expansion of the railroad network, and the invention of the steam-powered grain elevator (Lee, 1937; Leslie, 2022). Grain elevators revolutionized agricultural logistics by transforming discrete, sacked cargo into a continuous, mechanized bulk flow (Leslie, 2022). At the elevator, farmers dumped crops into hoppers, weighed them, and poured them directly into massive bins, where grain from many farms pooled before being shipped by railroad (Cronon, 1991; Lee, 1937). This effectively stripped the grain of its human origin—John's wheat became indistinguishable from Smith's. Because individual reputation could no longer guarantee quality in a shared bin, the Chicago Board of Trade established a standardized grading system that classified grain into abstract categories based on characteristics such as weight, variety, and moisture. The quality grade determined the grain price. Local grain prices became linked to trading in Chicago, where brokers responded to harvests and transport bottlenecks half a continent away (Cronon, 1991). In lieu of their physical crop, farmers received standardized warehouse receipts redeemable for any grain of the equivalent grade, converting agricultural produce into an impersonal, liquid financial asset (Cronon, 1991; Lee, 1937).

Once converted into a standardized, fungible asset, the farmer's crop passed through extensive trade networks driven by people they would never meet—commission merchants and out-of-town buyers who were increasingly agents of distant trading firms rather than local neighbors (Cronon, 1991). The grain of a typical Midwest farmer was sent to large primary regional markets, from which it entered national and international trading networks, ultimately reaching about 90 secondary markets throughout the country before final consumption by individuals far removed from the farmer's own community (Fogel, 1964).

At the same time, farmers increasingly encountered traveling agents, commercial “drummers”, and cattle buyers representing distant metropolitan firms (Cronon, 1991). Their connections to wider markets extended beyond selling crops to purchasing the tools and equipment needed to produce them. Home production gave way to purchased mass-produced goods. Mail-order houses such as Montgomery Ward and Sears, Roebuck—facilitated by the introduction of Rural Free Delivery—allowed farm families to bypass local stores and purchase goods from distant suppliers (Gordon, 2017).

Purchasing equipment and financing commercial production also changed farmers' credit relationships. Farmers bought factory-made machinery on credit, partly replacing reliance on shared tools and equipment (Clark, 1990; Bruegel, 2002). Book credit grounded in personal relationships

gave way to formal credit instruments (Clark, 1990). To finance land, farmers increasingly turned to farm mortgages funded by Eastern capital markets, often through local loan agents who assessed borrowers' creditworthiness (Snowden, 1987).

Market integration transformed the farmer from a participant in an intimate, self-reinforcing local exchange network into an economic actor whose livelihood was deeply dependent on distant and anonymous others. His grain was now pooled and consumed by individuals thousands of miles away; he provisioned his household with goods manufactured in distant places by people he would never see; and carried debts owed to financial institutions he would never visit.

Beyond agriculture, market integration drove a parallel shift toward interdependence with distant others in manufacturing. At the beginning of the nineteenth century, manufacturing in the U.S. was organized primarily around small artisan workshops centered on highly skilled craftsmen, sometimes employing a few assistants. The division of labor was limited, and production largely served local markets. By the mid-nineteenth century, large-scale factories employing substantial workforces and characterized by extensive division of labor began to emerge. Falling transport costs and rising market integration enabled these factories to sell their output and source raw materials from increasingly distant markets (Atack et al., 2011). In such settings, workers coordinated tasks with multiple co-workers and supervisors, while their livelihoods increasingly depended on demand from distant customers and inputs from distant suppliers.

3 Conceptual Framework

Our conceptual framework starts from a simple premise: culture and psychology can adapt to the social and economic environments in which people live. A large literature on cultural transmission emphasizes that cultural traits can respond to the incentives and social environments faced by individuals, rather than remaining fixed across contexts (Bisin and Verdier, 2011; Boyd and Richerson, 1985; Nunn, 2021), and that such adaptation can occur over relatively short time horizons as norms also travel horizontally (e.g., Boyd and Richerson, 2002; Bau, 2021; Raz, 2025).

A second pillar of our framework is the premise that people rely on cooperation and exchange to survive and solve individual and social problems. What differs across societies is the scope of cooperation and exchange and the mechanisms that support them, not their existence (Enke, 2019, 2024; Greif and Tabellini, 2017; Greif et al., 2025; Platteau, 2000). When economic life is localized, exchange relies heavily on repeated interactions with neighbors and kin, making reciprocity, interpersonal trust, and parochial attachments valuable. As market integration expands, economic livelihoods become increasingly dependent on exchange with strangers and on anonymous and distant others whom individuals may never meet. In such an environment, universalism,

generalized trust, impersonal prosociality, and tolerance of diverse norms become more valuable (Henrich et al., 2010; Henrich, 2020; Tabellini, 2008). By providing alternatives to local trade and support, these wider economic relationships can reduce dependence on kin and neighbors. When livelihoods depend less on maintaining local approval, departing from local norms becomes less costly. Market integration may therefore broaden social orientations both by changing people's attitudes towards distant others and by reducing the need to conform to local expectations.

As the economic environment changes, so do the returns to different orientations toward others, creating incentives for cultural adaptation. This perspective suggests that the appropriate way to think about the social consequences of markets is not simply to ask whether markets make people more or less prosocial, but toward whom prosociality is most valuable and how cooperation and exchange are organized in different economic environments.

Our historical setting captures a fundamental change in the structure of economic dependence—a transition from a predominantly localized economy to what Polanyi (1944) called a “market society.” Exchange, marketplaces, and commercial activity existed in virtually all societies, including those that remained predominantly local in their economic organization. What was distinctive about the nineteenth-century transformation was the increasing extent to which economic life became organized around impersonal exchange and dependence on distant markets.

We therefore conceptualize market integration as changing the structure of economic relationships along two closely related dimensions. First, it increases *economic interdependence with distant others*; an individual's material welfare increasingly depends on anonymous people and markets beyond the local community even in the absence of direct interaction with them. Evidence from financial markets suggests that these wider economic connections can shape social and political attitudes beyond the immediate exchange (Jha and Shayo, 2019; Jha et al., 2025; Margalit and Shayo, 2021). Second, it increases the extent to which *economic exchange takes place with strangers and non-local counterparties*. Repeated exposure to exchange with unknown others can increase the returns to norms of fairness, trust, and cooperation beyond the local community (Henrich et al., 2010; Baldassarri, 2020; Agneman and Chevrot-Bianco, 2023; Rustagi, 2026). These dimensions are conceptually distinct, but in the historical setting we study they move together: market integration simultaneously expands the set of people with whom individuals exchange and the geographic distance over which their economic fortunes are connected.

The central implication is that market integration should promote more universal (less parochial) social orientations, while reducing reliance on kin and local communities for economic support.

Prediction 1 (Market integration and universalism)

Greater market integration reduces local parochial orientation and reliance on kin for economic support, and promotes impersonal prosociality and tolerance of diverse norms and practices.

Who should adapt? The framework also generates an important prediction about who should respond most strongly to market integration. The force underlying adaptation is the changing nature of economic exchange and interdependence. The response should therefore be strongest where improved market access most expands exchange with strangers and dependence on distant markets. We expect these changes to be greater for farmers and workers in manufacturing and wholesale and retail trade, whose livelihoods depend on access to buyers and suppliers beyond the local community, than for workers in construction, personal services, and public administration, whose economic relationships remain predominantly local. This yields our second prediction:

Prediction 2 (Adaptation by market orientation)

The effect of market integration on parochial orientation should be stronger among individuals in market-oriented industries than among those in locally oriented industries.

This prediction is important because it helps distinguish our mechanism from several other consequences of market integration that could also affect culture, which we discuss below. Our mechanism specifically predicts that the response to market access should be stronger among those whose economic lives become more dependent on exchange with strangers and interdependent with distant others.

Alternative mechanisms. Market access did more than expand exchange and interdependence, and several of its other consequences could in principle produce the same cultural shift. Three rival explanations are most prominent, and our setting lets us weigh each against ours. First, market access raised incomes, and rising material security may foster universalism and tolerance (Inglehart and Welzel, 2005). If *income gains* were the main driver, we would expect stronger cultural responses among migrants whose economic circumstances improved more. Second, market access accelerated industrialization and the movement out of agriculture, a structural transformation long argued to reshape values (Inglehart and Baker, 2000). If the *transition out of agriculture* were the main driver, cultural responses should be stronger among those who left farming than among those who remained. Third, market access brought people into contact with more diverse populations, and intergroup contact is a classic source of reduced prejudice (Allport, 1954). If contact were the channel, the effect would track *local diversity and exposure to out-groups*. We examine these alternative explanations in Sections 5.2.2 and 6.

These are not the only features of the environment that market access reshaped; it also advanced urbanization, the spread of information, and the reach of formal legal institutions. We do not adjudicate each conceptually. Instead, we control for proxies of all of them in both the migrant-level and county-level specifications (Sections 5.2.2 and 6) and ask whether our estimates survive.

4 Data

We assemble a panel of U.S. counties spanning 1850 to 1920 that combines the full-count censuses (Ruggles et al., 2020), county-level census aggregates (Manson et al., 2020), historical presidential election returns (ICPSR, 1999), and a large-scale corpus of local newspapers. The data are harmonized to 1890 county boundaries following Hornbeck (2010). We present each measure concisely here; Appendix A gives full construction, spatial patterns, validation, and limitations.

No single historical proxy cleanly captures a population’s cultural orientation. Our measurement strategy is instead one of triangulation. We assemble measures of *universalism*—the degree to which prosocial regard, trust, and tolerance of unfamiliar ways extend to distant and anonymous others rather than being reserved for kin and neighbors and bound, through identity and conformity, to the local community—alongside an indicator of reliance on kin for economic support (Section 3). Drawing on first names, election returns, newspaper text, religious membership, and family structure, we ask whether market access shifts these outcomes in the directions predicted by our framework. The measures face different potential measurement problems. We discuss these limitations and how we address them in this section and in Appendix A.

4.1 Market Access

Our measure of market integration follows Donaldson and Hornbeck (2016). Building on a model of trade among counties, they define an empirical first-order approximation to county o ’s market access in year t as $MA_{ot} = \sum_{d \neq o} \tau_{odt}^{-\theta} N_{dt}$, where τ_{odt} is the bilateral trade cost between counties o and d in year t (relative to the average value of the transported goods P), N_{dt} is county d ’s population, which proxies for its market size, and θ is the trade elasticity. Intuitively, MA_{ot} rises when nearby or easily reached counties grow more populous, so it captures the size and accessibility of o ’s potential trading partners. We use the county-to-county cost matrices of Donaldson and Hornbeck (2016), which hold cost parameters and borders fixed at 1890 values and let costs vary over time as the railroad and canal network expands; population comes from Manson et al., 2020. Our baseline sets $\theta = 8.22$ and $P = 35$ (Donaldson and Hornbeck, 2016); we assess robustness to alternative calibrations, including that of Hornbeck and Rotemberg (2024) and a range of θ between 1 and 13 (Appendix A.2).

4.2 The Local Name Index

The workhorse of our analysis is the Local Name Index (LNI) of Raz (2025), a measure of parental parochial orientation based on the local distinctiveness of their children’s first names. For a name

n given to a child of gender g in county l and census year t ,

$$LNI_{nlgt} = 100 \times \frac{Pr(n \mid l, g, t)}{Pr(n \mid l, g, t) + Pr(n \mid -l, g, t)},$$

where $-l$ denotes all other counties and both probabilities are computed among children aged 0–10 (Fryer and Levitt, 2004; Raz, 2025). The county-level LNI averages this score across resident children aged 0–10. Higher values indicate that parents favor names distinctive to their own community rather than common nationally. Such choices can reflect parents’ attachment to the local community, pressure to follow its expectations, or both. We therefore interpret the LNI as a measure of locally oriented naming behavior, without requiring it to distinguish personal attachment from social pressure. Greater market access could reduce either: wider economic relationships may broaden parents’ identification, while alternatives to local exchange and support may make it easier to depart from local expectations.

The LNI is particularly well suited to anchor our analysis. It measures relative parochial orientation by focusing on the margin of social identity and norms directly affected by the transformation from local exchange networks to an integrated national market. Additionally, unusually for a cultural measure, it is available for individual families over time, enabling a difference-in-differences identification strategy, which compares siblings born to the same family before and after a move, focusing on the change in parochial orientation relative to the local communities in which families live (Section 5). Raz (2025) validates it against contemporary proxies for parochialism, including the density of social networks and survey-based moral universalism. We discuss its limitations in Appendix A.3. The appendix also defines two variants for the migrant design, where moving changes which county counts as “local”. The *origin-fixed* LNI scores all names relative to the county the family left, capturing how distinctive the chosen names remain of that community. It provides a proxy for continued orientation towards the origin community. The *benchmark-adjusted* LNI compares names chosen after the move with earlier same-sex sibling names, scoring both in the same destination county and census year.

4.3 County-Level Measures of Universalism

To establish that the shift is broad rather than an artifact of any one indicator, we assemble a battery of county-level measures of impersonal prosociality and tolerance, alongside an indicator of reliance on kin for economic support. These measures draw on different data sources and are potentially exposed to different confounds. Construction details and measure-specific limitations are in Appendix A.3; here we describe each briefly.

Impersonal prosociality. We use two measures. *Voter turnout* in presidential elections is a

standard proxy for civiness and impersonal cooperation (Putnam, 1995; Alesina and La Ferrara, 2000; Rupasingha et al., 2006; Rustagi, 2024); we combine historical returns (ICPSR, 1999) with full-count estimates of the eligible electorate, accounting for changes in women’s and Black suffrage. *Newspaper-based social trust* is a new measure built from the full-text local-newspaper corpus of *newspaperarchive.com*: following the contextualized-construct approach of Atari et al. (2023), we embed newspaper pages and standard trust items (e.g., “Most people can be trusted”) with Sentence-BERT, anchor the measure by subtracting an embedding of negatively worded items—which nets out the topic content the two poles share—and average the page-level cosine similarities to the county-decade. We assess its validity against General Social Survey trust in the post-1970 period (Appendix Figure A.6).

Tolerance. The *Norms Tolerance Index* (NTI) captures looseness in family norms (e.g., Gelfand et al., 2006, 2011): for married mothers aged 35–44 in each county-census-year, we take the first principal component of the coefficients of variation of mother’s age at the birth of her oldest co-resident child, the number of co-resident children linked to her, and the number of distinct families in the household, coded so that higher values denote greater dispersion—wider tolerance of diverse actions, on the logic that the central tendency of these distributions reflects the behavior norms prescribe and their dispersion how much deviation a community tolerates (Posch, 2021; Dimant et al., 2025). The *Religious Diversity Index* (RDI) captures tolerance of religious difference: one minus a Herfindahl-Hirschman index of denominational membership from the Censuses of Religious Bodies (Manson et al., 2020), so that higher values denote greater diversity. Both are community-level measures: a community that does not tolerate those who believe or live differently tends toward homogeneity—through exit, exclusion, or pressure to conform—so the heterogeneity these indices record is, at least in part, a signature of a community’s tolerance of difference, and not merely of who happens to live there.

Reliance on kin for economic support. Following Ghosh et al. (2023), we use the living arrangements of vulnerable individuals as an indicator of kin support. Our measure of *family care* is the share of vulnerable individuals—orphans (children under 16 without co-resident parents), people with disabilities, and the elderly (over 65)—living with a relative at home. Greater market access is predicted to *lower* family care, as wider economic relationships may weaken norms of kin obligation or reduce dependence on relatives and the need to conform to their expectations (Section 3). However, market access may also expand the supply of institutional care, reducing family care even if norms and the pressure to observe them remain unchanged. We probe this alternative by controlling for proxies of local economic development, such as per-capita income, manufacturing, and urbanization, as an indirect check on changes in the supply of care (Section 6).

The composite Universalism Index. To summarize the county-level evidence and reduce reliance

on separate hypothesis tests, we combine four outcomes in a composite Universalism Index:

$$\text{Universalism} = [z(\text{Turnout}) + z(\text{NTI})] - [z(\text{Family Care}) + z(\text{LNI})],$$

where each component is standardized within census year. The signs follow the framework’s predictions: market access raises turnout and norm tolerance and lowers family care and the local distinctiveness of names. We exclude newspaper-based trust, whose geographic coverage is limited, and the RDI, whose source data do not align one-for-one with the census decades (Appendix A.3). The index provides a single summary test of whether these outcomes shift, on average, in the predicted directions. We also report effects on the individual outcomes to show how the response varies across domains.

4.4 Mechanism Measures

Our conceptual framework distinguishes two channels through which market access may reshape culture: greater *exchange with strangers* and greater *interdependence with distant others* (Section 3). We probe these channels at two levels. At the county level, three measures let us examine whether exposure to each channel rose with market access (Section 7.2). At the individual level, we classify each worker’s industry as market-oriented or locally oriented and ask whether the cultural shift operates more strongly among those whose livelihoods ran through markets (Section 5). Because market-oriented work brings both more exchange with strangers and more dependence on distant others, this contrast tests the two mechanisms together rather than separating them.

Employment in wholesale and retail trade. Using industry codes in the full-count censuses (Ruggles et al., 2020), we compute the share of residents with a recorded industry who work in wholesale and retail trade, a simple indicator of the local prevalence of commerce. It captures exposure to market exchange broadly but does not distinguish the two channels.

The remaining two measures are built from the full-text local-newspaper corpus of *newspaperarchive.com*—the same corpus underlying our trust measure—using dictionaries of period words and phrases that signal each channel. For each newspaper-year we divide the number of matches to a channel’s terms by the word count, then average these frequencies within counties and census decades. These frequencies measure newspaper discussion of the relationships associated with each channel. We winsorize and standardize each resulting index. The full dictionaries and their construction and limitations are detailed in Appendix A.4.

Exchange with strangers. This dictionary covers mobile commercial counterparties, commodity intermediaries and commodity buyers, with terms such as “traveling salesman”, “commission

merchant” and “grain buyer”.

Interdependence with distant others. This dictionary covers credit dependence, distant-market orientation and the inspection or certification arrangements supporting trade with distant markets, with terms such as “Eastern capital”, “New York market” and “grain inspection”.

Market-oriented versus locally oriented industries. For the individual-level heterogeneity analysis, we sort each worker’s industry into two groups. *Market-oriented* industries either sell primarily to distant markets—such as agriculture and manufacturing—or are essential to the functioning of markets, such as wholesale and retail trade, transportation, and finance. *Locally oriented* industries mostly serve local markets and community needs, such as construction, utilities, personal services, entertainment, and public administration.⁵ We also report results using an alternative classification, defining market-orientation as migrants working in trade (wholesale and retail trade, except eating and drinking places) or “tradable” industries, based on a modern tradability classification of [Barkai and Karger \(2020\)](#), mapped to historical industries.

4.5 Other Demographic and Economic Measures

Our supporting analyses draw on a set of additional measures. Most are standard characteristics constructed from the full-count and county-level census data ([Ruggles et al., 2020](#); [Manson et al., 2020](#)): the occupational income score, real GDP per capita ([Fulford et al., 2020](#)), urbanization, the shares employed in agriculture, manufacturing and trade, the number of children, urban-versus-rural origin, nativity, Herfindahl-based indices of birthplace and ancestry diversity, and the share of multifamily households. Additionally, we employ the following measures, all detailed in Appendix A.5: intra-community marriage ([Raz, 2025](#))—the share of married couples with the same birthplace; kinship propinquity, which infers geographic proximity to kin from same-race same-surname households and their position in the census enumeration order ([Nelson, 2020](#)); shared ancestry, inferred probabilistically from surnames ([Posch et al., 2026](#)); and labor-force cooperation, capturing the extent of required cooperative traits on the job for different occupations, using O*NET work-style data.

5 Adaptation: Evidence from Domestic Migrants

In this section, we use individual-level data on domestic migrants to explore how and why changes in the degree of market access affected parochial orientation. To construct the sample of domestic

⁵Appendix A.4 lists the IPUMS IND1950 codes assigned to each group.

migrants, we use the Census Linking Project (Abramitzky et al., 2022)⁶, which enables us to link male heads of household across decennial censuses. To proxy the timing of migration, we limit our attention to families who migrated across states and had at least one child born in the origin state and one in the destination state, at least two years apart. We then proxy the year of migration by the midpoint between the birth year of the last child born in the origin state and the birth year of the first child born in the destination.⁷ We then link each household to the level of market access in both its origin and destination counties at the time of the later census, harmonizing county borders to 1890 (Hornbeck, 2010) to match the market access data.⁸

5.1 Market Integration Is Linked to Decline in Parochial Orientation

To test our theory that people adapt to increased market integration by becoming less parochial (Prediction 1), we examine changes in parental parochial orientation, as measured by their children’s LNI, using a dynamic difference-in-differences framework, following Raz (2025). Specifically, we compare LNI scores among siblings born before and after migration in families that moved to higher versus lower market access counties.⁹

Empirical strategy. We estimate the following equation:

$$LNI_i = \delta_{b(i)} + \theta_{f(i)} + \sum_{b \neq -1} \beta_b \cdot \mathbb{1}[b(i) = b] \cdot \mathbb{1}[MA_{d(i)} > MA_{o(i)}] + X_i \Omega + \epsilon_i \quad (1)$$

where LNI_i is the Local Name Index score for child i , currently residing in county $d(i)$ and born $b(i)$ years relative to the year the family $f(i)$ migrated from origin county $o(i)$ to destination county $d(i)$. The term $\delta_{b(i)}$ is a set of relative-year-of-birth fixed effects, absorbing any changes in locally distinctive naming relative to the year of migration among families that moved to a lower market access county. Crucially, $\theta_{f(i)}$ is a family fixed effect, removing any unobserved factor common to siblings, including the family’s migration path and permanent cultural and economic characteristics, so that identification comes from within-family changes over time. The vector X_i contains child characteristics—gender, birth order, and a set of 5-year cohort fixed effects—which we include in our robustness checks. The coefficients of interest, β_b , capture the impact of moving to a higher

⁶Appendix A.1 lists the census pairs, dataset DOIs, and version information.

⁷For example, consider a family living in Massachusetts in 1860 with two children: the first was born in 1854 in New York, and the second in 1856 in Massachusetts. In 1850, the father was living in New York. The proxy for this family’s year of migration from New York to Massachusetts is 1855.

⁸Specifically, households can be fuzzy-matched to multiple 1890-borders origin and destination counties, with probability weights determined by the geographical overlap between historical counties and the harmonized county units.

⁹In Section 6 we use county-level data, alternative identification strategies, and multiple cultural indicators to provide complementary evidence on the impact of market access on culture. The findings of both levels of analysis align.

versus a lower market access county on parental parochial orientation over time.¹⁰ We normalize β_{-1} to zero, so that β_b can be interpreted as the effect relative to the year just before migration.¹¹ We use two-way clustered standard errors ϵ_i at the county-of-origin and the county-of-destination, which together determine the treatment assignment (Bertrand et al., 2004; Abadie et al., 2023), but results are robust to clustering separately at the origin or destination. Because families differ in the number of children and our estimand of interest is the impact on parental parochial orientation, we use inverse-family size weights so that all families are given equal weight in the analysis. The results are robust to dropping those weights. Additionally, when families are matched to multiple 1890-borders counties in the process of county-border harmonization, we weight observations by the matching probabilities, so that the total weight per child sums to one.¹²

In this difference-in-differences, family fixed effects absorb persistent differences in levels, and the identifying assumption is parallel trends—absent differences in the change in market access, within-family LNI scores in families migrating to higher versus lower market access counties would have evolved similarly. We assess the plausibility of this assumption by examining the pre-trends ($\beta_{b<0}$). In Section 5.2.2, we further strengthen the plausibility of the identification assumption, showing that our findings are also robust to dynamically controlling for differences in multiple, potentially confounding socioeconomic characteristics between the origin and destination counties. While this design does not require migrants’ destinations to be randomly assigned, in Section 7.1 we further explore selective sorting of migrants on observed pre-migration characteristics and find limited evidence of it.

Results. We find that migration to a higher market access county rapidly decreases parochial orientation. Figure 1 shows estimates are mostly flat and close to zero in the years immediately before migration, supporting the identification assumption. In the first year after migration, the LNI of a child born to a family that moved to a higher market access county decreases by 2.22 points (p -value < 0.001) relative to a child born to a family whose move reduced market access, and this effect persists for at least ten years. The average post-migration effect is -2.18 points (p -value < 0.001).

¹⁰In our data, 37,532 families moved to a higher market access county while 67,837 moved to a lower market access county. Due to the process of harmonizing county borders to 1890, our data also contains 102 families that, although they migrated, are matched to the same county of origin and destination. Dropping these observations has a negligible impact on the results.

¹¹Our specification uses event time rather than calendar time, which means that all families are first treated between $b = -1$ and $b = 1$. As a result, concerns about negative weights in two-way fixed effects regressions with staggered treatment timing (e.g. Borusyak et al., 2024; de Chaisemartin and d’Haultfœuille, 2020; Goodman-Bacon, 2021; Sun and Abraham, 2021)—or more generally, in designs where groups experience different evolution of their exposure to treatment over time—do not arise.

¹²Our results are robust to restricting the sample to families that are matched to a single county or to the highest-probability match only.

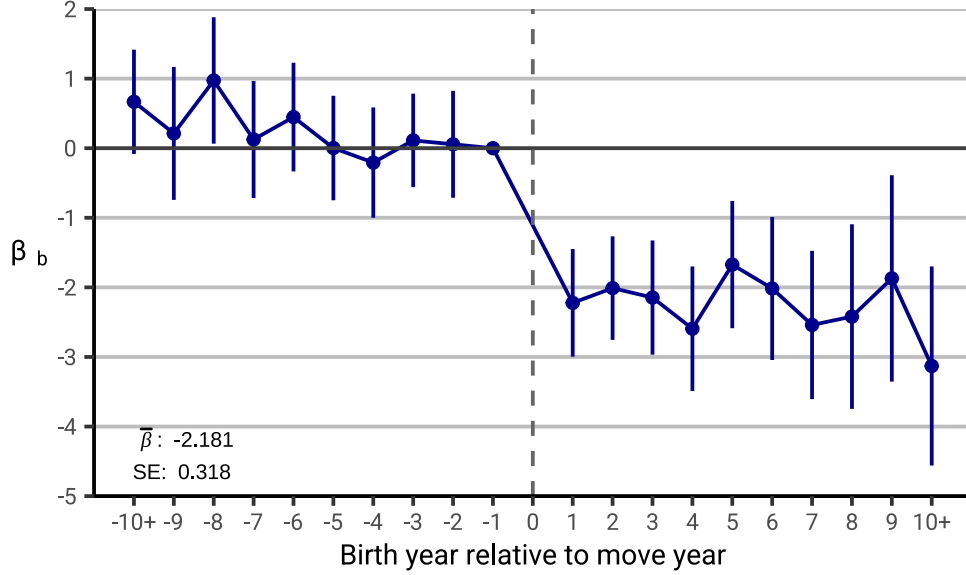


Figure 1: Migrants Become Less Parochially Oriented After Gaining Market Access

Note: This figure plots the estimates of β_b and 95% confidence intervals from the dynamic difference-in-differences equation (1). The dependent variable is children's LNI. $\bar{\beta}$ is the average treatment effect across post-migration periods weighted by the number of treated children born in each relative year.

This finding indicates that market integration weakens parochial orientation. Gaining market access, relative to losing it, reduces the tendency to give children locally distinctive names. This is consistent with a less locally bound social orientation, whether parents' attachments broaden or they become less constrained by local expectations. A separate possibility is that parents follow local naming practices to the same extent in every destination, but those practices are less locally distinctive in places with greater market access. We examine this interpretation, together with differences in exposure to non-local names, using the industry comparisons below.

The speed of the response is consistent with evidence of rapid cultural adjustment to changing environments (e.g., [Rao, 2019](#); [Bau, 2021](#); [Lowe, 2021](#); [Raz, 2025](#); [Ghosh et al., 2026a](#)). Two features of our setting matter for its interpretation. The estimate measures the *difference* in naming changes between migrants gaining and losing market access, rather than either group's absolute change. Moreover, these families have moved away from existing social networks into new economic environments, where learning about local expectations and the benefits of different forms of cooperation may be particularly rapid.

To examine whether this relative effect reflects responses to both gaining and losing market access, we present separate difference-in-differences comparisons for migrants from low- and high-market-access origins (Appendix Figure B.1). We classify origins and destinations separately using their respective census-year median market access across migrant family-route records. Values above the median are classified as high MA and the remaining values as low MA. Then, focusing on

migrants originating from low-MA counties, we estimate the impact of moving to a high-MA county relative to moving to another low-MA county. Similarly, focusing on migrants originating from high-MA counties, we estimate the impact of moving to a low-MA county relative to another high-MA county. We find opposite-signed responses in the two comparisons. Migrants moving from high- to low-MA counties show an increase in LNI relative to migrants moving to another high-MA county; those moving from low- to high-MA counties show a decrease. Migrants thus appear to respond to both gaining and losing market access.

Interpretation and Robustness. To explore robustness and tighten the interpretation of this finding, we examine how the naming response varies with the LNI measure, specification, treatment definition, sample, and inference (Appendix Figure C.1).

First, we employ two variants of the LNI to assess alternative explanations for its relative post-migration decline. We use the LNI to measure parental parochial orientation relative to the community in which the family lives, where its local economic relationships and expectations operate. The measure therefore treats the origin county as local for children born before migration and the destination county as local for children born afterwards. This change in reference community is part of the measure's intended interpretation. It also means that the same name can receive different scores in the origin and destination. For our difference-in-differences comparison, the concern is whether these scoring changes differ between families gaining and losing market access.

One possibility is that persistent attachment to the origin leads to a larger decline in LNI among families who gained market access than among those who lost it. Parents may continue to choose names characteristic of the community they left, but these names may be less distinctive of their destination. We examine this explanation using the *origin-fixed* LNI, which scores all children's names relative to the origin county. For families gaining rather than losing market access, we find that the origin-fixed LNI declines by approximately 0.5 points (p -value = 0.08). This suggests that market integration also weakened orientation towards the origin among families gaining market access relative to those losing it (Panel A).

Another possibility is that the change in what counts as "local" after migration lowers name scores more for families gaining rather than losing market access, even if parents continue to choose names in the same way. To assess this possibility, we use the *benchmark-adjusted* LNI, comparing the local distinctiveness of actual post-migration names with that of pre-migration same-sex sibling names, scoring both in the same destination county and census year. This actual-minus-benchmark difference is 0.87 points lower for families gaining rather than losing market access (p -value = 0.002; Panel B). On this comparison sample, the earlier-name benchmark therefore does not account for the full relative decline. This result suggests that changes in parents' naming behavior contribute to the relative decline in LNI, rather than the decline arising solely from the changes in

the local distribution of names.

Returning to the LNI, we find that the result is robust to controlling for gender, birth order, and 5-year cohort fixed effects; using a continuous treatment definition—the difference in log market access—instead of a binary indicator; measuring market access at alternative census dates; clustering by county of origin or the county of destination; longer event-time horizons; and excluding immigrants and non-Whites (Appendix Figure C.1, Panels C–L).

5.2 Evidence on the Mechanism

5.2.1 The Decline in Parochial Orientation is Stronger in Market-Oriented Industries

After identifying the impact of market access on migrants’ parochial orientation, we proceed to explore the mechanism behind it. We provide evidence suggesting that this reflects cultural adaptation in response to increased economic interdependence, which raised the returns to impersonal prosociality and tolerance relative to parochialism and kin-based prosociality (Prediction 2). Using the sample of domestic migrants and three related estimation frameworks, we show that the decline in parental parochial orientation is stronger among migrants whose livelihoods depend heavily on exchange with strangers and distant others.

Using the distinction between market-oriented and locally-oriented industry groups defined in Section 4.4 and Appendix A.4, we document a differential response to a change in market integration across migrants working in the two industrial classes.¹³ First, we use our difference-in-differences framework with a split-sample approach. We find that the decline in parochial orientation in response to higher market integration is stronger among families in market-oriented industries. Second, motivated by this heterogeneous treatment effect, we also estimate the following *Triple-Differences* specification:

$$\begin{aligned}
 LNI_i = & \delta_{b(i)} + \theta_{f(i)} + \sum_{b \neq -1} \alpha_b \cdot \mathbb{1}[b(i) = b] \cdot \mathbb{1}[\text{Market Oriented}_{f(i)}] + \\
 & \sum_{b \neq -1} \gamma_b \cdot \mathbb{1}[b(i) = b] \cdot [\text{MA}_{d(i)} > \text{MA}_{o(i)}] + \\
 & \sum_{b \neq -1} \beta_b \cdot \mathbb{1}[b(i) = b] \cdot \mathbb{1}[\text{Market Oriented}_{f(i)}] \cdot [\text{MA}_{d(i)} > \text{MA}_{o(i)}] + X_i \Omega + \epsilon_i
 \end{aligned} \tag{2}$$

¹³In interpreting our findings below, it is important to note that they are only meant to provide evidence on the likely channels, and do not necessarily represent causal estimates, as working in an industry highly dependent on exchange with strangers and distant others might itself be a result of the market access environment. We focus on migrants who remained in the same broad industry category before and after migration, so the comparison does not combine changes in location with switches between these categories. Remaining in a category is itself observed after migration, however, and the restriction does not eliminate selection into industry or identify a causal effect of industry exposure.

Where $\mathbb{1} [Market\ Oriented_{f(i)}]$ is a dummy variable that equals one if the father of a migrating family was working in a market-oriented industry both before and after the migration, and zero if he was working in a locally-oriented industry. All the other variables and specifications are the same as in Equation 1. The coefficients β_b identify the differential response to moving to a higher market access county across market-oriented and locally-oriented families in relative period b . As in the DD specification, the omitted period is $b = -1$. Our focus of interest is $\bar{\beta}$, the average post-migration treatment effect, weighted by the number of children born in each relative year.¹⁴

Our third framework compares families that moved from the same origin to the same destination in the same decade—and thus experienced the same change in environment—but differed in the industry group they worked in. We ask, did migrants who moved to a higher market access location and worked in a market-oriented industry were more likely to become less parochial, relative to those working in a locally-oriented industry? Specifically, we estimate:

$$Less\ Parochial_f = \gamma Market\ Oriented_f + \beta Market\ Oriented_f \cdot \mathbb{1} [MA_d > MA_o] + \delta_{dot} + X_f + \epsilon_f \quad (3)$$

where $Less\ Parochial_f$ is a dummy variable that equals one if the average LNI score of children born to migrating family f after the move is lower than that of children born before the migration. δ_{dot} is an origin-by-destination-by-year fixed effect, which also captures the main effect of moving to a county with higher versus lower market access. X_f is a vector of migrant characteristics, including age, race, birthplace, intra-communal marriage, and urban origin fixed effects, which we include as further robustness checks. The coefficient of interest is β , which captures the differential association between moving to a higher market-access location and becoming less parochial, driven by working in a market-oriented industry.¹⁵ Standard errors ϵ_f are two-way clustered at the counties of origin o and destination d .

Results. Figure 2 presents the results using the sample-split approach. For individuals in market-oriented industries, moving to a county with higher market access leads to an immediate and persistent decrease in parochial orientation, relative to moving to a lower market access county (Panel A). This pattern closely mirrors the findings from the full sample of migrants. For individuals working in locally-oriented industries, the estimates are close to zero and imprecise (Panel B). Nevertheless, we can reject equality of the average post-migration effects across the two groups. Specifically, the difference in the average post-migration treatment effect across the two groups is 2.29 points (p -value = 0.016).

¹⁴Because the split-sample DD and the DDD analysis are more statistically demanding, we aggregate the dynamic effects 5 or more years before and after migration, instead of 10, to increase power.

¹⁵In this framework, β is identified from several thousand cases in which there are multiple migrants within the same origin-destination-decade cell.

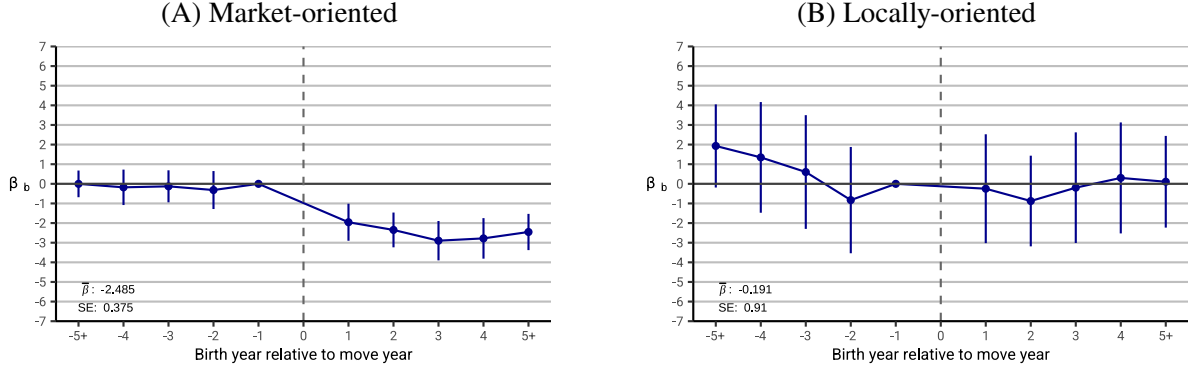


Figure 2: Market Access Has a Stronger Effect in Market-Oriented Industries

Note: This figure plots the estimates of β_b and 95% confidence intervals from the difference-in-differences equation (1). The dependent variable is children's LNI. In Panel A, the sample is restricted to households in which the father was working in a market-oriented industry before and after the migration. In Panel B, the sample is restricted to households in which the father was working in a locally-oriented industry before and after the migration. $\bar{\beta}$ is the average treatment effect across post-migration periods weighted by the number of treated children born in each relative year.

Column 2 of Panel A, Table 1, presents the DDD results. For families working in locally-oriented industries, the estimated effect of moving to a higher market access county is close to zero and imprecise. The response is significantly stronger for market-oriented families: a move that increases market access, compared with one that does not, reduces their LNI by an additional 2.28 points (p -value = 0.017) relative to locally-oriented families. This differential response is close in magnitude to, and statistically indistinguishable from, the baseline DD estimate in Column 1. Among families whose moves did not increase market access, the post-migration change in LNI is 2.48 points higher (p -value < 0.001) for market-oriented than for locally-oriented families. These findings suggest that adaptation to the market access environment is stronger among families working in market-oriented industries.

Finally, Column 2 of Panel B, Table 1, presents the results from our within origin-destination-decade comparison. Among migrants who moved from the same origin to the same destination in the same decade, the gap between market-oriented and locally-oriented families in the probability of becoming less parochially oriented is 6.28 percentage points (p -value = 0.029) higher when the move increases market access than when it does not. On moves that did not increase market access, market-oriented migrants were 5.01 percentage points (p -value = 0.021) less likely than locally-oriented migrants to become less parochially oriented. Appendix Table C.1 documents that this result remains stable when we add fixed effects for age, race, and birthplace, intra-communal marriage, and urban versus rural origin. As with the baseline estimate, both frameworks compare moves that increase market access with moves that do not. They therefore do not show whether market-oriented families respond more strongly to gaining access, to losing it, or to both;

Table 1: Cultural Adaptation is Stronger among Migrants Working in Market-oriented Industries

	(1)	(2)	(3)	(4)	(5)	(6)
Heterogeneity Variable:	Baseline DD	Market Orientation	Higher paying job	Out of Agriculture	Further From Kin	Fewer Shared Ancestry
<i>Panel A: Difference-in-Differences and Triple-Differences Designs</i> Dependent variable: LNI						
Post Migration × Higher Market Access	-2.181*** (0.318)	-0.204 (0.905)	-2.340*** (0.457)	-2.052*** (0.476)	-2.076*** (0.348)	-2.776*** (0.444)
Post Migration × Heterogeneity Variable		2.476*** (0.694)	0.289 (0.293)	-1.772*** (0.606)	0.349 (0.307)	-1.512*** (0.307)
Post Migration × Higher Market Access × Heterogeneity Variable		-2.282** (0.953)	0.234 (0.554)	0.274 (0.879)	-0.333 (0.565)	0.930* (0.491)
Observations	436,431	298,474	436,431	214,302	436,431	415,663
R ²	0.348	0.346	0.348	0.343	0.348	0.349
<i>Panel B: Within Origin-Destination-Decade Bins</i> Dependent variable: Less Parochial						
Heterogeneity Variable		-0.0501** (0.0216)	0.0067 (0.0113)	0.0463* (0.0279)	0.0004 (0.0105)	0.0187* (0.0113)
Higher Market Access × Heterogeneity Variable		0.0628** (0.0287)	-0.0080 (0.0176)	-0.0140 (0.0444)	-0.0092 (0.0198)	0.0105 (0.0148)
Observations		71,192	105,471	49,167	105,471	100,436
R ²		0.831	0.794	0.894	0.794	0.799

Note: Panel A reports the average treatment effect on children's LNI score in post-migration periods, weighted by the number of children born in each relative year, from the difference-in-differences framework (Eq. 1, column 1) and the triple-differences framework (Eq. 2, columns 2-6). Panel B reports estimates from the within origin-destination-decade bins framework (Eq. 3, columns 2-6). Each of columns 2-6 focuses on a different margin of heterogeneity across domestic migrants, indicated by the column headers. Standard errors two-way clustered at the county of destination and the county of origin in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Prediction 2 is consistent with each.

The stronger response among migrants working in market-oriented industries matters for our

interpretation in two ways. First, it weighs against the possibility that migrants simply adopt the naming conventions of their destination. Families moving along the same origin-destination route in the same decade encounter the same pool of local names and the same local information environment, yet market-oriented families' naming responds more to whether the move raised market access. The difference is consistent with two accounts that are difficult to separate: market-oriented families may have come to identify less with the local community and its norms, or their ties to distant markets may have made it less costly to depart from local expectations. In both, market ties loosen the hold of local expectations on families' choices. Second, it supports the hypothesis that direct exposure to exchange with strangers and economic interdependence with distant others, rather than county-level changes that affect all residents alike, shapes the degree of parochial orientation.

We conduct several additional tests to address possible concerns: First, we find that the effect is not primarily driven by farmers' families, as the response is similar for farmers and non-farmers in market-oriented industries (Appendix Figure B.3, Panel B). Second, random subsamples of the market-oriented group suggest that differences in sample size are highly unlikely to explain the null effect for locally-oriented jobs (Appendix Figure B.2). Third, while the groups differ in observed pre-migration characteristics (Appendix Table B.1), those differences are unlikely to explain the heterogeneous effects: the responses are similar across family size, urban origin, origin market access, and intra- versus inter-community marriages. Adaptation is stronger among families with higher prior LNI; however, even among those with low prior LNI, the estimated response is stronger for market-oriented than locally-oriented families (Appendix Figure B.3, Panels C–H). Finally, the results are similar under the alternative tradability classification described in Section 4.4 (Appendix Figure C.2 and Table C.2).

Taken together, these results indicate that the effect of market integration on parochial orientation is stronger among migrants whose livelihoods are more dependent on markets. This supports a cultural adaptation interpretation and suggests that the mechanism is linked to the direct and repeated experience of engaging in impersonal, market-based exchange with strangers and out-group members, or being dependent on them.

5.2.2 Alternative Explanations: Income, Modernization, and Intergroup Contact

Market integration could also shape naming choices through other channels, such as income growth, modernization, or changes in social contact. If these channels contributed to the naming response, the response should also vary with migrants' exposure to them. We use the same empirical

frameworks to assess these possibilities.¹⁶ Across these tests, we find no evidence that these channels account for the response to market access, although some are independently associated with less parochial naming.

A prominent alternative hypothesis is that market integration raised incomes, and that income growth, in turn, fostered cultural adaptation (e.g., [Inglehart, 1997](#); [Inglehart and Welzel, 2005](#)). We investigate this hypothesis through occupational mobility. Occupational income scores capture changes in occupational standing rather than earnings gains within occupations. Our comparison is informative about the income channel on the premise that, for a given change in market access, migrants moving down the occupational ladder experienced smaller income gains than those whose occupational score increased or remained unchanged. An income-driven explanation would then predict weaker cultural adaptation among downwardly mobile migrants.

Using the father’s occupational income score, we split the sample into two groups: those whose score rose or remained unchanged between the pre- and post-migration periods, and those whose score fell. We find similar cultural adaptation to an increase in market access in both groups. In the triple-differences framework, gaining market access reduces the LNI of downwardly mobile migrants by 2.34 points relative to losing market access (Table 1, Panel A, Column 3), close to the baseline estimate in Column 1; the response of migrants whose score rose or remained unchanged is slightly *weaker*, although the difference is statistically insignificant. The sample-split approach yields similar effects in both groups (Appendix Figure B.4, Panel B). The within-origin-destination-decade comparison sharpens the test, because migrants who moved from the same origin to the same destination in the same decade experienced the same change in market access. Among them, those whose occupational score rose or remained unchanged were not significantly more likely to become less parochial than those whose score fell, whether or not the move increased market access (Table 1, Panel B, Column 3). If a rise in income were the key driver of cultural adaptation, we would expect weaker responses among downwardly mobile migrants, which we do not observe. This weighs against an explanation operating primarily through income growth.

We next examine a modernization account operating through the transition out of agriculture. Leaving agricultural employment may change migrants’ routines, social relationships, and exposure to new norms. If this is central to adaptation, the response to greater market access should be stronger among migrants who leave agriculture than among those who remain farmers.¹⁷

Focusing on migrants who previously worked in agriculture, we find similar adaptation to a higher market access environment among those who transitioned out of agriculture and those who

¹⁶As with the industry comparison, these tests provide evidence on likely channels rather than causal estimates: changes in occupational standing, leaving agriculture, and living near kin or people of shared ancestry are observed after migration or depend on the destination chosen, and may themselves respond to the move.

¹⁷We have already shown that the effects on farmers and on non-farmers in market-oriented industries are similar (Appendix Figure B.3, Panel B).

remained farmers (Table 1, Column 4). Among those who remained farmers, gaining market access reduces the LNI by 2.05 points, close to the baseline estimate in Column 1 despite the narrower sample. The sample-split approach yields the same conclusion (Appendix Figure B.4, Panel C). Leaving agriculture does, however, predict less locally distinctive naming separately from market access: among families whose moves decrease market access, the post-migration change in LNI is 1.77 points lower for families who left farming than for those who remained farmers (p -value < 0.01), and this gap does not vary significantly with the change in market access. In the within-origin-destination-decade comparison, migrants who left agriculture are 4.6 percentage points more likely to become less parochial (p -value < 0.10), again with no significant difference by the change in market access (Table 1, Panel B, Column 4). The similar responses to market access weigh against agricultural exit as the principal route of cultural adaptation.

Another central alternative hypothesis is that market integration induced migration and increased population diversity, thereby reducing individuals' day-to-day social contact with a narrow in-group (e.g., family or shared ancestry) and increasing their contact with socially distant, diverse populations, which in turn shaped culture (Allport, 1954). To investigate this possibility, we first consider migrants' proximity to kin. Research on kinship links intensive family networks to a narrower scope of trust and cooperation (Schulz et al., 2019; Enke, 2019). Nearby relatives may allow everyday interaction to remain concentrated within family networks. We therefore expect moving away from kin to amplify the shift towards interaction with non-kin associated with greater market access. This provides a test of the contact explanation: if such interaction drives cultural adaptation, the naming response should be stronger among migrants who move away from kin. We examine this prediction using the kinship-proximity measure of Nelson (2020). We find similar responses to migrating to a higher market access environment among the subpopulation who moved away from kin—those who lived close to kin before migrating but not after—and for those who did not (Table 1, Column 5, and Appendix Figure B.4, Panel E). We also find similar responses among the subpopulation living close to kin after migration and among those who did not (Appendix Figure B.4, Panel D).

We next consider opportunities for contact across ancestry groups. Moving to a place with a smaller share of people of one's own ancestry than in the origin may expand opportunities for interaction with other groups. On the premise that this change amplifies the increase in intergroup contact associated with greater market access, the contact explanation predicts stronger adaptation among these migrants. Following Posch et al. (2026), we infer ancestry probabilistically from surnames, using the distribution of foreign countries of birth of each surname. We then compute for each migrating family's head of household the expected share of household heads in his origin and destination, defined as either the county or the enumeration district, who are of his ancestry.

The estimated adaptation response is not stronger among migrants who moved to places with a

smaller share of people of their ancestry (Table 1, Column 6, and Appendix Figure B.4, Panels F-G). In fact, the point estimate for the differential effect of moving to an area with a lower own-ancestry share than their origin, measured at the enumeration district (Table 1, Column 6), is positive, suggesting, if anything, a slightly weaker adaptation response for this group. If reduced contact with kin and people of shared ancestry were the channel through which market access operates, the response to market access should be stronger among migrants who moved away from kin or to places with a smaller share of people of their ancestry. We do not find this pattern. Moving to a place with less shared ancestry is nonetheless associated with lower LNI scores: among families whose moves decrease market access, the post-migration change in LNI is 1.51 points lower for those who moved to an enumeration district with a smaller own-ancestry share (p -value < 0.01 ; Table 1, Column 6). Among migrants whose own-ancestry share did not fall, the response to gaining market access remains large, at 2.78 points. Living farther from kin shows no comparable association (Table 1, Column 5).

Because market integration may shape multiple features of the socioeconomic environment, other pathways are possible. For example, railroad connectivity and market integration might have facilitated the spread of new ideas and information.¹⁸ Places with better access to information might have been more likely to encounter diverse populations and cultural norms, making them more open-minded to different ways of life.¹⁹ Market integration may also have contributed to the emergence and strengthening of local legal institutions that address the “fundamental problem of [impersonal] exchange” (Greif, 1993, 2000; Cantoni and Yuchtman, 2014). External enforcement by public-order institutions may foster impersonal prosociality and broaden the scope of cooperation (Tabellini, 2008; Gorodnichenko and Roland, 2017; Henrich, 2020; Eruchimovitch et al., 2024).

Beyond these conceptual alternatives, there are also empirical concerns about competing channels and mediators. For example, market integration may have enabled urbanization, and urbanization may in turn have brought income growth, economic interdependence, and more social contact with distant others. In that case, the adaptation response could reflect the socioeconomic changes accompanying urbanization. A related concern is that the LNI might pick up factors such as urbanization, economic growth, or the transition out of agriculture.

To address these concerns, we examine sensitivity to controls for changes in income, agricultural exit, industrialization, urbanization, population diversity, access to information, and local legal institutions. These controls assess whether the estimated naming response remains after accounting for observed differences in migrants’ socioeconomic environments.²⁰ We then augment the

¹⁸In our historical context, railroad connectivity was related to communication technologies such as mail, the telegraph, and newspaper circulation.

¹⁹Recall that our results are not sensitive to including or excluding immigrants and non-whites from the sample (Appendix Figure C.1).

²⁰Specifically, we assemble county controls for: the share of urban population, the share of immigrants, birthplace

difference-in-differences framework (Eq. 1) to account for the dynamic impact of moving to a county with higher versus lower levels of these competing channels and mediators. Specifically, we interact event-time indicators with an indicator for whether each measure is higher in the destination county than in the origin county and then examine how controlling for these differences affects the estimated impact of moving to a higher market access county on locally distinctive naming.²¹

We find that the estimated impact of moving to a higher market access county on parochial orientation remains negative and statistically significant when these controls are included separately or jointly (Appendix Figure B.5). With the full set of controls, the estimate is -1.43 LNI points (p -value < 0.001), compared with -2.07 without these controls estimated in the same sample. This reduction is statistically significant (p -value < 0.001). The contrast between market-oriented and locally-oriented industries also retains its sign, although it is less precisely estimated with the full set of controls (p -value $= 0.051$; Appendix Figure B.6).

In summary, we find no evidence that the naming response is stronger among migrants who maintain or improve their occupational standing, leave agriculture, or move away from kin. Leaving agriculture and moving to places with less shared ancestry are associated with less locally distinctive naming, but the naming response to market access also remains present in these comparisons. The response remains negative after accounting for differences in economic conditions, population composition, access to information, and legal institutions, although its magnitude is attenuated. Together with the stronger response in market-oriented industries, these findings point to a role for exchange with strangers and economic interdependence with distant others.

6 County-level Analysis

We now turn to our county-level analysis, which complements our migrant-level analysis in three important ways: First, it broadens the analysis beyond naming to impersonal prosociality, tolerance, and reliance on kin for economic support. Studying these outcomes allows us to examine whether the naming response is part of a wider change in social orientation as places become more market-integrated. Second, it focuses on the entire population rather than domestic migrants alone. Third, it strengthens causal inference by relying on a different framework and identification assumptions. We leverage variation in county-level market access over time and use four complementary empirical designs: (i) a baseline specification that relates changes in market access to cultural and behavioral changes conditional on a rich set of fixed effects and controls; (ii) a recentering approach following

diversity, ancestry diversity, log of one plus the number of manufacturing establishments, the share working in manufacturing, the share working in agriculture, the mean occupational income score, log real GDP per capita, the number of information workers per 1,000 workers, and the number of lawyers and judges per 1,000 workers.

²¹The additional controls take the following form: $\sum_{b \neq -1} \gamma_b \cdot \mathbb{1}[b(i) = b] \cdot \mathbb{1}[Mediator_{d(i)} > Mediator_{o(i)}]$.

Borusyak and Hull (2023); (iii) flexibly controlling for local railroad connectivity and population to focus on changes in the wider network; and (iv) two instrumental-variable strategies employed in previous work (Donaldson and Hornbeck, 2016; Hornbeck and Rotemberg, 2024).

6.1 Empirical Strategy

We estimate the effect of market access at the county level, following the empirical strategy in Donaldson and Hornbeck (2016). Our regression equation is:

$$outcome_{ot} = \beta \ln(MA_{ot}) + \delta_o + \delta_{s(o)t} + f(x_o, y_o) \times \delta_t + X'_{ot}\gamma + \epsilon_{ot} \quad (4)$$

where o indexes counties and t indexes years. Our key explanatory variable is the log of market access, $\ln(MA_{ot})$, and the coefficient of interest is β . The specification includes county fixed effects (δ_o) to absorb all time-invariant differences across counties and state-by-year fixed effects ($\delta_{s(o)t}$) to capture common state-level trends and shocks over time. It also flexibly accounts for smooth geographic trends by including cubic polynomials in longitude and latitude interacted with year fixed effects ($f(x_o, y_o) \times \delta_t$). Finally, X_{ot} denotes additional time-varying controls that we introduce below to strengthen a causal interpretation and to probe competing mechanisms and mediators. Because the treatment is spatially assigned, we cluster standard errors using arbitrary 100-mile square spatial grids to account for spatial autocorrelation (Bester et al., 2011). Our results are highly robust to using grid sizes between 50 and 250 miles and to clustering at the state level.

The identifying assumption is conditional exogeneity: conditional on the controls in (4), changes in market access are as good as random with respect to other unobserved determinants of the outcomes we study. County fixed effects imply that identification comes from within-county changes in market access over time, after netting out all average cross-sectional differences. The state-by-year fixed effects remove all time variation common to counties within the same state, including state-level policies and shocks related to both geography and institutions, so that identification comes from *excess* changes in counties' market access relative to other counties in the same state. The cubic spatial polynomial interacted with year fixed effects further removes time-varying smooth spatial patterns, so that the remaining variation in market access reflects county-specific deviations from both their state and broad national spatial trends.

Even with this rich set of controls, two main concerns remain. First, reverse causality: counties that became less parochial might have been more successful in lobbying for, or attracting, local railroad construction, which in turn increased their market access. Second, omitted variable bias: unobserved factors—such as income growth or prospective economic development—might have simultaneously influenced both railroad construction and our outcomes of interest.

We address these concerns in three ways. First, we implement the recentering approach recom-

mended by [Borusyak and Hull \(2023\)](#), accounting for the component of treatment associated with its “expected” value. Following [Hornbeck and Rotemberg \(2024\)](#), we define the expected treatment as a measure of market access that excludes railroads but includes the network of canals that might have been built instead, as proposed by [Fogel \(1964\)](#).²² We then recenter the treatment by adding log expected market access as a control.

Second, we follow [Donaldson and Hornbeck \(2016\)](#) and [Hornbeck and Rotemberg \(2024\)](#) and leverage the fact that a county’s market access depends not only on railroad construction within or near that county, but also on developments in distant counties throughout the broader transportation network, which are more plausibly exogenous to local conditions. By including flexible time-varying controls for the presence and extent of local railroads in and around each county, we isolate variation in market access that is orthogonal to the included measures of local railroad connectivity.²³ Following the same logic, we also include flexible controls for local population in and around each county. This further mitigates concerns that local population shifts—which might respond endogenously to local culture or behavior—could bias our estimates.²⁴ With these controls in place, identification relies on plausibly exogenous changes in the broader rail and population network.²⁵ In addition, we combine both approaches by jointly controlling for expected market access and for local railroad connectivity and population.

Third, for completeness, we implement two instrumental-variable strategies used in the literature. The first applies the alternative recentering IV suggested by [Borusyak and Hull \(2023\)](#), instrumenting market access with $\ln(\overline{MA_{ot}}) = \ln(MA_{ot}) - \ln(\mu_{ot})$, where $\ln(\mu_{ot})$ is log expected market access. The second follows [Donaldson and Hornbeck \(2016\)](#) and [Hornbeck and Rotemberg \(2024\)](#) and uses preexisting low-cost waterway transportation to instrument for growth in market access.²⁶ We report these IV results in Appendix Table B.2; they are consistent with the main estimates.

²²This expected treatment measures market access in a counterfactual scenario in which railroad technology did not exist, but extensive investment in network canals occurred instead to facilitate low-cost transportation and efficient trade flows, given the counterfactual technology. We share the view of [Hornbeck and Rotemberg \(2024\)](#) that effectively random placement and timing of railroads is not plausible in this setting. The canal counterfactual is therefore not an expectation over randomly assigned railroad lines; we use it as the closest available approximation to the market access counties could have expected.

²³Specifically, we add a dummy for railroad presence in the county, a cubic polynomial in total railroad mileage in the county, and controls for railroad presence and a cubic polynomial in railroad mileage within a 10, 20, 30, and 40-mile buffer around each county.

²⁴Note that a county’s own population size does not enter the market access measure; see Section 4.1.

²⁵Local railroad connectivity and the population of surrounding counties are components of market access itself rather than conventional covariates. Controlling for them removes the local component of the treatment, which is most exposed to concerns about reverse causality and local shocks, so that the estimate relies on changes in market access driven by developments in more distant parts of the network. The attenuation of the coefficient when these controls are added therefore reflects, at least in part, a change in the variation used for identification, and does not by itself measure selection on unobservables in the sense of [Oster \(2019\)](#).

²⁶Specifically, we use log water market access in 1850 interacted with year fixed effects to predict changes in log market access in subsequent decades.

6.2 Results

We report the results in Table 2. The first outcome is the composite universalism index. In our baseline specification (column 1), a 10 percent increase in market access raises the composite Universalism Index by 0.053 points (p -value < 0.001), which corresponds to about 2.5 percent of a standard deviation.²⁷ Appendix Figure B.7 shows that this relationship is approximately linear and not driven by outliers. Recentering market access has little impact on the estimate (column 2). With the full local railroad and population controls (column 3), the coefficient is smaller but remains economically meaningful, at about 1.5 percent of a standard deviation, and highly statistically significant (p -value < 0.001). Combining these controls with recentering yields a similar estimate (column 4). We obtain comparable coefficients when we use the two instrumental-variables strategies described above (Appendix Table B.2). In column 5, we examine the robustness of our results to accounting for the potential effects of several competing channels and possible mediators, in a parallel strategy to the one implemented in the migrant-level analysis. Specifically, we add measures of economic development, population composition, access to information, and legal institutions, some of which may themselves respond to market access, to assess the sensitivity of our estimates. Including these controls has little impact on the estimate. Appendix Figure B.8 further reports that a negative and economically and statistically significant coefficient on the universalism index is maintained for all possible combinations of these controls. Our finding is thus remarkably robust to conditioning on these measured factors.

The remaining rows present results for parochialism, impersonal prosociality, reliance on family, and tolerance. In our baseline specification (column 1), a 10 percent increase in market access reduces the average LNI by 0.088 points (p -value < 0.001), or about 1.3 percent of a standard deviation. The relationship is linear and not driven by outliers (Appendix Figure B.9, Panel A). A similar effect is estimated when market access is recentered (column 2). With the railroad and population controls (column 3), the effect remains sizable at about 0.6 percent of a standard deviation (p -value = 0.019). Combining both recentering and local controls yields very similar findings (column 4).

The family-care estimates show that the share of vulnerable individuals living with relatives at home declines with market access. A 10 percent increase in market access reduces this share by 0.12 percentage points (p -value < 0.001), an effect size of about 1.1 percent of a standard deviation. Recentering and adding local railroad and population controls leave the estimates largely unchanged. The relationship is not driven by outliers (Appendix Figure B.9, Panel B).

The NTI and RDI estimates show strong effects on cultural tolerance. A 10 percent increase in market access raises the Norm Tolerance Index (NTI) and the Religious Diversity Index (RDI)

²⁷ $\log(1.1) \times 0.5601 \approx 0.053$.

Table 2: Market Access Fosters Universalism, Impersonal Prosociality, and Tolerance

Dependent variable	Baseline	Recentering	Local controls	Both	Both + extra ctrls.
	(1)	(2)	(3)	(4)	(5)
Composite universalism index	0.5602***	0.5508***	0.3378***	0.3344***	0.3170***
Mean = 0.106; SD = 2.139	(0.0642)	(0.0653)	(0.0705)	(0.0717)	(0.0783)
Observations	16,574	16,558	16,574	16,558	11,603
Local Name Index	-0.9250***	-0.8958***	-0.4051**	-0.3839**	-0.6635***
Mean = 67.82; SD = 6.6	(0.1610)	(0.1614)	(0.1724)	(0.1755)	(0.2311)
Observations	18,182	18,154	18,182	18,154	12,250
Share in family care	-0.0123***	-0.0114***	-0.0105***	-0.0095**	-0.0116***
Mean = 0.778; SD = 0.111	(0.0032)	(0.0032)	(0.0040)	(0.0040)	(0.0043)
Observations	18,173	18,145	18,173	18,145	12,245
Norms Tolerance Index	0.1704***	0.1673***	0.1388***	0.1347***	0.1556***
Mean = 0; SD = 1	(0.0280)	(0.0279)	(0.0314)	(0.0312)	(0.0401)
Observations	17,112	17,087	17,112	17,087	11,948
Religious Diversity Index	0.2678***	0.2538***	0.1893***	0.1750***	0.1089**
Mean = 0; SD = 1	(0.0346)	(0.0345)	(0.0384)	(0.0379)	(0.0481)
Observations	17,303	17,283	17,303	17,283	9,349
Presidential voter turnout	0.0375***	0.0371***	0.0205***	0.0202***	0.0219***
Mean = 0.63; SD = 0.24	(0.0061)	(0.0062)	(0.0064)	(0.0065)	(0.0078)
Observations	45,308	45,268	45,308	45,268	31,511
Social trust	0.1201**	0.1194**	0.1291**	0.1274**	0.0208
Mean = 0.002; SD = 0.998	(0.0484)	(0.0484)	(0.0632)	(0.0632)	(0.0751)
Observations	6,821	6,812	6,821	6,812	4,244

Note: This table reports estimates of equation (4). Cells contain the coefficient on log market access, its standard error in parentheses, and the number of observations. All columns include county fixed effects, state-by-year fixed effects, and a cubic spatial polynomial interacted with year fixed effects. Column 1 is the baseline. Column 2 controls for expected log market access using [Borusyak and Hull \(2023\)](#)'s recentering approach and [Fogel \(1964\)](#)'s proposed canal. Column 3 adds local railroad and population controls to the baseline; column 4 combines these with recentering. Railroad controls comprise an any-railroad indicator and a cubic polynomial in railroad length, calculated within the county and within 10-, 20-, 30-, and 40-mile buffers. Population controls are cubic polynomials in total population within the county and the same buffers. Column 5 additionally controls for the urban and immigrant shares; birthplace and ancestry diversity; log of one plus manufacturing establishments; manufacturing and agriculture employment shares; mean occupational income score; log real GDP per capita; and information workers and lawyers and judges per 1,000 workers. Standard errors are clustered in 100-mile-square grid cells ([Bester et al., 2011](#)). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

by 0.016 and 0.026 standard deviations, respectively (p -value < 0.001 in both cases, column 1). These are sizable effects. The relationships are linear (Appendix Figure [B.9](#), Panels C–D) and robust to recentering (column 2), to the full local railroad and population controls (column 3), and

to combining both approaches (column 4).

The turnout estimate in column 1 shows that a 10 percent increase in market access increases turnout by 0.36 percentage points, or 1.5 percent of a standard deviation (p -value < 0.001). This effect remains economically and statistically significant when the treatment is recentered and when we add local railroad and population controls (columns 2–4). The relationship is plotted in Appendix Figure B.9, Panel E, and again appears linear and not driven by outliers.

The final row reports the effects on social trust. We find that a 10 percent increase in market access increases social trust by 0.011 standard deviations (p -value = 0.014). The estimate is similar across columns 1–4, but is smaller and statistically insignificant in column 5, which adds controls for competing channels and possible mediators. This decline almost entirely reflects the smaller sample available for these controls.²⁸ Appendix Figure B.9, Panel F, visualizes the relationship.

Robustness. We conduct a range of robustness checks. Appendix Table C.3 shows that the results are not sensitive to including or excluding immigrants and non-whites when constructing county-level cultural indicators, suggesting that they are not mechanically driven by changing demographics or population diversity. Appendix Figure C.3 documents robustness to alternative methods for clustering standard errors to account for spatial autocorrelation, including clustering at the state level. Appendix Table C.4 shows that our results are robust to alternative values of the trade elasticity (θ) and the value of transported goods (P) when calculating market access, including those used by Hornbeck and Rotemberg (2024).²⁹ Appendix Table C.5 confirms that no single region of the country drives the results. Appendix Table C.6 documents robustness to winsorizing family care at different percentiles. Finally, we find a positive impact of market access on all three components of the NTI, capturing the dispersion of specific norms: the coefficients of variation of the mother’s age at the birth of her oldest co-resident child, the number of linked co-resident children, and the number of distinct families in the household.³⁰

Overall, our county-level analysis reinforces the findings from the migrant-level analysis and supports Prediction 1. Market integration had a broad and robust impact on both culture and behavior: it increased tolerance and impersonal prosociality while reducing reliance on kin for economic support.

²⁸The coefficient on log market access in column 4 is 0.127 (p -value = 0.045), based on 6,812 observations. Estimating the same specification on the 4,244 observations used in column 5 gives a coefficient of 0.022 (p -value = 0.780). Adding the column-5 controls yields a nearly identical coefficient of 0.021 (p -value = 0.782).

²⁹In our baseline analysis, we follow Donaldson and Hornbeck (2016) in setting the trade elasticity to $\theta = 8.22$ and the average value of transported goods per ton to $P = 35$. Our results are robust to using the alternative parameter values in Hornbeck and Rotemberg (2024)— $P = 38.7$ and $\theta = 3.05$ —and to any value of θ between 1 and 13. While the point estimates naturally vary with the parameters, we consistently find effects with the predicted signs that are statistically significant.

³⁰Results are not reported for brevity.

7 More Evidence on Mechanisms

In this section, we provide additional evidence on the mechanisms. First, we find limited evidence for selective sorting of incoming domestic migrant families on observed pre-migration characteristics (Section 7.1). Second, we document an increase in proxy measures for the extent of commerce, exchange with strangers, and economic interdependence with distant others in response to market integration (Section 7.2). Finally, we find that becoming less parochial after migrating is more strongly associated with child survival and real-property holdings when the move increased market access, consistent with positive returns to adaptation (Section 7.3).

7.1 Selective Sorting

One possibility is that people with stronger impersonal prosocial traits and related characteristics were more likely to move to areas that were becoming more market-integrated. If so, the positive relationship between market access and impersonal prosociality might simply reflect selective sorting with respect to where people chose to move, rather than cultural adaptation.

We explore this possibility using over 100,000 observations on migrating families. Specifically, using a family-level version of Equation (4), we test whether counties that experienced excess increases in market access attracted migrants with lower pre-migration parochialism, as measured by the average Local Name Index (LNI) of children born before migration and Intra-Community Marriage (ICM).³¹ We also explore selection more broadly using multiple relevant pre-migration attributes: the tendency to engage in broad, beneficial social interactions, as measured by a labor-force cooperation score and residence with non-kin; kinship propinquity and the percentage of residents from shared ancestry; the number of children; urban versus rural origin; nativity; occupational income score; and working in the agricultural, manufacturing, and trade sectors. For comparability, we standardize all outcomes to z-scores.

Figure 3 plots the results. We find little evidence that counties experiencing higher market access growth attracted a selective group of migrating families. Panel A plots the coefficients from the baseline specification (in parallel to column 1 of Table 2), while Panel B strengthens causal inference by recentering market access and additionally controlling for local railroad connectivity

³¹Our estimation equation takes the following form:

$$trait_{f,t-1} = \beta \ln(MA_{dt}) + \delta_d + \delta_{s(d)t} + f(x_d, y_d) \times \delta_t + X'_{dt} \gamma + \epsilon_{ft},$$

where f denotes a family that migrated into destination county d between period $t-1$ and period t . As in Equation (4), δ_d are destination-county fixed effects, $\delta_{s(d)t}$ are state-by-year fixed effects, $f(x_d, y_d) \times \delta_t$ are cubic polynomials in longitude and latitude interacted with year fixed effects, and X_{dt} are additional time-varying controls that we introduce to strengthen a causal interpretation. This specification, therefore, relates pre-migration traits to the same notion of excess changes in market access as in the county-level analysis. We use inverse migrant-family count weights so that all counties are given the same weight in the regression, regardless of the number of families migrating to them.

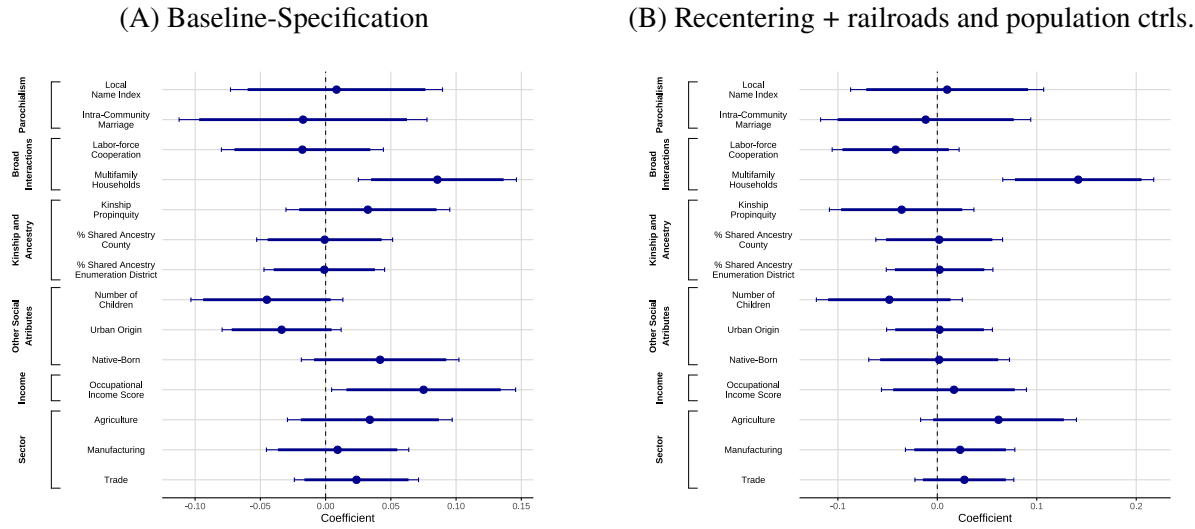


Figure 3: Market Access and the Pre-Migration Characteristics of Incoming Domestic Migrants

Note: This figure plots the estimates of β and 95% confidence intervals from a family-level version of Equation (4), estimated on a dataset of incoming domestic migrants into counties. The dependent variables are: mean LNI of children born before migration, ICM, labor-force cooperation, residing with a non-kin, kinship propinquity, percentage of residents from shared ancestry (at the enumeration district and at the county), the number of children, urban vs rural origin, nativity, occupational income score, and dummy variables for working in agriculture, manufacturing, and trade. All outcomes are measured pre-migration and standardized into z -scores. Panel A uses the baseline specification, in parallel to column 1 of Table 2. Panel B strengthens causal interpretation by additionally recentering market access and flexibly controlling for local railroad connectivity and population, in parallel to column 4 of Table 2. Standard errors are clustered at arbitrary grid cells of 100 miles square (Bester et al., 2011).

and population growth (in parallel to column 4 of Table 2). We find no relationship between excess changes in market access and prior levels of universalism among incoming migrants. Migrants to counties experiencing higher growth in market access are also no more or less likely to work in occupations that rely on impersonal interactions, although they are somewhat more likely to have previously lived with a non-kin. In addition, they do not differ systematically in kinship propinquity, ancestral enclave, number of children, urban versus rural origin, nativity, or the sectors in which they worked. We find evidence for positive selection on occupational income scores in Panel A, which does not survive the controls added to strengthen causal inference.

These results show little systematic association between changes in market access and the observed pre-migration characteristics of incoming domestic migrant families. Together with the within-family results, this points to adjustment among existing families rather than sorting, although the exercise does not decompose the county-level associations into adaptation and migration.

Table 3: Market Access and Commercial Relationships

	Trade share (mean = 0.055 , sd = 0.036)		Exchange with strangers (mean = 0.0 , sd = 1.0)		Distant-market dependence (mean = 0.0 , sd = 1.0)	
	(1)	(2)	(3)	(4)	(5)	(6)
Log market access	0.0051*** (0.0008)	0.0017* (0.0009)	0.2203*** (0.0531)	0.2498*** (0.0708)	0.2057*** (0.0439)	0.2119*** (0.0571)
Observations	18,266	18,238	6,809	6,800	6,809	6,800

Note: Coefficients on log market access from equation (4). Outcomes are the wholesale and retail employment share (columns 1–2), and standardized newspaper indices of exchange with strangers (3–4) and interdependence with distant others (5–6). All columns include county and state-by-year fixed effects and year-specific cubic spatial polynomials. Even columns also include expected market access and the full local railroad and population controls described in Section 6. Standard errors clustered in 100-mile-square grid cells are in parentheses (Bester et al., 2011). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

7.2 Commerce, Exchange with Strangers, and Interdependence with Distant Others

Using our county-level framework (Eq. 4), we show that counties with greater market access had higher employment shares in wholesale and retail trade and more newspaper content related to exchange with strangers and interdependence with distant others (Table 3). Our baseline specification in column 1 suggests that a 10 percent increase in market access raises the wholesale and retail employment share by 0.05 percentage points (p -value < 0.001), or about 1.3 percent of a standard deviation. When we additionally recenter market access (Borusyak and Hull, 2023) and control for railroad connectivity and population growth in and around each county, the effect attenuates to a 0.02 percentage point increase, significant at the 10% level (p -value = 0.075).

Columns 3–6 focus on exchange with strangers and economic interdependence with distant others, as measured from historical local newspapers. Columns 3 and 5 suggest that a 10 percent increase in market access raises the indices of newspaper discussion of exchange with strangers and interdependence by 0.02 standard deviations (p -value < 0.001). Recentering market access and controlling for local railroad connectivity and population growth have little impact on the effect (columns 4 and 6).

These relationships are approximately linear and not driven by outliers (Appendix Figure B.10). Appendix Table B.3 reports the results separately for the three components of each newspaper index. The impact in all six component estimates is positive. With the full set of controls, the estimates for commodity intermediaries, commodity buyers, and distant-market orientation are significant

at least at the 5 percent level. Credit dependence is significant at the 10 percent level, while the estimates for mobile commercial counterparties and standardization are imprecise. These results document changes in commercial employment and newspaper discussion consistent with the two channels.

Robustness. These results are robust to alternative clustering (Appendix Figure C.4), population samples (Appendix Table C.7), market-access parameters (Appendix Table C.8), regional exclusions (Appendix Table C.9), and treatments of outliers and skewness (Appendix Tables C.10 and C.11).

7.3 The Positive Returns to Adaptation

Finally, we present evidence suggesting that adapting to the market-integrated environment benefited migrants and their families. Using our within origin-destination-decade bins framework (Eq. 3), we compare families that moved from the same origin to the same destination in the same decade—and thus experienced the same change in environment—but differed in whether they became less parochial after the move. While this test is informative, it provides only suggestive evidence and does not causally identify the returns to adaptation. We estimate:

$$success_f = \gamma Less\ Parochial_f + \beta Less\ Parochial_f \cdot \mathbb{1}[MA_d > MA_o] + \delta_{dot} + X_f \Omega + \epsilon_f \quad (5)$$

where $success_f$ denotes two measures of success available in the historical censuses: the survival rate of children (available for 1900–1910),³² and the value of real property owned (available for 1850–1870),³³ and the rest of the variables are the same as in Equation 3.

We find that the association between becoming less parochial and family outcomes is more positive among migrants whose moves increased market access. Column 1 of Table 4 shows that the association between a decline in LNI and children’s survival is 1.82 percentage points larger among families whose moves increased market access than among families whose moves did not (p -value = 0.026), or about 11% of a standard deviation. This difference remains similar when we add fixed effects for age, race, and birthplace (column 2) and when we further control for ICM and urban versus rural origin (column 3). Column 4 shows a corresponding difference of approximately \$561 in the association with real property (p -value = 0.053), roughly 21% of a standard deviation. This difference remains positive when we include individual-level controls (columns 5 and 6).³⁴

³²To calculate children’s survival rate, we rely on IPUMS variables CHBORN and CHSURV.

³³We use IPUMS variable REALPROP, available for 1850–1870. Code zero combines blank responses with recorded zeros. We treat it as missing, and analyze real-property holdings conditional on positive recorded values.

³⁴We find no comparable effect on personal property. When we use total property value (real plus personal property) as the dependent variable, we find positive but insignificant effects.

Table 4: Cultural Adaptation and Family Outcomes

	Children Survival Rate (mean = 0.879 , sd = 0.162)			Real Property Value (mean = 2333.2 , sd = 2716.6)		
	(1)	(2)	(3)	(4)	(5)	(6)
Less Parochial	-0.0015 (0.0063)	-0.0002 (0.0066)	-0.0003 (0.0066)	-84.71 (128.2)	-77.87 (123.0)	-79.10 (122.6)
Higher Market Access × Less Parochial	0.0182** (0.0082)	0.0177** (0.0087)	0.0177** (0.0087)	560.8* (290.1)	607.7** (255.8)	599.1** (251.6)
Observations	24,797	24,797	24,797	23,769	23,769	23,769
Individual Ctrl. (demographics)		Yes	Yes		Yes	Yes
Individual Ctrl. (traits)			Yes			Yes

Note: This table reports estimates of equation (5). The dependent variables are different measures of success: children's survival rate (columns 1–3) and real property value (columns 4–6). All columns include origin-by-destination-by-year fixed effects. Individual demographic controls include age, race, and birthplace fixed effects. Individual traits controls include fixed effects for ICM and an urban origin dummy. Standard errors two-way clustered at the county of destination and the county of origin in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

These patterns are consistent with the idea that people adapt their cultural traits to reduce costly mismatch between their values and their changing economic environment (Nunn, 2021). Market integration increased economic interdependence and created more opportunities for beneficial exchanges with socially distant individuals. Those who adapted appear to have been better positioned to exploit these opportunities.

Robustness. The real-property comparison is similar under alternative winsorization thresholds and treatments of the long right tail (Appendix Tables C.12 and C.13).

8 Conclusions

We study how U.S. market integration between 1850 and 1920 reshaped social orientation and reliance on relatives for economic support. Within domestic migrant families, gaining rather than losing market access led to a rapid and persistent shift towards less locally distinctive naming. The county analysis extends the evidence to civic participation, trust, tolerance and family support arrangements. The naming response is stronger among families in market-oriented industries. Alongside increases in commercial employment and newspaper discussion of exchange with strangers and distant-market dependence, this pattern is consistent with adaptation to wider economic relationships. Several tests of potential alternative channels and socioeconomic controls

support this interpretation. Among families making the same migration journey, becoming less parochial is also more positively associated with child survival and real-property holdings when the move increased market access, providing suggestive evidence of benefits from adaptation.

Our findings speak to the long-standing debate about markets and social life. Broader social orientations can develop alongside reduced reliance on family-based support. The historical experience of the United States suggests that markets can reshape the cultural boundaries of cooperation as well as patterns of production and consumption. Market integration may therefore help create the cultural foundations for cooperation beyond the local community, even as economic life becomes less dependent on family and local ties.

References

- Abadie, A., S. Athey, G. W. Imbens, and J. M. Wooldridge, “When Should You Adjust Standard Errors for Clustering?,” *The Quarterly Journal of Economics*, 2023, 138 (1), 1–35.
- Abramitzky, R., L. Boustan, K. Eriksson, M. Rashid, and S. Pérez, *Census Linking Project: Census-to-Census Crosswalks* 2022. Dataset. Harvard Dataverse crosswalk releases; <https://censuslinkingproject.org>.
- , L. P. Boustan, and K. Eriksson, “A nation of immigrants: Assimilation and economic outcomes in the age of mass migration,” *Journal of Political Economy*, 2014, 122 (3), 467–506.
- Agneman, G. and E. Chevrot-Bianco, “Market Participation and Moral Decision-Making: Experimental Evidence from Greenland,” *The Economic Journal*, February 2023, 133 (650), 537–581.
- Alesina, A. and E. L. Ferrara, “Participation in heterogeneous communities,” *The Quarterly Journal of Economics*, 2000, 115 (3), 847–904.
- Allport, G. W., *The Nature of Prejudice*, Cambridge, MA: Addison-Wesley, 1954.
- Anelli, M., M. Morelli, and M. Pappaletta, “Railroads and the Silver Shoes of Populism: The Rise and Fall of the People’s Party in 19th-Century America,” *BAFFI CAREFIN Centre Working Paper No. 269*, Bocconi University, 2026.
- Atack, J., F. Bateman, M. Haines, and R. A. Margo, “Did railroads induce or follow economic growth?: Urbanization and population growth in the American Midwest, 1850–1860,” *Social Science History*, 2010, 34 (2), 171–197.
- , M. Haines, and R. A. Margo, “Railroads and the Rise of the Factory: Evidence for the United States, 1850–1870,” in P. W. Rhode, J. L. Rosenbloom, and D. F. Weiman, eds., *Economic Evolution and Revolution in Historical Time*, Stanford University Press, 2011, pp. 162–179.
- Atari, M., A. Omrani, and M. Dehghani, “Contextualized Construct Representation: Leveraging Psychometric Scales to Advance Theory-Driven Text Analysis,” February 2023. PsyArXiv preprint.
- Atherton, L. E., *The Frontier Merchant in Mid-America*, Columbia, MO: University of Missouri Press, 1971.
- Baldassarri, D., “Market integration accounts for local variation in generalized altruism in a nationwide lost-letter experiment,” *Proceedings of the National Academy of Sciences*, February 2020, 117 (6), 2858–2863.

- Barkai, S. and E. Karger**, “Classifying Industries into Tradable/Nontradable via Geographic Radius Served,” *Working paper*, 2020. Available at SSRN: <https://ssrn.com/abstract=4306269> or <http://dx.doi.org/10.2139/ssrn.4306269>.
- Bartling, B., E. Fehr, and Y. Özdemir**, “Does Market Interaction Erode Moral Values?,” *The Review of Economics and Statistics*, January 2023, 105 (1), 226–235.
- Bau, N.**, “Can Policy Change Culture? Government Pension Plans and Traditional Kinship Practices,” *American Economic Review*, June 2021, 111 (6), 1880–1917.
- Bazzi, S., M. Fiszbein, and M. Garcia**, “The Moral Values of “Rugged Individualism,”” Working Paper 32433, National Bureau of Economic Research May 2024.
- , —, and **M. Gebresilas**, “Frontier Culture: The Roots and Persistence of “Rugged Individualism” in the United States,” *Econometrica*, 2020, 88 (6), 2329–2368.
- Bertrand, M., E. Duflo, and S. Mullainathan**, “How much should we trust Differences-In-Differences estimates?,” *The Quarterly Journal of Economics*, 2004, 119 (1), 249–275.
- Bester, A. C., T. G. Conley, and C. B. Hansen**, “Inference with dependent data using cluster covariance estimators,” *Journal of Econometrics*, 2011, 165 (2), 137–151.
- Bisin, A. and T. Verdier**, “The Economics of Cultural Transmission and Socialization,” in “Handbook of Social Economics,” Vol. 1, Elsevier, 2011, pp. 339–416.
- Borusyak, K. and P. Hull**, “Nonrandom exposure to exogenous shocks,” *Econometrica*, 2023, 91 (6), 2155–2185.
- , **X. Jaravel, and J. Spiess**, “Revisiting event-study designs: robust and efficient estimation,” *Review of Economic Studies*, 2024, 91 (6), 3253–3285.
- Boyd, R. and P. J. Richerson**, *Culture and the Evolutionary Process*, University of Chicago Press, 1985.
- and **P. J. Richerson**, “Group Beneficial Norms Can Spread Rapidly in a Structured Population,” *Journal of Theoretical Biology*, April 2002, 215 (3), 287–296.
- Bruegel, M.**, *Farm, Shop, Landing: The Rise of a Market Society in the Hudson Valley, 1780–1860*, Durham, NC: Duke University Press, 2002.
- Burszty, L., T. Chaney, T. A. Hassan, and A. Rao**, “The Immigrant Next Door,” *American Economic Review*, February 2024, 114 (2), 348–384.
- Cantoni, D. and N. Yuchtman**, “Medieval universities, legal institutions, and the commercial revolution,” *The Quarterly Journal of Economics*, 2014, 129 (2), 823–887.
- Chan, J.**, “Farming Output, Concentration, and Market Access: Evidence from the 19th-Century American Railroad Expansion,” *Journal of Development Economics*, June 2022, 157, 102878.
- Clark, C.**, *The Roots of Rural Capitalism: Western Massachusetts, 1780–1860*, Ithaca, NY: Cornell University Press, 1990.
- Cronon, W.**, *Nature’s Metropolis: Chicago and the Great West*, New York: W. W. Norton, 1991.
- d. S. Montesquieu, C.**, *The Spirit of Laws* Eighteenth Century Collections Online, the second edition corrected and considerably improved. ed., London: Printed for J. Nourse, and P. Vaillant, 1748.
- de Chaisemartin, C. and X. d’Haultfœuille**, “Two-way fixed effects estimators with heterogeneous treatment effects,” *American Economic Review*, 2020, 110 (9), 2964–2996.
- Dimant, E., M. Gelfand, A. Hochleitner, and S. Sonderegger**, “Strategic Behavior with Tight, Loose, and Polarized Norms,” *Management Science*, March 2025, 71 (3), 2245–2263.
- Donaldson, D. and R. Hornbeck**, “Railroads and American Economic Growth: A “Market Access” Approach,” *The Quarterly Journal of Economics*, May 2016, 131 (2), 799–858.

- Enke, B.**, “Kinship, Cooperation, and the Evolution of Moral Systems,” *The Quarterly Journal of Economics*, May 2019, 134 (2), 953–1019.
- , “Market Exposure and Human Morality,” *Nature Human Behaviour*, 2023, 7 (1), 134–141.
- , “Moral Boundaries,” *Annual Review of Economics*, August 2024, 16 (1), 133–157.
- Eruchimovitch, I., M. Michaeli, and A. Sarid**, “On the coevolution of individualism and institutions,” *Journal of Economic Growth*, 2024, 29 (3), 391–432.
- Falk, A. and N. Szech**, “Morals and Markets,” *Science*, May 2013, 340 (6133), 707–711.
- Fogel, R. W.**, *Railroads and American Economic Growth*, Johns Hopkins Press Baltimore, 1964.
- Fouka, V. and T. Serlin**, “Industry and Identity: The Migration Linkage Between Economic and Cultural Change in 19th Century Britain,” Technical Report w33114, National Bureau of Economic Research, Cambridge, MA November 2024.
- Francois, P., T. Fujiwara, and T. van Ypersele**, “The origins of human prosociality: Cultural group selection in the workplace and the laboratory,” *Science Advances*, 2018, 4 (9), eaat2201.
- Fryer, R. and S. Levitt**, “The causes and consequences of distinctively Black names,” *The Quarterly Journal of Economics*, 2004, 119 (3), 767–805.
- Fulford, S. L., I. Petkov, and F. Schiantarelli**, “Does it matter where you came from? Ancestry composition and economic performance of US counties, 1850–2010,” *Journal of Economic Growth*, 2020, 25 (3), 341–380.
- Gelfand, M. J., J. L. Raver, L. Nishii, L. M. Leslie, J. Lun, B. C. Lim, L. Duan, A. Al-maliach, S. Ang, J. Arnadottir, Z. Aycan, K. Boehnke, P. Boski, R. Cabecinhas, D. Chan, J. Chhokar, A. D’Amato, M. Ferrer, I. C. Fischlmayr, R. Fischer, M. Fülöp, J. Georgas, E. S. Kashima, Y. Kashima, K. Kim, A. Lempereur, P. Marquez, R. Othman, B. Overlaet, P. Panagiotopoulou, K. Peltzer, L. R. Perez-Florizno, L. Ponomarenko, A. Realo, V. Schei, M. Schmitt, P. B. Smith, N. Soomro, E. Szabo, N. Taveesin, M. Toyama, E. V. de Vliert, N. Vohra, C. Ward, and S. Yamaguchi**, “Differences Between Tight and Loose Cultures: A 33-Nation Study,” *Science*, May 2011, 332 (6033), 1100–1104.
- Gelfand, M. J., L. H. Nishii, and J. L. Raver**, “On the nature and importance of cultural tightness-looseness,” *Journal of applied psychology*, 2006, 91 (6), 1225–1244.
- Ghosh, A., P. Kundu, M. Lowe, and G. Nellis**, “Creating Cohesive Communities: A Youth Camp Experiment in India,” *The Review of Economic Studies*, 2026, 93 (1), 438–475.
- , **S. I. M. Hwang, and M. Squires**, “Economic consequences of kinship: Evidence from US bans on cousin marriage,” *The Quarterly Journal of Economics*, 2023, 138 (4), 2559–2606.
- , —, and —, “Industrialization and Integration in America: Evidence from Marriage Networks,” *Unpublished Manuscript*, 2026.
- Goodman-Bacon, A.**, “Difference-in-differences with variation in treatment timing,” *Journal of Econometrics*, 2021, 225 (2), 254–277.
- Gordon, R.**, *The rise and fall of American growth: The US standard of living since the civil war*, Princeton university press, 2017.
- Gorodnichenko, Y. and G. Roland**, “Culture, Institutions, and the Wealth of Nations,” *The Review of Economics and Statistics*, 2017, 99 (3), 402–416.
- Greif, A.**, “Contract Enforceability and Economic Institutions in Early Trade: The Maghribi Traders’ Coalition,” *The American Economic Review*, 1993, 83 (3), 525–548.
- , “The fundamental problem of exchange: a research agenda in historical institutional analysis,” *European Review of Economic History*, 2000, 4 (3), 251–284.
- **and G. Tabellini**, “The clan and the corporation: Sustaining cooperation in China and Europe,”

- Journal of Comparative Economics*, 2017, 45 (1), 1–35.
- , **J. Mokyr**, and **G. Tabellini**, *Two Paths to Prosperity: Culture and Institutions in Europe and China, 1000–2000*, Princeton University Press, 2025.
- Grosfeld, I., S. O. Sakalli, and E. Zhuravskaya**, “Middleman Minorities and Ethnic Violence: Anti-Jewish Pogroms in the Russian Empire,” *The Review of Economic Studies*, January 2020, 87 (1), 289–342.
- Henrich, J.**, *The WEIRDest People in the World: How the West Became Psychologically Peculiar and Particularly Prosperous*, London: Penguin, 2020.
- , **J. Ensminger, R. McElreath, A. Barr, C. Barrett, A. Bolyanatz, J. C. Cardenas, M. Gurven, E. Gwako, N. Henrich, C. Lesorogol, F. Marlowe, D. Tracer, and J. Ziker**, “Markets, Religion, Community Size, and the Evolution of Fairness and Punishment,” *Science*, March 2010, 327 (5972), 1480–1484.
- , **R. Boyd, S. Bowles, C. Camerer, E. Fehr, and H. Gintis**, eds, *Foundations of Human Sociality: Economic Experiments and Ethnographic Evidence from Fifteen Small-Scale Societies*, Oxford University Press, March 2004.
- Hirschman, A. O.**, “Rival Interpretations of Market Society: Civilizing, Destructive, or Feeble?,” *Journal of Economic Literature*, 1982, 20 (4), 1463–1484.
- Hornbeck, R.**, “Barbed wire: Property rights and agricultural development,” *The Quarterly Journal of Economics*, 2010, 125 (2), 767–810.
- and **M. Rotemberg**, “Growth Off the Rails: Aggregate Productivity Growth in Distorted Economies,” *Journal of Political Economy*, 2024, 132 (11), 3547–3602.
- Hume, D.**, *Essays, Moral, Political, and Literary*, London: A. Millar, 1758.
- Inglehart, R.**, *Modernization and Postmodernization: Cultural, Economic, and Political Change in 43 Societies*, Princeton University Press, May 1997.
- and **C. Welzel**, *Modernization, Cultural Change, and Democracy: The Human Development Sequence*, Cambridge, UK: Cambridge University Press, 2005.
- and **W. E. Baker**, “Modernization, Cultural Change, and the Persistence of Traditional Values,” *American Sociological Review*, 2000, 65 (1), 19–51.
- Inter-university Consortium for Political and Social Research**, “United States Historical Election Returns, 1824–1968 [dataset],” 1999. Inter-university Consortium for Political and Social Research [distributor].
- Jha, S.**, “Trade, Institutions, and Ethnic Tolerance: Evidence from South Asia,” *American Political Science Review*, November 2013, 107 (4), 806–832.
- and **M. Shayo**, “Valuing peace: the effects of financial market exposure on votes and political attitudes,” *Econometrica*, 2019, 87 (5), 1561–1588.
- , —, and **C. M. Weiss**, “Financial market exposure increases generalized trust,” *Journal of Public Economics*, 2025, 242, 105303.
- Lee, G. A.**, “The Historical Significance of the Chicago Grain Elevator System,” *Agricultural History*, 1937, 11 (1), 16–32.
- Leslie, T.**, “Chicago’s Other Skyscrapers: Grain Elevators and the City, 1838–1957,” *Journal of Urban History*, 2022, 48 (1), 3–34.
- Lowe, M.**, “Types of Contact: A Field Experiment on Collaborative and Adversarial Caste Integration,” *American Economic Review*, June 2021, 111 (6), 1807–1844.
- Lowes, S., N. Nunn, J. A. Robinson, and J. L. Weigel**, “The Evolution of Culture and Institutions: Evidence From the Kuba Kingdom,” *Econometrica*, 2017, 85 (4), 1065–1091.

- Manson, S., J. Schroeder, D. V. Riper, T. Kugler, and S. Ruggles**, “IPUMS National Historical Geographic Information System: Version 15.0 [dataset],” 2020. Minneapolis, MN: IPUMS.
- Margalit, Y. and M. Shayo**, “How markets shape values and political preferences: A field experiment,” *American Journal of Political Science*, 2021, 65 (2), 473–492.
- Nelson, M. A.**, “The decline of patrilineal kin propinquity in the United States, 1790–1940,” *Demographic research*, 2020, 43, 501–532.
- Ngoma, M., C. Sievert, X. Jaravel, N. Nunn, and J. L. Weigel**, “How Market Access Shapes Wellbeing and Values: Experimental Evidence from the D.R. Congo,” *Unpublished Manuscript*, 2025.
- Nunn, N.**, “History as Evolution,” in A. Bisin and G. Federico, eds., *The Handbook of Historical Economics*, Academic Press, January 2021, pp. 41–91.
- **and L. Wantchekon**, “The Slave Trade and the Origins of Mistrust in Africa,” *American Economic Review*, December 2011, 101 (7), 3221–3252.
- Oster, E.**, “Unobservable selection and coefficient stability: Theory and evidence,” *Journal of Business & Economic Statistics*, 2019, 37 (2), 187–204.
- Platteau, J.-P.**, *Institutions, Social Norms and Economic Development* Fundamentals of Development Economics, Amsterdam, The Netherlands: Harwood Academic Publishers, 2000.
- Polanyi, K.**, *The Great Transformation*, New York: Rinehart & Co., inc, 1944.
- Posch, M.**, “Do disasters affect the tightness of social norms,” *Unpublished Manuscript*, 2021.
- **, J. Schulz, and J. Henrich**, “How Cultural Diversity Drives Innovation: Surnames and Patents in U.S. History,” *Journal of Political Economy*, 2026, 134 (4), 1245–1305.
- Putnam, R. D.**, “Tuning in, tuning out: The strange disappearance of social capital in America,” *PS: Political science & politics*, 1995, 28 (4), 664–683.
- Rao, G.**, “Familiarity Does Not Breed Contempt: Generosity, Discrimination, and Diversity in Delhi Schools,” *American Economic Review*, March 2019, 109 (3), 774–809.
- Raz, I. T.**, “Soil Heterogeneity, Social Learning, and the Formation of Close-Knit Communities,” *Journal of Political Economy*, 2025, 133 (8), 2643–2691.
- Ruggles, S., S. Flood, R. Goeken, J. Grover, E. Meyer, J. Pacas, and M. Sobek**, “IPUMS USA: Version 10.0 [dataset],” 2020. Minneapolis, MN: IPUMS.
- Rupasingha, A., S. J. Goetz, and D. Freshwater**, “The production of social capital in US counties,” *The journal of socio-economics*, 2006, 35 (1), 83–101.
- Rustagi, D.**, “Historical Self-Governance and Norms of Cooperation,” *Econometrica*, 2024, 92 (5), 1473–1502.
- **, “Market Exposure, Civic Values, and Rules,”** 2026. Working paper.
- Schulz, J. F., D. Bahrami-Rad, J. P. Beauchamp, and J. Henrich**, “The Church, Intensive Kinship, and Global Psychological Variation,” *Science*, November 2019, 366 (6466), eaau5141.
- Smith, A.**, *An Inquiry into the Nature and Causes of the Wealth of Nations*, London: W. Strahan and T. Cadell, 1776.
- Snowden, K. A.**, “Mortgage Rates and American Capital Market Development in the Late Nineteenth Century,” *Journal of Economic History*, 1987, 47 (3), 671–691.
- Sun, L. and S. Abraham**, “Estimating dynamic treatment effects in event studies with heterogeneous treatment effects,” *Journal of Econometrics*, 2021, 225 (2), 175–199.
- Tabellini, G.**, “The Scope of Cooperation: Values and Incentives,” *The Quarterly Journal of Economics*, August 2008, 123 (3), 905–950.

Online Appendices

A Data and Measurement	45
A.1 Linked Census Data	45
A.2 Market Access	45
A.3 Outcome Measures	46
A.4 Mechanism Measures	54
A.5 Other Demographic and Economic Measures	60
B Further Figures, Analysis and Results	62
B.1 Migrant-level analysis	62
B.2 County-level analysis	68
B.3 More Evidence on Mechanisms	71
Appendix References	73

A Data and Measurement

A.1 Linked Census Data

We use Census Linking Project crosswalks for 1850–1860, 1860–1870, 1870–1880, 1900–1910, and 1910–1920. The crosswalk records cited here are the 2022 Harvard Dataverse V2 releases: 1850–1860, <https://doi.org/10.7910/DVN/K05J44>; 1860–1870, <https://doi.org/10.7910/DVN/TXNANS>; 1870–1880, <https://doi.org/10.7910/DVN/OCWCFR>; 1900–1910, <https://doi.org/10.7910/DVN/XUXYSR>; and 1910–1920, <https://doi.org/10.7910/DVN/Q2QJ2V>. We drop households living in group quarters (keeping GQ=1 or 2).

A.2 Market Access

Section 4.1 details the construction of MA_{ot} ; here we document its variation over space and time and its measurement properties. Appendix Figure A.1 plots the expansion of the railroad network

decade by decade, and Appendix Figure A.2 the resulting spatial distribution of log market access, generally highest in the Northeast and Midwest and lowest in the West. Panel A of Appendix Figure A.3 traces the rise of average log market access over the sample period and separates its two sources: recomputing MA_{ot} with transport costs held at their 1850 levels shows that the falling cost of moving goods—the expanding network—accounts for roughly four-fifths of the increase, and population growth for the remainder; because population growth itself partly responds to improved access, its share is, if anything, an overstatement. Panel B maps the change between 1850 and 1920; counties with low initial market access—typically less settled and less developed areas—experienced the largest increases in log market access.

Market access is a model-based measure of *potential* rather than realized integration: it abstracts from within-county transport frictions, proxies market size with population, and holds cost parameters fixed at their 1890 values, ignoring subsequent changes in railroad technology, pricing, and competition. These simplifications introduce measurement error. Our design addresses the concern in three ways: county fixed effects absorb time-invariant gaps between potential and realized integration; we re-estimate all main specifications under the parameter values of [Hornbeck and Rotemberg \(2024\)](#) and for $\theta \in \{1, 2, \dots, 13\}$; and Section 7.2 tests directly whether market access raised realized exposure to the two channels of our framework, as measured by employment in wholesale and retail trade and the newspaper-based indices of exchange with strangers and interdependence with distant others (Appendix A.4).

A.3 Outcome Measures

This section details the construction, validation, and limitations of the outcome measures. We build the census-based cultural measures from those of [Raz \(2025\)](#). We use the Local Name Index directly, so that higher values denote greater *parochial orientation*; the tolerance indices are oriented so that higher values denote greater *tolerance*—the Norms Tolerance Index from the dispersion of family norms and the Religious Diversity Index from denominational diversity.³⁵ The Local Name Index and Norms Tolerance Index drop households living in group quarters.

Local Name Index (LNI). The LNI is based on the relative frequency of children’s first names in the full-count censuses. Let $Pr(n \mid l, g, t)$ denote the probability that name n is given to a child of gender g aged 0–10 in county l in census year t , and $Pr(n \mid -l, g, t)$ the corresponding probability in all other counties. Following [Fryer and Levitt \(2004\)](#) and [Raz \(2025\)](#),

$$LNI_{nlgt} = 100 \times \frac{Pr(n \mid l, g, t)}{Pr(n \mid l, g, t) + Pr(n \mid -l, g, t)}.$$

³⁵The corresponding measures in [Raz \(2025\)](#) are the Tight Norms Index and the Religious Homogeneity Index.

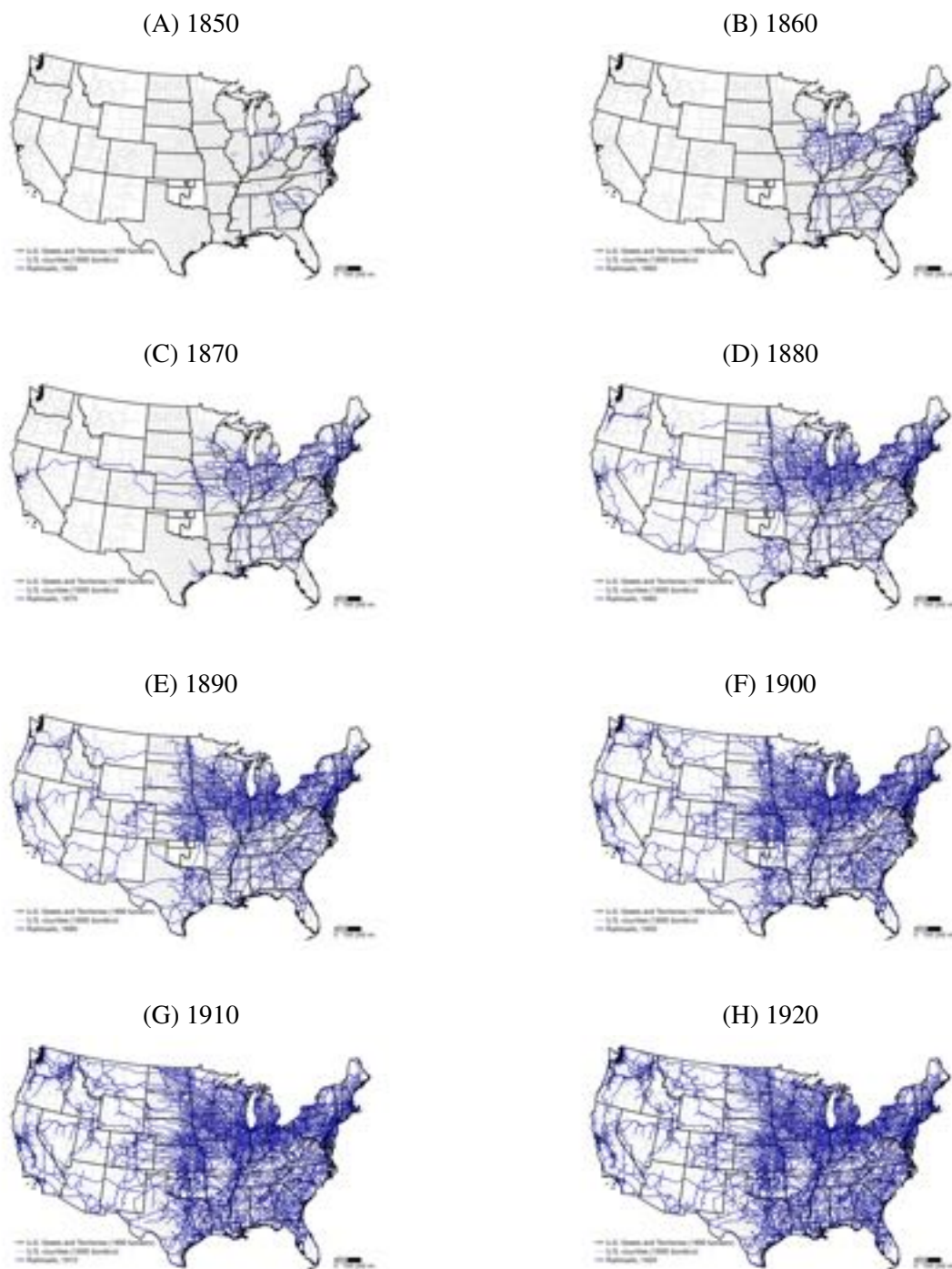


Figure A.1: The Railroad Network, 1850-1920

Note: This figure plots the railroad network for each decade between 1850-1920.

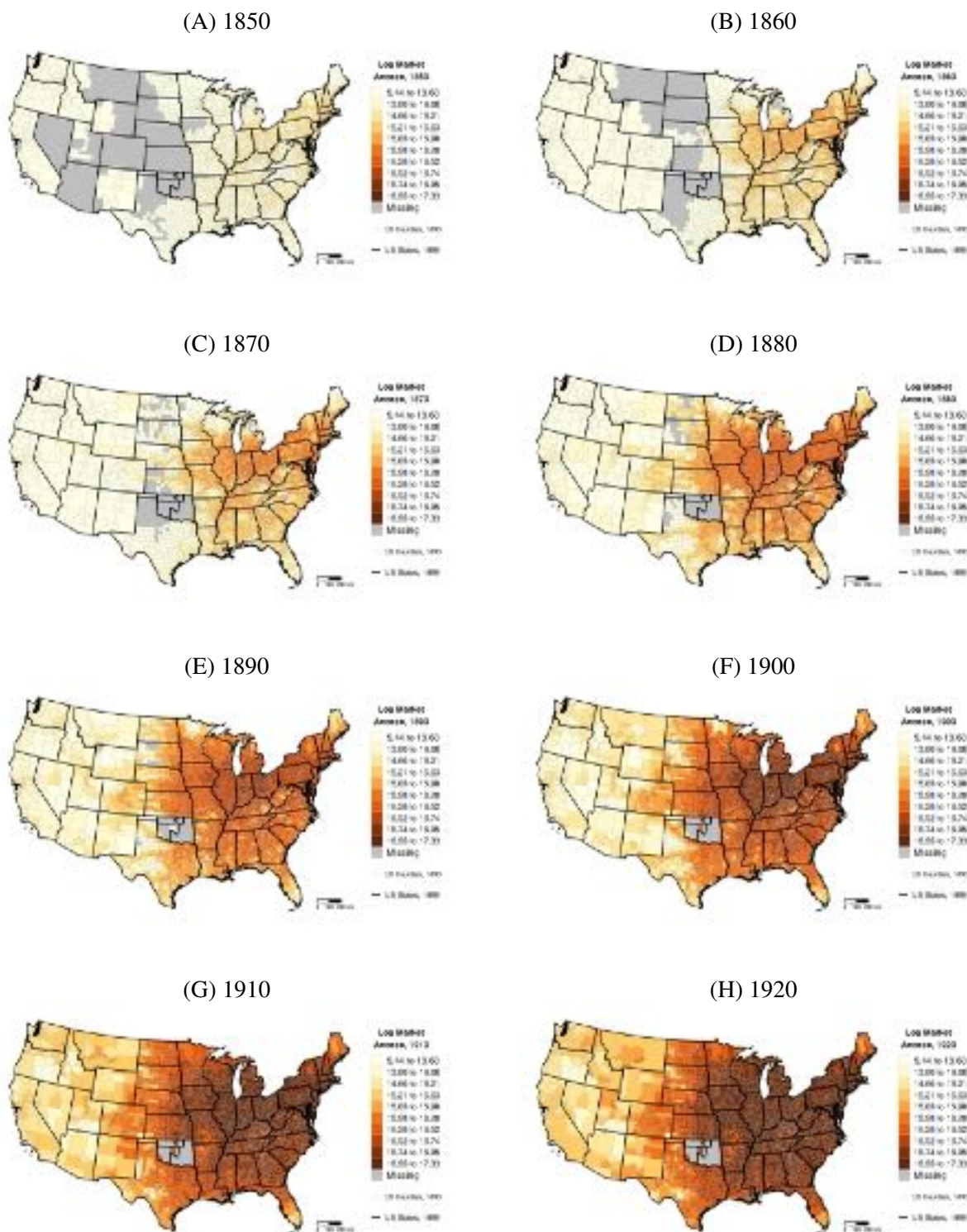


Figure A.2: Log Market Access, 1850-1920

Note: This figure plots the spatial distributions of log market access for each decade between 1850-1920. A darker color implies a higher market access.

Year	United States (log market access)	United Kingdom (log market access)
1850	13.6	13.6
1860	14.1	13.6
1870	14.7	13.6
1880	15.5	13.8
1890	15.9	13.9
1900	16.2	14.1
1910	16.6	14.3
1920	17.1	14.4

Change in Long-Term Average Annual Precipitation (mm) 1950-1999

0.00 to 2.71
2.71 to 5.42
5.42 to 8.13
8.13 to 10.84
10.84 to 13.55
13.55 to 16.26
16.26 to 18.97
18.97 to 21.68
21.68 to 24.39
24.39 to 27.10
27.10 to 29.81
29.81 to 32.52
32.52 to 35.23
35.23 to 37.94
37.94 to 40.65
40.65 to 43.36
43.36 to 46.07
46.07 to 48.78
48.78 to 51.49
51.49 to 54.20
54.20 to 56.91
56.91 to 59.62
59.62 to 62.33
62.33 to 65.04
65.04 to 67.75
67.75 to 70.46
70.46 to 73.17
73.17 to 75.88
75.88 to 78.59
78.59 to 81.30
81.30 to 84.01
84.01 to 86.72
86.72 to 89.43
89.43 to 92.14
92.14 to 94.85
94.85 to 97.56
97.56 to 100.27
100.27 to 102.98
102.98 to 105.69
105.69 to 108.40
108.40 to 111.11
111.11 to 113.82
113.82 to 116.53
116.53 to 119.24
119.24 to 121.95
121.95 to 124.66
124.66 to 127.37
127.37 to 130.08
130.08 to 132.79
132.79 to 135.50
135.50 to 138.21
138.21 to 140.92
140.92 to 143.63
143.63 to 146.34
146.34 to 149.05
149.05 to 151.76
151.76 to 154.47
154.47 to 157.18
157.18 to 159.89
159.89 to 162.60
162.60 to 165.31
165.31 to 168.02
168.02 to 170.73
170.73 to 173.44
173.44 to 176.15
176.15 to 178.86
178.86 to 181.57
181.57 to 184.28
184.28 to 186.99
186.99 to 189.70
189.70 to 192.41
192.41 to 195.12
195.12 to 197.83
197.83 to 200.54
200.54 to 203.25
203.25 to 205.96
205.96 to 208.67
208.67 to 211.38
211.38 to 214.09
214.09 to 216.80
216.80 to 219.51
219.51 to 222.22
222.22 to 224.93
224.93 to 227.64
227.64 to 230.35
230.35 to 233.06
233.06 to 235.77
235.77 to 238.48
238.48 to 241.19
241.19 to 243.90
243.90 to 246.61
246.61 to 249.32
249.32 to 252.03
252.03 to 254.74
254.74 to 257.45
257.45 to 260.16
260.16 to 262.87
262.87 to 265.58
265.58 to 268.29
268.29 to 271.00
271.00 to 273.71
273.71 to 276.42
276.42 to 279.13
279.13 to 281.84
281.84 to 284.55
284.55 to 287.26
287.26 to 290.00
290.00 to 292.71
292.71 to 295.42
295.42 to 298.13
298.13 to 300.84
300.84 to 303.55
303.55 to 306.26
306.26 to 308.97
308.97 to 311.68
311.68 to 314.39
314.39 to 317.10
317.10 to 319.81
319.81 to 322.52
322.52 to 325.23
325.23 to 327.94
327.94 to 330.65
330.65 to 333.36
333.36 to 336.07
336.07 to 338.78
338.78 to 341.49
341.49 to 344.20
344.20 to 346.91
346.91 to 349.62
349.62 to 352.33
352.33 to 355.04
355.04 to 357.75
357.75 to 360.46
360.46 to 363.17
363.17 to 365.88
365.88 to 368.59
368.59 to 371.30
371.30 to 374.01
374.01 to 376.72
376.72 to 379.43
379.43 to 382.14
382.14 to 384.85
384.85 to 387.56
387.56 to 390.27
390.27 to 392.98
392.98 to 395.69
395.69 to 398.40
398.40 to 401.11
401.11 to 403.82
403.82 to 406.53
406.53 to 409.24
409.24 to 411.95
411.95 to 414.66
414.66 to 417.37
417.37 to 420.08
420.08 to 422.79
422.79 to 425.50
425.50 to 428.21
428.21 to 430.92
430.92 to 433.63
433.63 to 436.34
436.34 to 439.05
439.05 to 441.76
441.76 to 444.47
444.47 to 447.18
447.18 to 449.89
449.89 to 452.60
452

Note: Panel A plots the average log market access by decade. The blue curve represents market access calculated using the contemporaneous transportation costs and population, and the dashed dark red curve fixes transportation costs to 1850. Panel B maps the difference between log market access in 1920 and 1850.

Locally distinctive naming can reflect attachment to the local community, conformity to its expectations, or both (Section 4.2). The index does not distinguish these sources of parochial orientation. Raz (2025) demonstrates robustness to alternative samples, phonetic standardization, and the trimming of rare names, and validates the index against contemporary proxies for parochialism, including the density of social networks and survey-based moral universalism. Similar name-based indices have been used to study immigrant assimilation and nation-building (e.g., Bazzi et al., 2019; Fouka, 2020; Abramitzky et al., 2020); in line with the variation we study—local versus nationally integrated markets—our focus is on the local-versus-national component of in-group orientation rather than the racial, ethnic, or national identities those indices target. Panels C-D in Appendix Figure A.4 map the LNI in 1850 and 1920, and Panel B in Appendix Figure A.5 its change: naming is most distinctively local in frontier counties in 1850; by 1920 the South, the wheat belt, and Utah show the most locally distinctive naming and the West Coast, the Northeast, and the large cities the least; the largest declines occur in the counties that were least developed in 1850.

³⁶For the migrant analysis (Section 5), the reference counts are constructed from US-born children aged 0–10 with valid cleaned names and successful links to both co-resident parents, each aged at least 18. The baseline reference pool includes all races and does not require the parents to be US-born. Children aged 0–10 at the later census are scored using that census’s naming distribution; children aged 11–20 use the earlier census’s distribution. We assign the origin or destination county using the child’s state of birth and the father’s linked census counties.

transcription error; and the index exists only for parents whose children survive to be recorded at ages 0–10. The within-family migrant design addresses much of this, absorbing persistent differences across families (Section 5).

Origin-fixed LNI. We also employ a variant of the LNI measure which scores all children’s names in the family’s origin county using the same census-year rule as the baseline. It holds the county fixed while allowing its naming patterns to change over time. This measure captures how characteristic chosen names are of the community the family left, providing a proxy for continued orientation towards that community.

Benchmark-adjusted LNI. This variant assesses how much of the relative decline in LNI remains after accounting for how the family’s pre-migration naming choices would score in the destination. We use earlier same-sex sibling names as a benchmark for these choices. For each child born after migration, we draw one name at random from the distinct names of same-sex siblings born before the move, giving each distinct name equal probability. We score this name in the same destination county and census year as the child’s actual name. This is the *sibling-name benchmark*. Subtracting it from the actual LNI gives the benchmark-adjusted LNI: how much more or less local the child’s name is than the earlier name drawn for comparison. For children born before the move, the benchmark equals the actual LNI, so the adjusted measure is zero by construction.

Both variants use available name–sex–county–census scores. A name absent from the relevant county’s same-sex reference children has no score and is left missing rather than assigned zero. Origin-fixed LNI therefore requires an origin score for the chosen name; benchmark-adjusted LNI after migration requires an eligible earlier sibling name and a destination score for the name drawn. We do not redraw when that score is missing. Section 5 reports the estimates and compares them with baseline LNI on the same samples.

Voter turnout in presidential elections. We compute county-level turnout for all presidential elections from 1850 to 1920 using historical returns (ICPSR, 1999) and estimates of the eligible voting population, assigning each election to the nearest census decade. For the latter, we use the 1850–1920 full-count censuses (Ruggles et al., 2020) and 1890 county-level data on the male population aged 21 and over (Manson et al., 2020) to estimate the eligible-voter stock in census years, taking into account historical changes in women’s and Black suffrage, and linearly interpolate between census years. Because the true voting population can exceed our estimate, our proxy can exceed 100%; some estimates are implausibly high (e.g., 500%), suggesting coding error, so we drop 338 county-election observations (0.7% of the sample) with turnout above 125%. Panels K and L in Appendix Figure A.4 map turnout in 1852 and 1920.

Turnout is nonetheless an imperfect proxy for civicism. It responds to political competition

and mobilization, and, in the nineteenth-century United States, to suppression: poll taxes, literacy tests, and intimidation, above all in the post-Reconstruction South. Our eligible-electorate estimates account for changes in who could legally vote; county fixed effects absorb permanent local differences; and state-by-year fixed effects absorb electoral rules and practices to the extent that they varied at the state level, where most formal restrictions were set. What none of these absorb is local, time-varying competition or suppression.

Newspaper-based social trust. U.S. newspapers historically had very local readership—the median paper sold more than 80 percent of its copies in its home county ([Gentzkow et al., 2011](#))—and their content tracked their readers’ values and politics ([Gentzkow et al., 2014](#); [Gentzkow and Shapiro, 2010](#)), potentially making them a useful source for cultural and psychological traits. We use full-text local newspapers from *newspaperarchive.com*, covering about 241 million pages across U.S. states from 1736 to 2023, assigning each page to a county by the newspaper’s place of publication and to a year by publication date.

We employ the contextualized-construct method of [Atari et al. \(2023\)](#), which measures how closely newspaper texts match psychometric questionnaire items used to quantify cultural constructs. We embed text using Sentence-BERT (SBERT), a transformer-based model ([Reimers and Gurevych, 2019](#)) designed for sentence-level embeddings, and compute cosine similarity between embeddings of questionnaire items and newspaper texts.³⁷ We implement the method in three steps. First, each newspaper page is embedded with the compact all-MiniLM-L6-v2 version of SBERT; pages exceeding the model’s context window are split into sections, each embedded separately and then averaged. Second, we embed survey items measuring trust: two positive statements—“Most people can be trusted” (GSS) and “Most people would treat you fairly, even if they had the opportunity to take advantage of you” (adapted from the World Values Survey)—whose embeddings we average, and two negative statements—“You can’t be too careful in dealing with people” (GSS) and “Most people would try to take advantage of you if they got a chance” (WVS)—whose averaged embedding we subtract. This yields an “anchored” measure of trust ([Kozlowski et al., 2019](#)). The subtraction is what isolates trust: the trusting and distrusting statements concern the same subject matter and differ only in their valence, so differencing the two embeddings nets out the topic content they share. A page dense with crime reporting, for instance, might score highly on both poles in the raw measures but neutral in the anchored measure. Relying on canonical survey items rather than terms of our own choosing also ties our hands: the language defining the construct was fixed decades in advance by the survey literature, not selected with our results in mind. Third, we

³⁷SBERT creates embeddings that place semantically similar sentences close together in vector space; it performs substantially better than dictionary-based or word-embedding approaches (e.g., word2vec) and matches advanced large language models across cultural and psychological traits ([Atari et al., 2023](#); [Abdurahman et al., 2024](#)), while remaining transparent and computationally efficient.

average newspaper-year similarity scores within each county and census decade. Publication years are assigned to the nearest census decade. Higher values indicate more trust relative to distrust in newspaper language.

We assess the measure against the GSS (Appendix Figure A.6): aggregated by year, it closely tracks the well-documented post-1970s decline in trust (correlation $\rho = 0.77$, Panel B), and aggregated by state, it is positively and significantly correlated with GSS trust (Panel C). The national-level aggregate data shows a sharp decline in trust during the Civil War and World War I, as expected (Panel A). Appendix Figure A.4 maps the measure in 1850 and 1920.

The measure reaches trust only indirectly. Printed content reflects readers' attitudes only jointly with the choices of editors, owners, and advertisers; the method captures the prevalence of trust-related language, not attitudes elicited from individuals, and the anchoring holds topic constant only across the two poles of the trust items, so a page's broader subject matter can still color its score; and the corpus's coverage is uneven across space and time, leaving some communities underrepresented. County fixed effects absorb permanent differences in the local press, and state-by-year fixed effects absorb changes in the local press that are common within the same state and year. Changes within a county's press over time may remain a source of measurement error.

Norms Tolerance Index (NTI). We focus on married mothers aged 35–44 who live with their spouses, are household heads or spouses, and have at least one child linked to them in the household. For each county-census-year, we compute border-weighted coefficients of variation of (i) mothers' age at the birth of their oldest co-resident child, used as a proxy for age at first birth; (ii) the number of co-resident children linked to the mother; and (iii) the number of distinct families in the household. Separately for each census year, we take the first principal component of the three standardized variables, orient its sign so that higher values indicate greater dispersion, and standardize it to z -scores within census year.³⁸

Dispersion is not tolerance itself. Family outcomes also spread when communities experience economic shocks, mortality, or heavy in-migration. The development and composition controls—including the immigrant share and birthplace and ancestry diversity—account for measured differences in economic conditions and population composition (Section 6).

Religious Diversity Index (RDI). Using county-level membership in religious denominations from the Censuses of Religious Bodies (Manson et al., 2020), we compute $RDI_{ot} = 1 - \sum_j s_{ojt}^2$, where s_{ojt} is the share of members in denomination j in county o and year t , and standardize to z -scores within each census wave. The underlying data cover 1850, 1860, 1870, and 1890, and the Censuses

³⁸We retain mothers whose implied age at first birth is between 12 and 50. Each component requires at least two valid mother records with positive weights summing to at least ten. These are county-border harmonization weights. The principal-component analysis uses counties with all three components observed; the index is left missing if the three loadings do not share the same sign.

of Religious Bodies of 1906, 1916, and 1926, each of the latter assigned to the preceding census decade (1906 to 1900, 1916 to 1910, 1926 to 1920); no data exist for 1880. Panels I-J in Appendix Figure A.4 and Panel E in Appendix Figure A.5 map its variation.

Membership counts miss the unaffiliated and track the supply of churches as much as demand; and a diverse membership profile may reflect who happens to live in a county rather than how residents regard those who worship differently. To address the last concern, we include the immigrant share and the diversity of birthplaces and ancestries as controls in some of our specifications (Section 6).

Family care. Following Ghosh et al. (2023), we use the living arrangements of vulnerable individuals as an indicator of kin support. We construct the measure from the full-count censuses for three groups: *orphans* (children under 16 whose mother and father are not present, via MOMLOC and POPLOC); *people with disabilities* (coded deaf, blind, “idiotic,” “insane,” sick, or temporarily disabled on the day of enumeration, via DEAF, BLIND, IDIOTIC, INSANE, and SICKNESS, where available); and the *elderly* (over 65). Using FAMSIZE and GQ, we classify a vulnerable individual as living with relatives at home if they reside in a family household (not group quarters) with at least one relative; for orphans, we do not count co-residence with a sibling orphan. Disability coverage varies across censuses: we use deafness, blindness, and the recorded mental-disability categories in 1850 and 1860; selected sickness and disability categories in 1880; and deafness and blindness in 1910. Individuals qualifying for more than one vulnerable group enter the denominator once; for orphaned children with a recorded disability, we use the orphan care classification. For each county-year we compute the share of vulnerable individuals in this category. Vulnerable individuals living in group quarters remain in the denominator and are classified as not living with relatives at home. Our baseline specification winsorizes family care at the bottom 2.5%. We verify robustness to alternative treatments of skewness (Appendix C). Panels E-F in Appendix Figure A.4 map family care in 1850 and 1920.

Co-residence is a partial record of kin support: it misses financial transfers and help from kin who live elsewhere; is shaped by housing and economic constraints; and responds to the supply of institutional alternatives as well as to the strength of kin obligation. Section 4.3 discusses the interpretation of institutional substitution and our empirical response.

Composite Universalism Index. We summarize the outcomes in a single index,

$$\text{Universalism} = [z(\text{Turnout}) + z(\text{NTI})] - [z(\text{Family Care}) + z(\text{LNI})],$$

with each component standardized to a z -score within census year; family care enters in its winsorized form, and turnout enters as the county’s average across the presidential elections of the census decade. We exclude newspaper-based trust, whose geographic coverage is limited, and the

RDI, whose waves do not map one-for-one onto the census decades (no 1880 data; the 1906–1926 censuses are assigned to preceding decades). The index is available for every census year except 1890, whose full-count records were destroyed and with them the LNI, the NTI, and family care. Panels A-B in Appendix Figure A.4 map the index in 1850 and 1920.

Triangulation. The measures draw on different sources and face different measurement problems, so agreement across them is less vulnerable to a weakness specific to one indicator. However, economic development and population composition can affect several outcomes together; we examine sensitivity to controls for these factors (Sections 5 and 6). We interpret the individual estimates alongside the pattern across the battery and its summary in the composite. Construct validity is additionally supported by Raz (2025), which shows that the name and tolerance indices correlate with kin propinquity, contemporary social-network density, and universalistic moral values.

A.4 Mechanism Measures

This section details the measures of local exposure to the two mechanisms introduced in Section 4.4.

Employment in wholesale and retail trade. Using IND1950 codes in the 1850–1920 full-count censuses (Ruggles et al., 2020), we compute the share of residents with a recorded industry who work in wholesale and retail trade. The denominator excludes observations with missing IND1950, code 0, or codes above 976; the numerator includes codes 606–699. Our baseline specification winsorizes wholesale and retail employment at the top 2.5%. We verify robustness to alternative treatments of skewness (Appendix C). Appendix Figure A.7 maps the share in 1850 and 1920.

Employment in trade tracks commerce only roughly. The share rises with local development generally—income, urbanization—and not only with exchange, a concern the development controls are meant to address (Section 6); and it counts commerce of any reach, including purely local dealing, whereas market access measures integration between counties. The latter is one reason it serves as a broad indicator only: the channel-specific evidence comes from the dictionaries, which capture commercial counterparties, distant markets, and institutions supporting impersonal trade.

Newspaper mechanism dictionaries. The two channel-specific measures are built from the full-text local-newspaper corpus of *newspaperarchive.com*—the same corpus underlying our trust measure—using dictionaries of period words and phrases that signal each mechanism. We compiled the dictionaries iteratively with the assistance of large language models (Claude Opus 4.7 and GPT-5.5), which we prompted to propose nineteenth-century words and phrases signaling each channel. For *exchange with strangers* the dictionary comprises, in full: *mobile commercial counterparties*—“commercial traveler,” “traveling salesman,” “commercial drummer,” “traveling drummer”; *commodity intermediaries*—“commission merchant,” “produce commission merchant,”

(A) Composite Universalism Index, 1850



(B) Composite Universalism Index, 1920



(C) Local Name Index, 1850



(D) Local Name Index, 1920



(E) Family Care, 1850



(F) Family Care, 1920



Figure A.4: Universalism, Impersonal Prosociality, and Tolerance, 1850 and 1920

Note: This figure plots the spatial distributions of our measures: the composite index, the LNI, the share in family care, the NTI, the RDI, voter turnout (1852 and 1920), and social trust, in 1850 (left column) and in 1920 (right column). A lighter color implies a higher prevalence of universalism. The figure continues to the next page.

“grain commission merchant,” “livestock commission merchant”; and *commodity buyers*—“grain buyer,” “cattle buyer,” “hog buyer,” “wool buyer,” “cotton buyer,” “tobacco buyer.” For *interdependence with distant others* it comprises terms for credit dependence, distant-market orientation,

North American Monsoon Index, 1990

0.000 to 0.200
0.200 to 0.400
0.400 to 0.600
0.600 to 0.800
0.800 to 1.000
1.000 to 1.200
1.200 to 1.400
1.400 to 1.600
1.600 to 1.800
1.800 to 2.000
2.000 to 2.200
2.200 to 2.400
2.400 to 2.600
2.600 to 2.800
2.800 to 3.000
3.000 to 3.200
3.200 to 3.400
3.400 to 3.600
3.600 to 3.800
3.800 to 4.000
4.000 to 4.200
4.200 to 4.400
4.400 to 4.600
4.600 to 4.800
4.800 to 5.000
5.000 to 5.200
5.200 to 5.400
5.400 to 5.600
5.600 to 5.800
5.800 to 6.000
6.000 to 6.200
6.200 to 6.400
6.400 to 6.600
6.600 to 6.800
6.800 to 7.000
7.000 to 7.200
7.200 to 7.400
7.400 to 7.600
7.600 to 7.800
7.800 to 8.000
8.000 to 8.200
8.200 to 8.400
8.400 to 8.600
8.600 to 8.800
8.800 to 9.000
9.000 to 9.200
9.200 to 9.400
9.400 to 9.600
9.600 to 9.800
9.800 to 10.000
10.000 to 10.200
10.200 to 10.400
10.400 to 10.600
10.600 to 10.800
10.800 to 11.000
11.000 to 11.200
11.200 to 11.400
11.400 to 11.600
11.600 to 11.800
11.800 to 12.000
12.000 to 12.200
12.200 to 12.400
12.400 to 12.600
12.600 to 12.800
12.800 to 13.000
13.000 to 13.200
13.200 to 13.400
13.400 to 13.600
13.600 to 13.800
13.800 to 14.000
14.000 to 14.200
14.200 to 14.400
14.400 to 14.600
14.600 to 14.800
14.800 to 15.000
15.000 to 15.200
15.200 to 15.400
15.400 to 15.600
15.600 to 15.800
15.800 to 16.000
16.000 to 16.200
16.200 to 16.400
16.400 to 16.600
16.600 to 16.800
16.800 to 17.000
17.000 to 17.200
17.200 to 17.400
17.400 to 17.600
17.600 to 17.800
17.800 to 18.000
18.000 to 18.200
18.200 to 18.400
18.400 to 18.600
18.600 to 18.800
18.800 to 19.000
19.000 to 19.200
19.200 to 19.400
19.400 to 19.600
19.600 to 19.800
19.800 to 20.000
20.000 to 20.200
20.200 to 20.400
20.400 to 20.600
20.600 to 20.800
20.800 to 21.000
21.000 to 21.200
21.200 to 21.400
21.400 to 21.600
21.600 to 21.800
21.800 to 22.000
22.000 to 22.200
22.200 to 22.400
22.400 to 22.600
22.600 to 22.800
22.800 to 23.000
23.000 to 23.200
23.200 to 23.400
23.400 to 23.600
23.600 to 23.800
23.800 to 24.000
24.000 to 24.200
24.200 to 24.400
24.400 to 24.600
24.600 to 24.800
24.800 to 25.000
25.000 to 25.200
25.200 to 25.400
25.400 to 25.600
25.600 to 25.800
25.800 to 26.000
26.000 to 26.200
26.200 to 26.400
26.400 to 26.600
26.600 to 26.800
26.800 to 27.000
27.000 to 27.200
27.200 to 27.400
27.400 to 27.600
27.600 to 27.800
27.800 to 28.000
28.000 to 28.200
28.200 to 28.400
28.400 to 28.600
28.600 to 28.800
28.800 to 29.000
29.000 to 29.200
29.200 to 29.400
29.400 to 29.600
29.600 to 29.800
29.800 to 30.000
30.000 to 30.200

Poverty: 1992

Percentage of the population in poverty, 1992

0.00 to 0.20
0.20 to 0.40
0.40 to 0.60
0.60 to 0.80
0.80 to 1.00
Missing

0 100 Miles

Note: This figure plots the spatial distributions of our measures: the composite index, the LNI, the share in family care, the NTI, the RDI, voter turnout (1852 and 1920), and social trust, in 1850 (left column) and in 1920 (right column). A lighter color implies a higher prevalence of universalism. The figure continues to the next page.

56

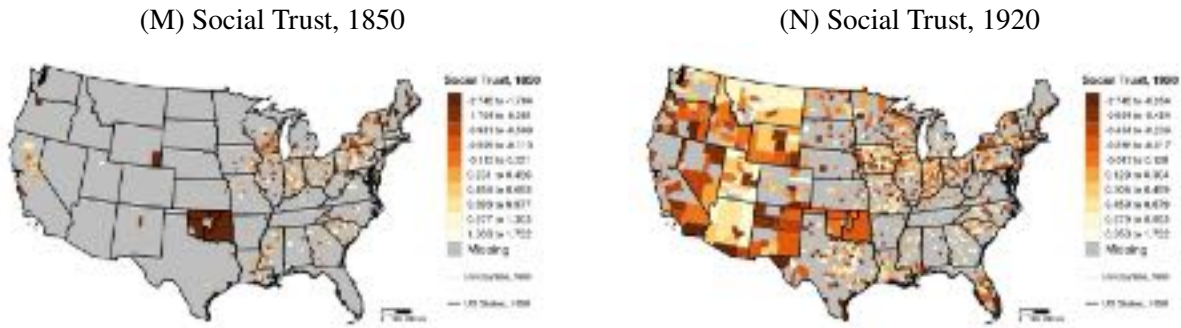


Figure A.4: Universalism, Impersonal Prosociality, and Tolerance, 1850 and 1920 (cont.)

Note: This figure plots the spatial distributions of our measures: the composite index, the LNI, the share in family care, the NTI, the RDI, voter turnout (1852 and 1920), and social trust, in 1850 (left column) and in 1920 (right column). A lighter color implies a higher prevalence of universalism.

erpool market,” “Eastern markets,” “market for our products,” “channels of trade,” “commercial relations,” “commercial intercourse”; and *standardization*—“grain inspection,” “inspected and graded,” “inspection certificate,” “public weigher,” “weighmaster,” “sworn inspector,” “standard weights and measures,” “assay office.” Each term is matched as a word-bounded, case-insensitive regular expression, with plural or spelling variants where specified in the dictionary (compiled with `re2`); for every page, we count matches per construct. For each newspaper-year, we divide the number of matches by the supplied word count, which counts tokens of at least three characters. We assign newspaper locations to 1890 county boundaries, assign newspaper-years to the nearest census decade, and average the frequencies within each county-decade. For each index, we sum the raw frequencies of its components, winsorize this sum at the 2.5th and 97.5th percentiles, and standardize it across the merged county panel before regression exclusions. The component outcomes in Appendix Table B.3 are separately winsorized and standardized in the same way. The indices therefore measure how frequently the relevant language appears in local print.

Interpretation and limitations. Dictionary methods can record terms used in unintended senses—“commercial drummer” as a vaudeville character, “Eastern capital” as a capital city, “foreign capital” as a foreign seat of government. Inspection and certification terms can also refer to settings other than commodity trade. These measures capture discussion of commercial relationships and institutions, rather than directly measuring the extent to which individuals depend on distant others.

Market-oriented versus locally-oriented industries. Using IPUMS IND1950 codes, we classify as *market-oriented*: agriculture, forestry, and fishing (105–126), mining (206–239), manufacturing (306–499), transportation (506–568), telecommunications (578–579), wholesale trade (606–627),

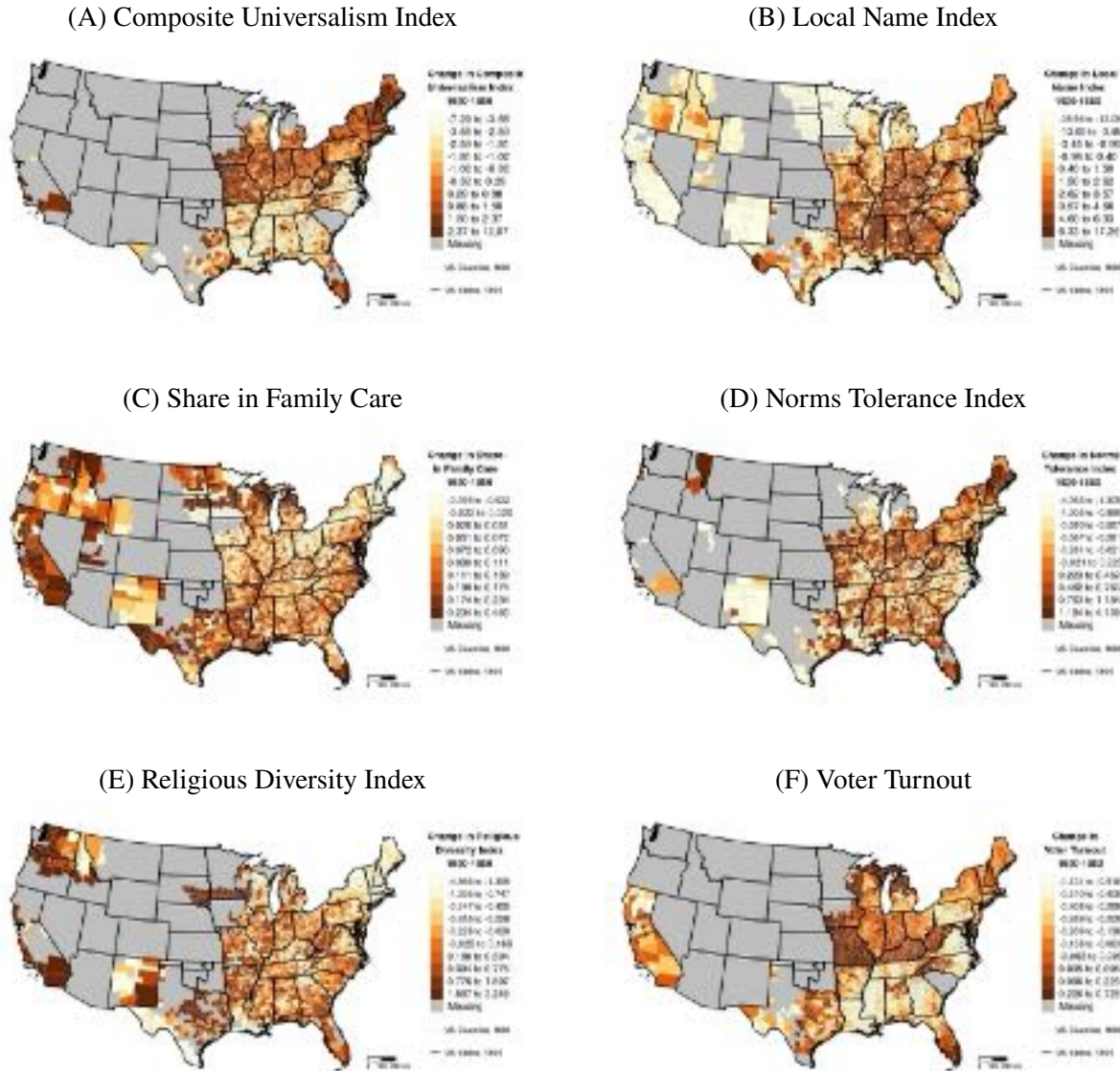


Figure A.5: The Change in Universalism, Impersonal Prosociality, and Tolerance

Note: This figure plots the difference in our outcome measures between 1920 and 1850: the composite universalism index, the LNI, the share in family care, the NTI, the RDI, and voter turnout (1852 and 1920).

retail trade (636–699, except eating and drinking places, 679), banking and credit agencies (716), security and commodity brokerage and investment companies (726), insurance (736), advertising (806), accounting, auditing, and bookkeeping services (807), and miscellaneous business services (808); and as *locally oriented*: construction (246), utilities and sanitary services (586–598), eating and drinking places (679), real estate and related offices (746, 756), repair services (816–817), personal services (826–849), entertainment and recreation services (856–859), professional and related services (868–899), and public administration (906–946). We assess robustness to the

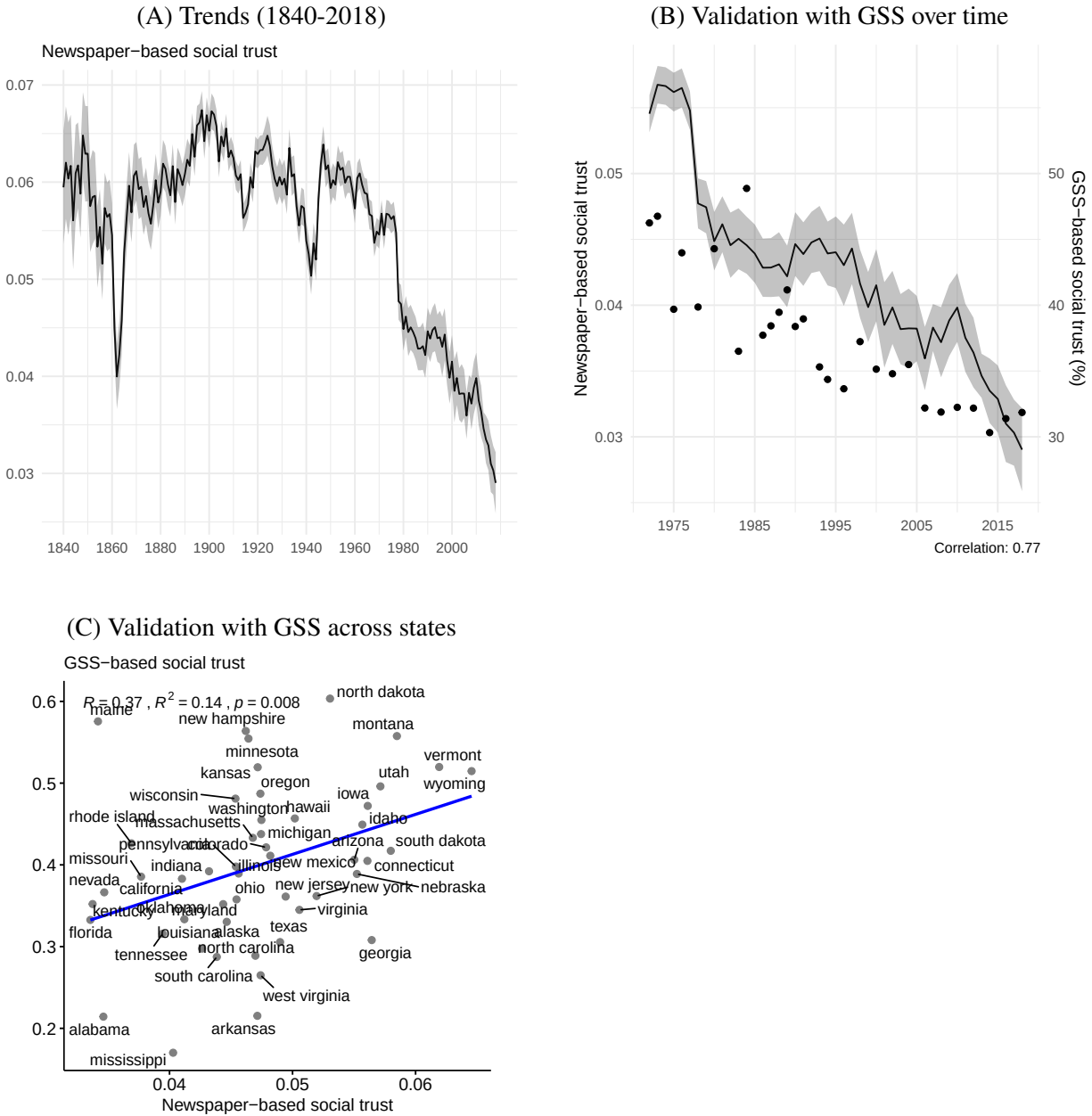


Figure A.6: Newspaper-based Social Trust: Trends and Validation

Note: This figure presents descriptive and validation results for the newspaper-based measure of social trust. Top row: national-trends in trust from 1840 to 2018 (left plot) and 1972 to 2018 (right plot). The dots depict trust data from the GSS (labeled on right y-axis). Bottom row: State-level correlation between newspaper and GSS-based trust from 1972 to 2014, net of year fixed effects before aggregation.

tradable/non-tradable classification of [Barkai and Karger \(2020\)](#). We map modern NAICS categories to IND1950 and include wholesale and retail trade, excluding eating and drinking places, with tradable industries; this assumes that modern geographic market reach approximates historical

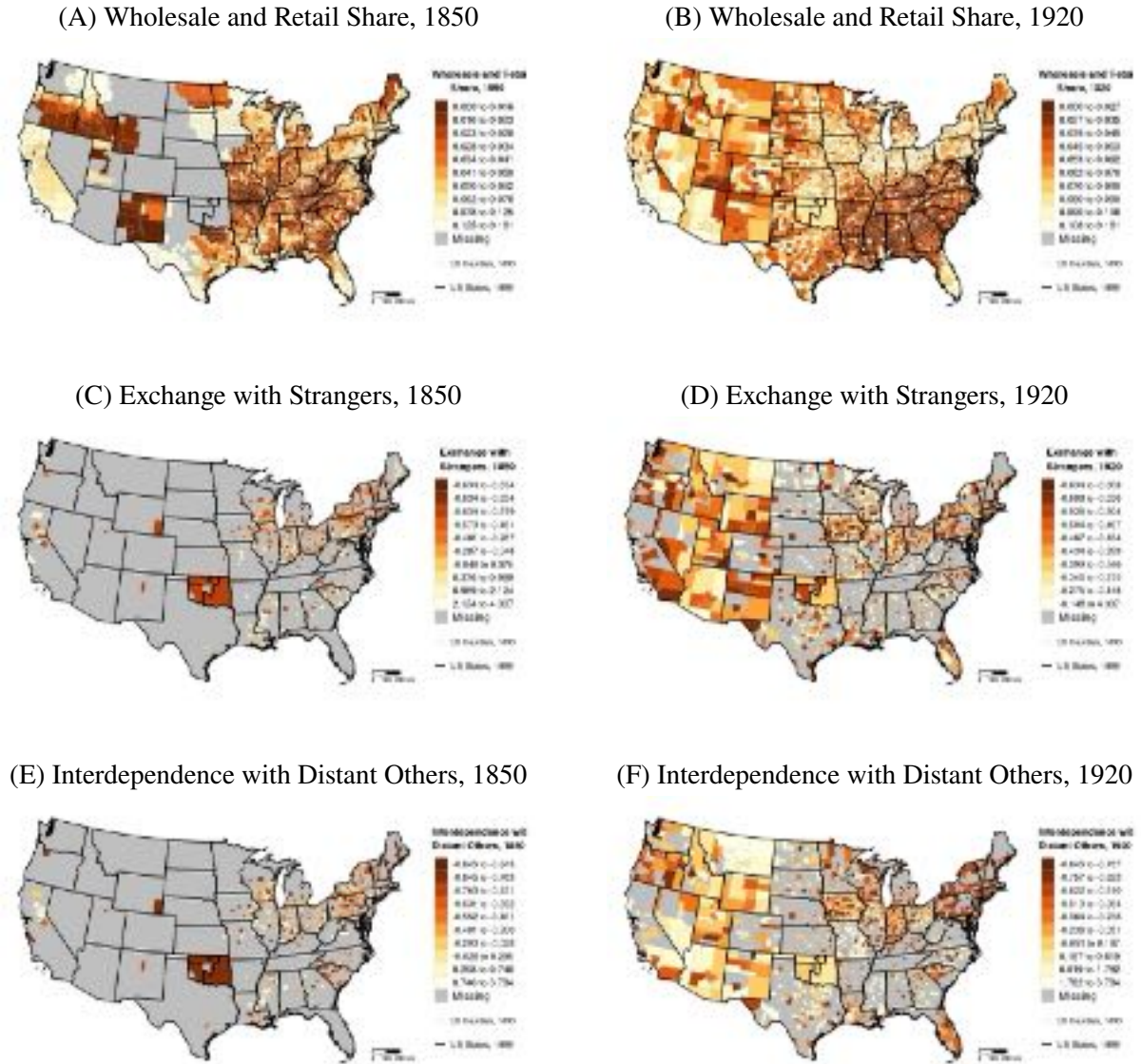


Figure A.7: Measures of Channels, 1850 and 1920

Note: This figure plots the spatial distribution of our measures of the channels: the share of employed residents working in wholesale and retail trade, exchange with strangers in local newspapers, and economic interdependence with distant others in local newspapers, in 1850 (left column) and 1920 (right column). A lighter color implies a higher share.

exposure.

A.5 Other Demographic and Economic Measures

To probe mechanisms and robustness, and to test for selective migration (Section 7.1), we use several additional county- and individual-level variables, constructed from the full-count and county-level census data (Ruggles et al., 2020; Manson et al., 2020) and the sources noted below.

Intra-community marriage (ICM). Following [Raz \(2025\)](#), we use birthplaces to define communities: for each married couple not in group quarters with birthplace information for both partners, we record spouses’ birthplaces (state for U.S.-born, country for foreign-born) and set an indicator equal to one if the couple share a birthplace. The county-level ICM is the share of couples with same birthplaces. Birthplace is an imperfect proxy for social group: couples from the same state or country may differ in ethnicity, religion, or class, and cross-birthplace marriages may occur within homogeneous subgroups; the measure also records only formal marriages captured by the census; and in sparsely populated frontier counties such marriages may reflect necessity rather than openness. These weaknesses are among the reasons we use the ICM only as a supporting measure.

Kinship propinquity. We measure kinship propinquity using data from [Nelson \(2020\)](#), which exploits the fact that distance between households on the census enumeration form correlates with geographic distance. When a same-race, same-surname household appears within the same enumeration district, the probability that the match reflects kinship rather than chance is $P(K_{rie}) = \left(1 - \frac{N_{rie}-1}{N_{re}-1}\right)^{D_i}$, where N_{rie} is the number of same-race (r) same-surname (i) households in enumeration district e , N_{re} is the total number of same-race households in the district, and D_i is the number of different-surname households as close as the nearest same-surname household. We classify a household as exhibiting kinship propinquity when this probability exceeds 50%.

Shared ancestry and ancestry diversity. We infer ancestry probabilistically from surnames, following [Posch et al. \(2026\)](#). Pooling the 1850–1930 full-count censuses, we compute, for each surname, the distribution of countries of birth among its foreign-born holders, and assign these shares to every household head bearing that surname. Aggregating across household heads yields the expected ancestry composition of a county or enumeration district, from which we construct two measures: a household head’s *shared ancestry*—the expected share of household heads in his county or enumeration district who are of his ancestry—and a county-level *ancestry-diversity index*, one minus the Herfindahl index over the county’s ancestry shares. Unlike birthplace-based proxies, these measures distinguish ancestries within the native-born population, for whom birthplace no longer records origin.

Labor-force cooperation. This measure captures the degree to which success in the local labor market regularly required interacting with others in a cooperative, pleasant manner. We rely on contemporary O*NET work-style “cooperation” rating, which measures the extent to which the job requires being pleasant with others and displaying a cooperative attitude. We map O*NET’s SOC codes to historical IPUMS OCC1950 codes, assuming that the relative differences in cooperation requirements across occupations are stable over time. This allows us to assign a labor-force cooperation score for every working individual in the full count censuses 1850-1920. We then compute an employment-weighted average at the county-decade level.

Standard demographic and economic variables. We also use: occupational income scores (the mean of IPUMS OCCSCORE, occupation-level earnings based on 1950); log real GDP per capita at the 1980 county-group level (Fulford et al., 2020); the share employed in agriculture (IND1950 code 105) and in manufacturing (codes 306–499); the number of manufacturing establishments (Manson et al., 2020); the share of population in urban areas (incorporated places of at least 2,500); the share of foreign-born residents; a birthplace-diversity index, one minus the Herfindahl index over birthplace shares (from BPL), treating the U.S.-born as a single group and the foreign-born by country of birth; an analogous ancestry-diversity index over the surname-based ancestry shares described above; the number of children, urban-versus-rural origin, and nativity (used as pre-migration characteristics in Section 7.1); information workers per 1,000 workers with a recorded occupation³⁹; and lawyers and judges per 1,000 workers with a recorded occupation (OCC1950 code 55), proxying local legal-institutional development.

B Further Figures, Analysis and Results

B.1 Migrant-level analysis

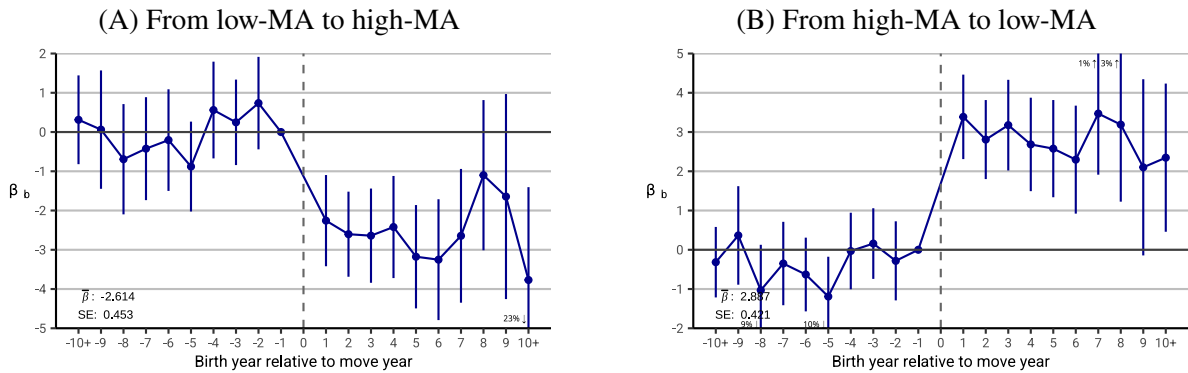


Figure B.1: Naming Responses to Gains and Losses in Market Access

Note: This figure plots the estimates of β_b and 95% confidence intervals from two different dynamic difference-in-differences estimations. The dependent variable is children's LNI. In Panel A, the sample is restricted to households originating from a below-median MA county, and treatment is defined as migrating to an above-median MA destination, as opposed to migrating to another below-median MA destination. In Panel B, the sample is restricted to households originating from an above-median MA county, and treatment is defined as migrating to a below-median MA destination. $\bar{\beta}$ is the average treatment effect across post-migration periods weighted by the number of treated children born in each relative year in the relevant sample.

³⁹OCC1950 codes: editors and reporters (36), postmasters (270), express messengers and railway mail clerks (325), mail carriers (335), telegraph messengers (360), telegraph operators (365), newsboys (460), pressmen and plate printers (575), and printing-trade apprentices (613).

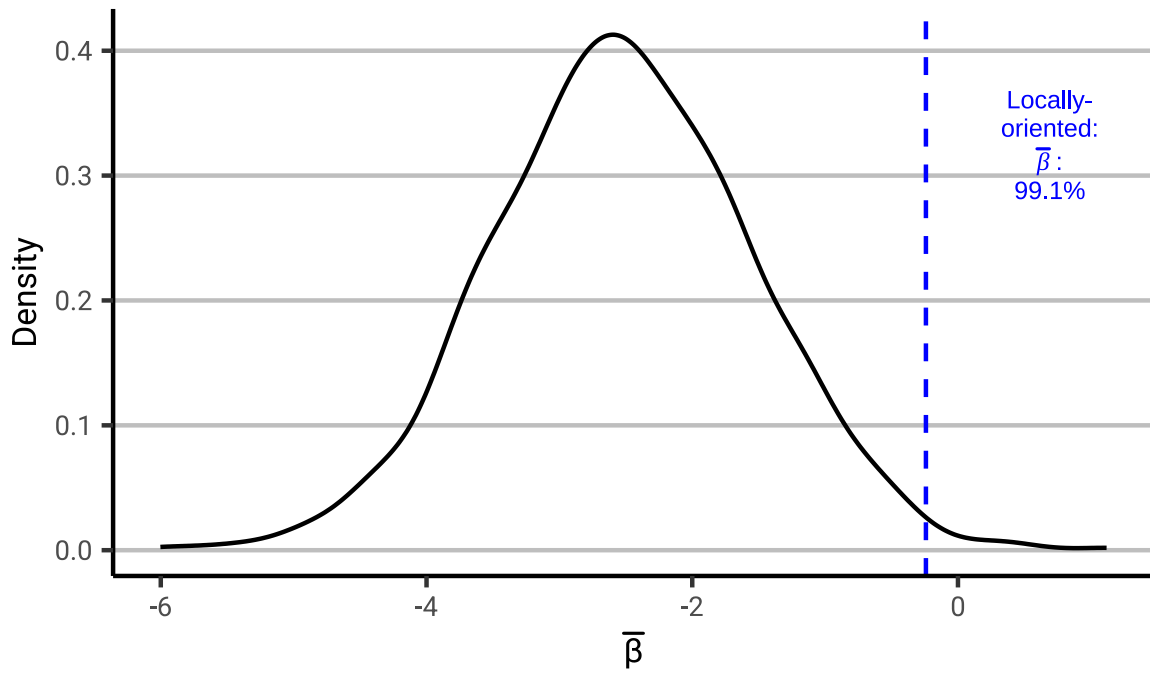


Figure B.2: Assessing the Role of Differences in Sample Size

Note: This figure plots the estimates of $\bar{\beta}$ —the average post-migration treatment effect—from the difference-in-differences equation (1), estimated on 1,000 random sample draws without replacement of 5,292 families from the market-oriented group (the same number of families as in the locally-oriented group). The blue dashed vertical line plots the estimation of $\bar{\beta}$ for the locally-oriented group (Figure 2, Panel B).

Table B.1: Characteristics of Families in Market-Oriented and Locally-Oriented Categories

	Category:		
	Locally-oriented N=5,292 (1)	Market-oriented N=55,842 (2)	Difference p-value (3)
Father's Age	31.8 (6.21)	31.7 (6.86)	0.619
Number of Children	4.04 (1.61)	4.61 (1.79)	<0.001
Origin is Urban	0.44 (0.50)	0.20 (0.40)	<0.001
High MA Origin	0.63 (0.48)	0.53 (0.50)	<0.001
Intra-Community Marriage	0.52 (0.50)	0.58 (0.49)	<0.001
Low Pre-migration LNI	0.52 (0.50)	0.51 (0.50)	0.057

Note: This table reports the mean characteristics of domestic out-of-state migrant households in the market-oriented and locally-oriented categories. Standard deviations in parentheses.

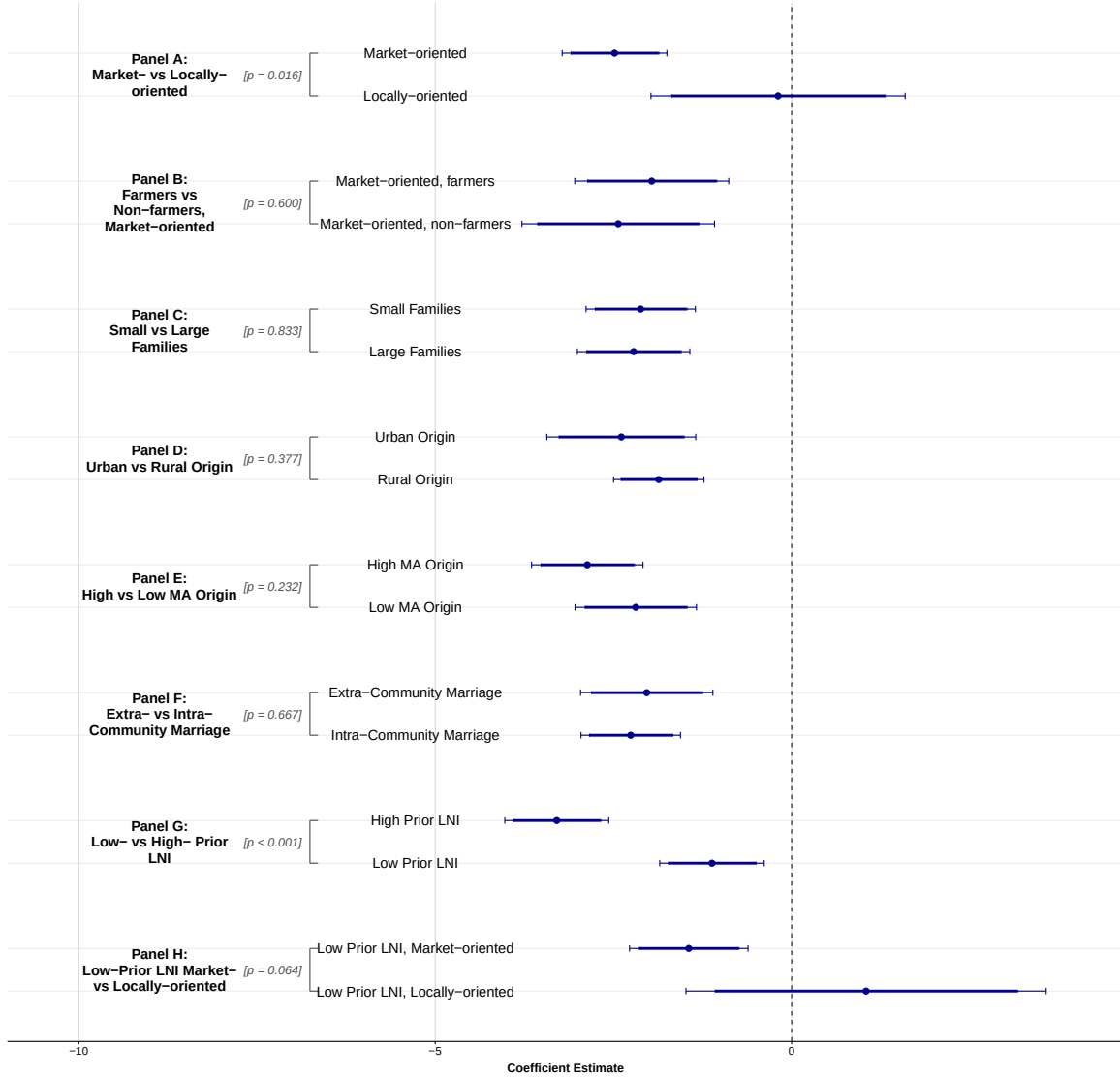


Figure B.3: Cultural Adaptation Across Family Characteristics

Note: This figure plots the estimates of $\bar{\beta}$ —the average post-migration treatment effect weighted by the number of treated children born in each relative year in the relevant sample—and 95% confidence intervals from the difference-in-differences equation 1, when the sample is split along different social dimensions. The market-oriented and locally-oriented groups contain 55,842 and 5,292 families, respectively. Farmers account for about 46% of the market-oriented sample. The farmer/non-farmer comparisons use 25,800 and 12,237 families; their sum is smaller because some individuals switch occupations.

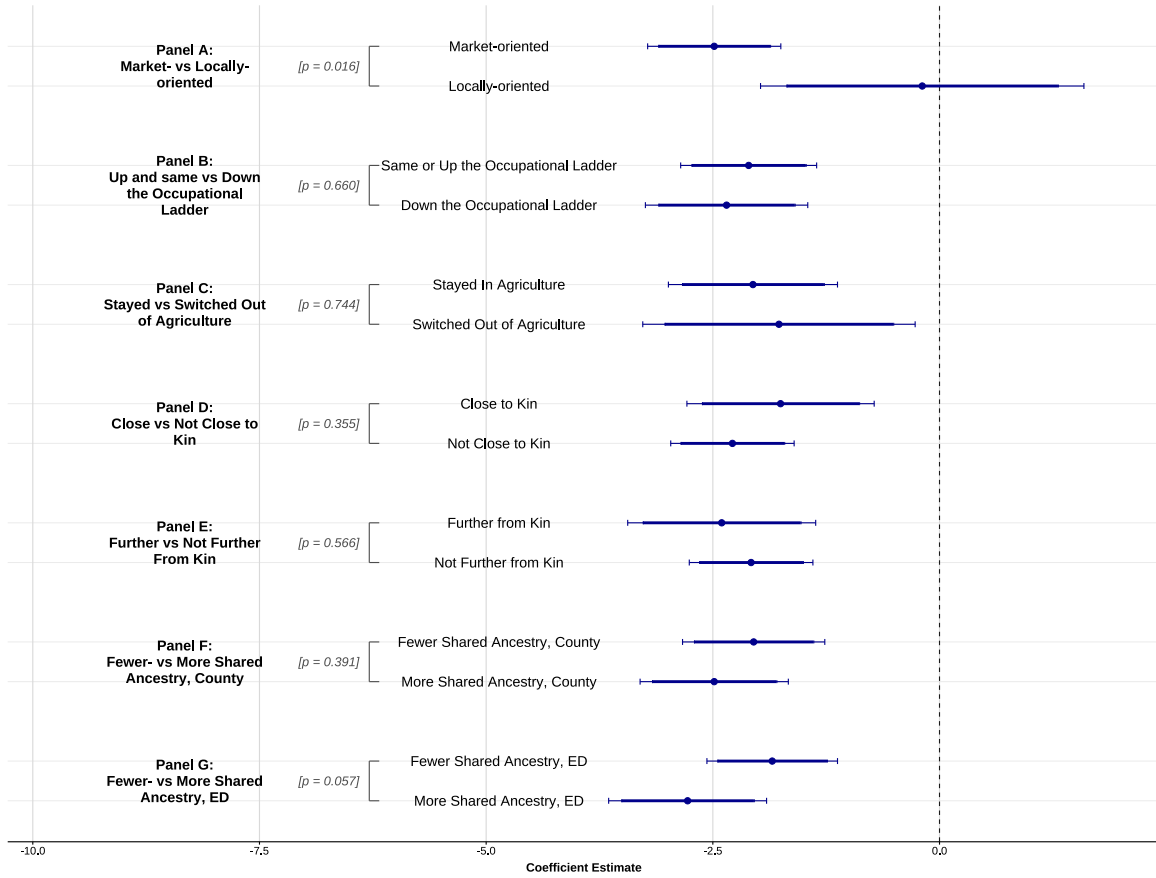


Figure B.4: Cultural Adaptation Across Occupational and Social Groups

Note: This figure plots the estimates of $\bar{\beta}$ —the average post-migration treatment effect weighted by the number of treated children born in each relative year in the relevant sample—and 95% confidence intervals from the difference-in-differences equation 1, when the sample is split along different potential channels.

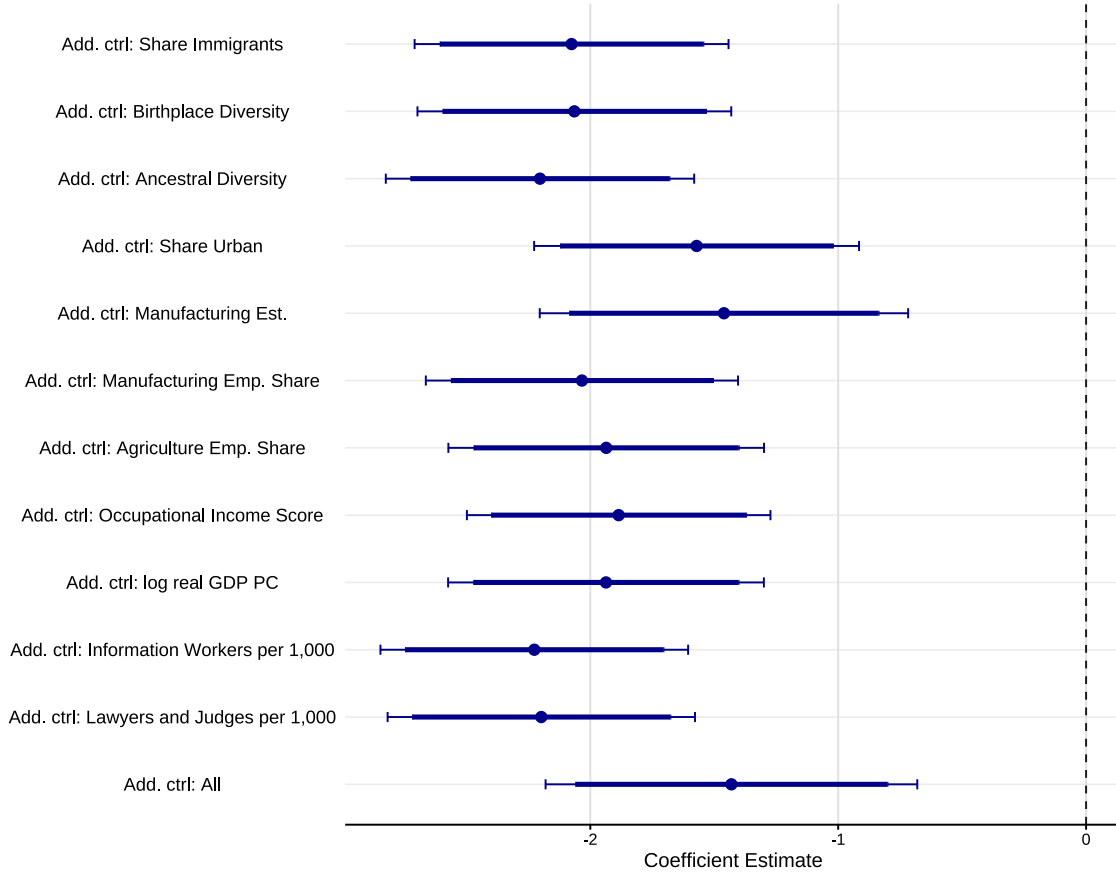


Figure B.5: Cultural Adaptation with Controls for Competing Channels

Note: This figure plots the estimates of $\bar{\beta}$ —the average post-migration treatment effect weighted by the number of treated children born in each relative year—and 95% confidence intervals from an augmented version of the difference-in-differences equation (1) that also controls for the dynamic impact of potential confounders and mediators, defined as the differences in the levels of the potential confounder or mediator between origin county o and destination county d : the share of urban population, the share of immigrants, birthplace diversity, ancestry diversity, log of one plus the number of manufacturing establishments, the share working in manufacturing, the share working in agriculture, the mean occupational income score, log real GDP per capita, the number of information workers per 1,000 workers, and the number of lawyers and judges per 1,000 workers. On the 315,535 observations available for all controls, the estimate without controls is -2.07 , and the estimate with all controls is -1.43 . The difference, 0.64 LNI points (SE = 0.18), is statistically significant (p -value < 0.001); the test accounts for the covariance between the two estimates.

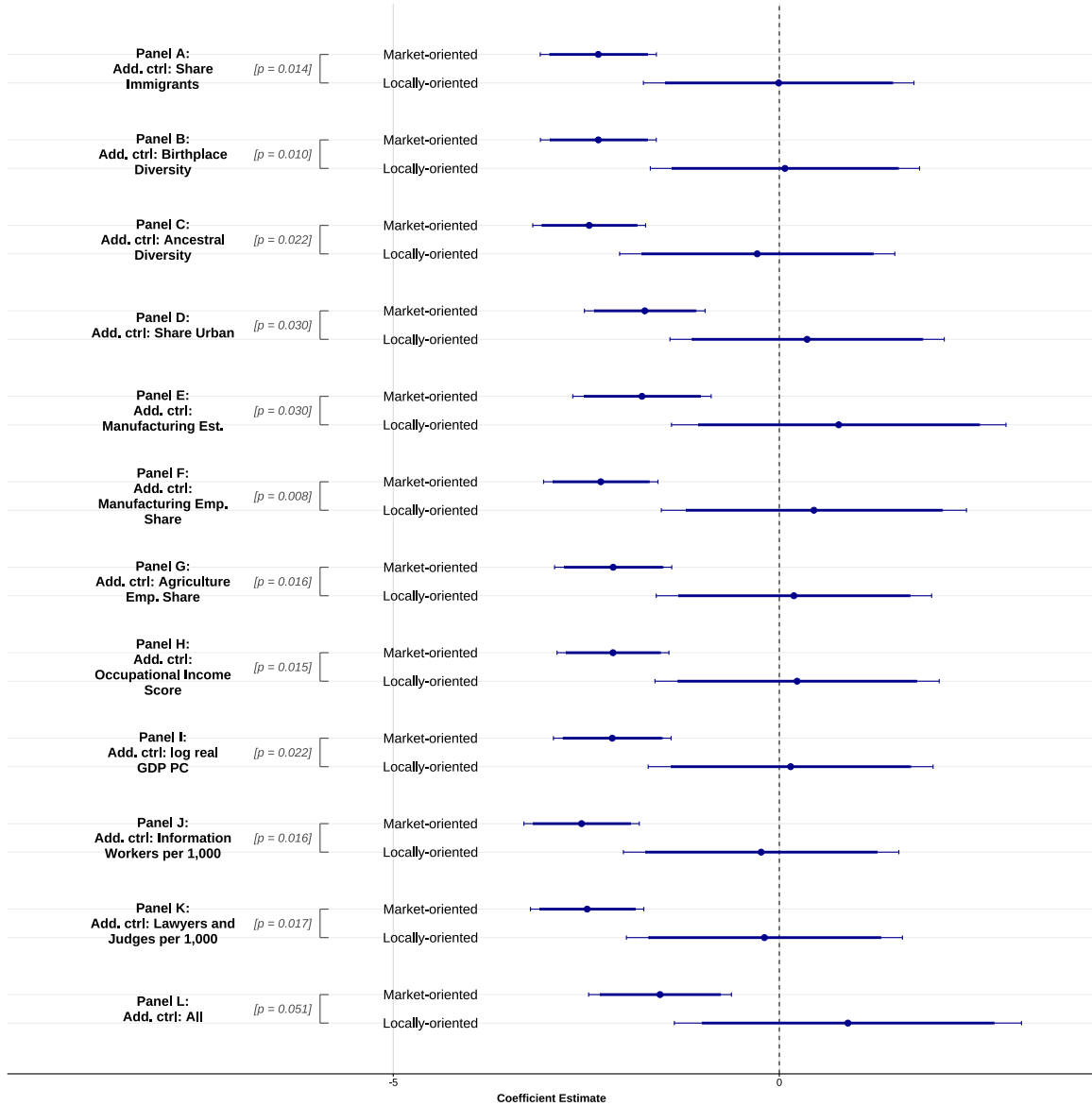


Figure B.6: Industry Differences with Controls for Competing Channels

Note: This figure plots the estimates of $\bar{\beta}$ —the average post-migration treatment effect weighted by the number of treated children born in each relative year in the relevant sample—and 95% confidence intervals from an augmented version of the difference-in-differences equation (1) that also controls for the dynamic impact of potential confounders and mediators, defined as the differences in the levels of the potential confounder or mediator between origin county o and destination county d , and the sample is split between market-oriented and locally-oriented families.

B.2 County-level analysis

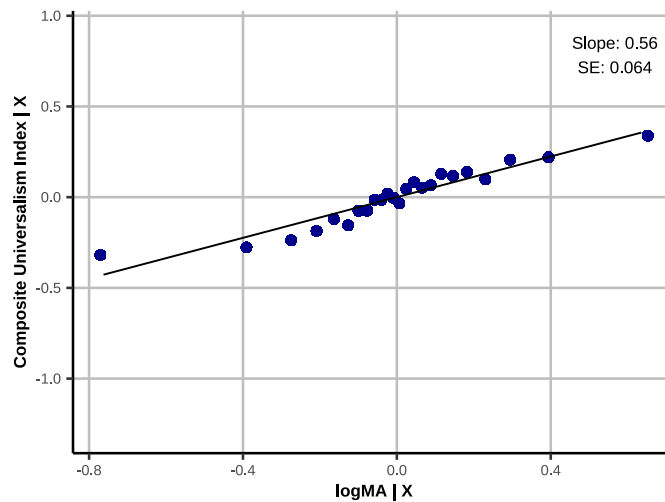


Figure B.7: Market Access and the Composite Universalism Index

Note: This figure presents bin scatter plots of the relationship between log market access and the composite universalism index, using equation (4). The plots present the conditional relationship after partialling out the baseline controls, corresponding to column 1 in Table 2. All bins have the same number of observations.

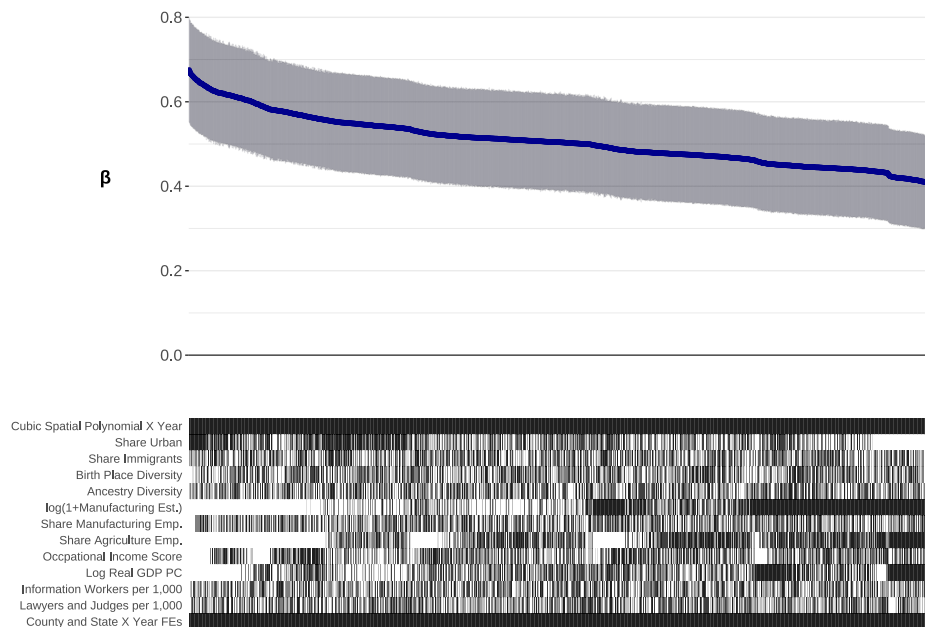


Figure B.8: The Impact is Robust to Controlling for Competing and Mediating Factors

Note: This figure plots estimates of β and 95% confidence intervals from Equation (4) when the dependent variable is the composite universalism index. In addition to the baseline controls, the regressions control for all possible combinations of the considered competing and mediating factors: urban and immigrant shares; birthplace and ancestry diversity; log of one plus manufacturing establishments; manufacturing and agriculture employment shares; mean occupational income score; log real GDP per capita; and information workers and lawyers and judges per 1,000 workers. Standard errors are clustered at arbitrary grid cells of 100 miles square (Bester et al., 2011).

Table B.2: Composite Universalism Index: Instrumental Variable

	Dependent variable:			
	1 st stage	2 nd stage	1 st stage	2 nd stage
	Log market access	Composite Universalism Index	Log market access	Composite Universalism Index
	(1)	(2)	(3)	(4)
Log market access		0.5525*** (0.0650)		0.4135*** (0.1073)
Recentered log market access	1.002*** (0.0027)			
Log water market access in 1850 $\times$ year = 1860			-0.3089*** (0.0354)	
Log water market access in 1850 $\times$ year = 1870			-0.5072*** (0.0400)	
Log water market access in 1850 $\times$ year = 1880			-0.7537*** (0.0347)	
Log water market access in 1850 $\times$ year = 1900			-0.8396*** (0.0332)	
Log water market access in 1850 $\times$ year = 1910			-0.8522*** (0.0325)	
Log water market access in 1850 $\times$ year = 1920			-0.8637*** (0.0324)	
Wald (1st stage)	135,263.5		128.2	
Observations	16,558	16,558	14,703	14,703
R ²	1.00	0.805	0.973	0.805

Note: This table reports the results of two instrumental variable strategies. Columns 1-2 report the results of a recentered IV (Borusyak and Hull, 2023) in which the log expected market access is based on the network of proposed canals from Fogel (1964). Columns 3-4 report the results of a waterways IV (Donaldson and Hornbeck, 2016; Hornbeck and Rotemberg, 2024) in which log water market access in 1850, interacted with year fixed effects is used to predict changes in log market access. All columns control for county fixed effects, state-by-year fixed effects, and a cubic spatial polynomial interacted with year fixed effects. Standard errors clustered at arbitrary grid cells of 100 miles square in parentheses (Bester et al., 2011). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

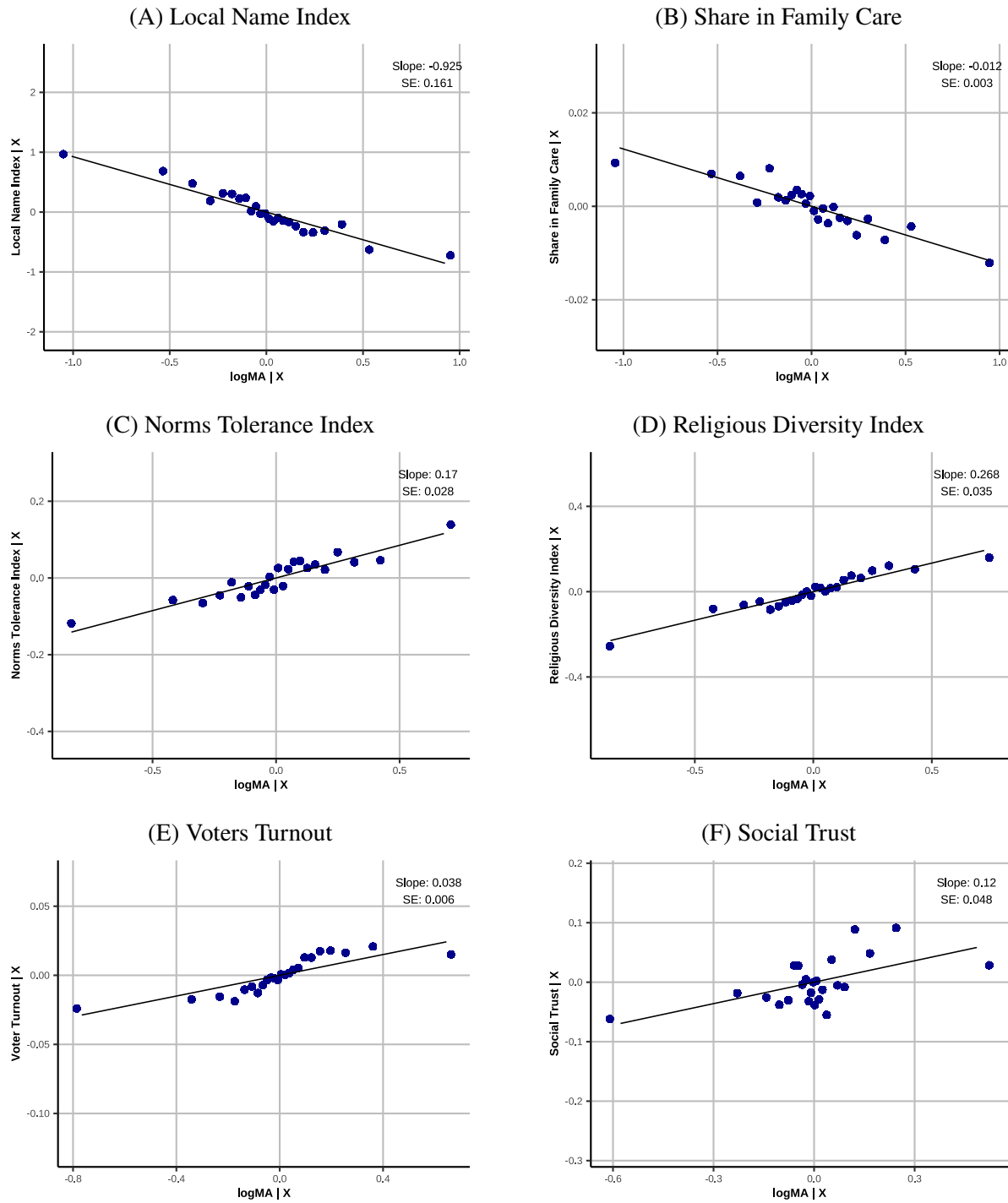


Figure B.9: Market Access and County-level Outcomes: Bin Scatter Plots

Note: This figure presents bin scatter plots of the relationship between log market access and the individual county-level outcomes, using equation (4). The plots present the conditional relationship after partialling out the baseline controls, corresponding to column 1 in Table 2. All bins have the same number of observations.

B.3 More Evidence on Mechanisms

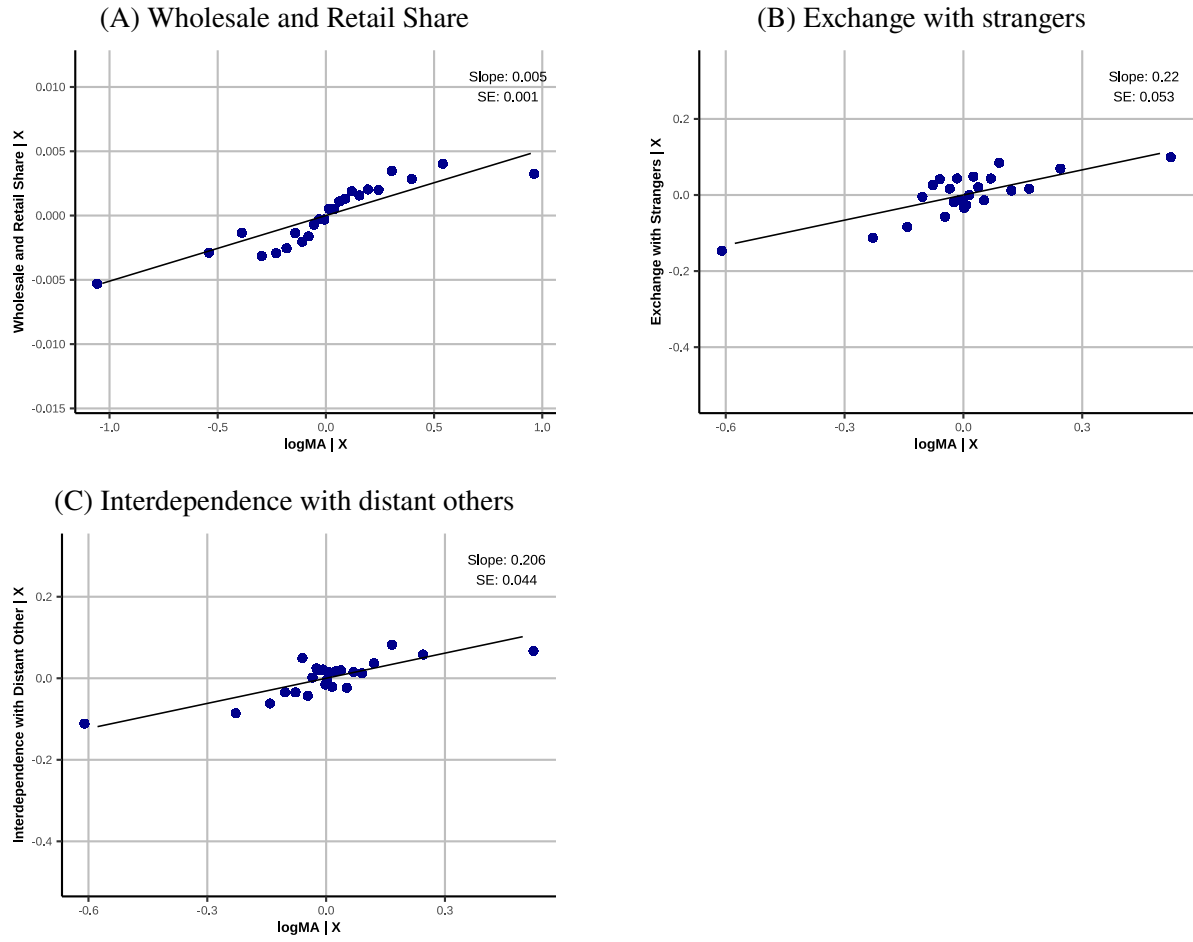


Figure B.10: Market Access, Commerce, Exchange with Strangers, and Economic Interdependence: County-level Bin Scatter Plots

Note: This figure presents bin scatter plots of the relationship between log market access and each of the three mechanism measures, using equation (4). The plots present the conditional relationship after partialling out the baseline controls, corresponding to columns 1, 3, and 5 in Table 3. All bins have the same number of observations.

Table B.3: Market Access and the Components of Exchange with Strangers and Economic Interdependence

	Dependent variable:					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Exchange with strangers</i>						
	Mobile Commercial Counterparties		Commodity Intermediaries		Commodity Buyers	
Log market access	0.0807* (0.0461)	0.0663 (0.0538)	0.2050*** (0.0511)	0.2456*** (0.0693)	0.1140*** (0.0426)	0.1034** (0.0467)
Observations	6,809	6,800	6,809	6,800	6,809	6,800
R ²	0.571	0.575	0.682	0.688	0.635	0.637
<i>Panel B: Interdependence with distant others</i>						
	Credit Dependence		Distant-Market Orientation		Standardization	
Log market access	0.1145*** (0.0368)	0.0909* (0.0524)	0.2301*** (0.0540)	0.3062*** (0.0763)	0.0602 (0.0578)	0.0646 (0.0662)
Observations	6,809	6,800	6,809	6,800	6,809	6,800
R ²	0.662	0.664	0.509	0.513	0.713	0.718
Expected log market access	Yes		Yes		Yes	
Any railroad	Yes		Yes		Yes	
Railroads within nearby buffer	Yes		Yes		Yes	
Railroads within further buffers	Yes		Yes		Yes	
Population within further buffers	Yes		Yes		Yes	

Note: This table reports estimates of equation (4). The dependent variables are the components of the exchange with strangers in local newspapers (Panel A), and economic interdependence with distant others in local newspapers (Panel B). All columns control for county fixed effects, state-by-year fixed effects, and a cubic spatial polynomial interacted with year fixed effects. Even columns also control for the expected log market access and local railroads and population. Standard errors clustered at arbitrary grid cells of 100 miles square in parentheses (Bester et al., 2011). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Appendix References

- Abdurahman, S., M. Atari, F. Karimi-Malekabadi, M. J. Xue, J. Trager, P. S. Park, P. Golazizian, A. Omrani, and M. Dehghani**, “Perils and opportunities in using large language models in psychological research,” *PNAS Nexus*, July 2024, 3 (7), pgae245.
- Abramitzky, R., L. Boustan, and K. Eriksson**, “Do immigrants assimilate more slowly today than in the past?,” *American Economic Review: Insights*, March 2020, 2 (1), 125–41.
- Atari, M., A. Omrani, and M. Dehghani**, “Contextualized Construct Representation: Leveraging Psychometric Scales to Advance Theory-Driven Text Analysis,” February 2023. PsyArXiv preprint.
- Barkai, S. and E. Karger**, “Classifying Industries into Tradable/Nontradable via Geographic Radius Served,” *Working paper*, 2020. Available at SSRN: <https://ssrn.com/abstract=4306269> or <http://dx.doi.org/10.2139/ssrn.4306269>.
- Bazzi, S., A. Gaduh, A. D. Rothenberg, and M. Wong**, “Unity in diversity? How intergroup contact can foster nation building,” *American Economic Review*, 2019, 109 (11), 3978–4025.
- Bester, A. C., T. G. Conley, and C. B. Hansen**, “Inference with dependent data using cluster covariance estimators,” *Journal of Econometrics*, 2011, 165 (2), 137–151.
- Borusyak, K. and P. Hull**, “Nonrandom exposure to exogenous shocks,” *Econometrica*, 2023, 91 (6), 2155–2185.
- Donaldson, D. and R. Hornbeck**, “Railroads and American Economic Growth: A “Market Access” Approach,” *The Quarterly Journal of Economics*, May 2016, 131 (2), 799–858.
- Fogel, R. W.**, *Railroads and American Economic Growth*, Johns Hopkins Press Baltimore, 1964.
- Fouka, V.**, “Backlash: The unintended effects of language prohibition in U.S. schools after World War I,” *The Review of Economic Studies*, 2020, 87 (1), 204–239.
- Fryer, R. and S. Levitt**, “The causes and consequences of distinctively Black names,” *The Quarterly Journal of Economics*, 2004, 119 (3), 767–805.
- Fulford, S. L., I. Petkov, and F. Schiantarelli**, “Does it matter where you came from? Ancestry composition and economic performance of US counties, 1850–2010,” *Journal of Economic Growth*, 2020, 25 (3), 341–380.
- Gentzkow, M. and J. M. Shapiro**, “What Drives Media Slant? Evidence From U.S. Daily Newspapers,” *Econometrica*, 2010, 78 (1), 35–71.
- , **J. M. Shapiro, and M. Sinkinson**, “The Effect of Newspaper Entry and Exit on Electoral Politics,” *American Economic Review*, December 2011, 101 (7), 2980–3018.
- , **J. M. Shapiro, and M. Sinkinson**, “Competition and Ideological Diversity: Historical Evidence from US Newspapers,” *American Economic Review*, October 2014, 104 (10), 3073–3114.
- Ghosh, A., S. I. M. Hwang, and M. Squires**, “Economic consequences of kinship: Evidence from US bans on cousin marriage,” *The Quarterly Journal of Economics*, 2023, 138 (4), 2559–2606.
- Hornbeck, R. and M. Rotemberg**, “Growth Off the Rails: Aggregate Productivity Growth in Distorted Economies,” *Journal of Political Economy*, 2024, 132 (11), 3547–3602.
- Inter-university Consortium for Political and Social Research**, “United States Historical Election Returns, 1824–1968 [dataset],” 1999. Inter-university Consortium for Political and Social Research [distributor].
- Kozlowski, A. C., M. Taddy, and J. A. Evans**, “The Geometry of Culture: Analyzing the Meanings of Class through Word Embeddings,” *American Sociological Review*, 2019, 84 (5), 905–949.
- Manson, S., J. Schroeder, D. V. Riper, T. Kugler, and S. Ruggles**, “IPUMS National Historical

- Geographic Information System: Version 15.0 [dataset],” 2020. Minneapolis, MN: IPUMS.
- Nelson, M. A.**, “The decline of patrilineal kin propinquity in the United States, 1790–1940,” *Demographic research*, 2020, 43, 501–532.
- Posch, M., J. Schulz, and J. Henrich**, “How Cultural Diversity Drives Innovation: Surnames and Patents in U.S. History,” *Journal of Political Economy*, 2026, 134 (4), 1245–1305.
- Raz, I. T.**, “Soil Heterogeneity, Social Learning, and the Formation of Close-Knit Communities,” *Journal of Political Economy*, 2025, 133 (8), 2643–2691.
- Reimers, N. and I. Gurevych**, “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” in “Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)” Association for Computational Linguistics Hong Kong, China November 2019, pp. 3982–3992.
- Ruggles, S., S. Flood, R. Goeken, J. Grover, E. Meyer, J. Pacas, and M. Sobek**, “IPUMS USA: Version 10.0 [dataset],” 2020. Minneapolis, MN: IPUMS.

Supplementary Material: Additional Robustness Checks (Not for Online Publication)

C Robustness Checks	75
C.1 Migrant-level analysis	75
C.2 County-level analysis	80
C.3 More Evidence on Mechanisms	88
Supplementary Material References	95

C Robustness Checks

C.1 Migrant-level analysis

The baseline bins event times ten or more years before and after migration; the alternative uses the full available horizon of nineteen years on either side. LNI is computed for children aged 0–10 in each census ([Bazzi et al., 2020](#); [Raz, 2025](#)).

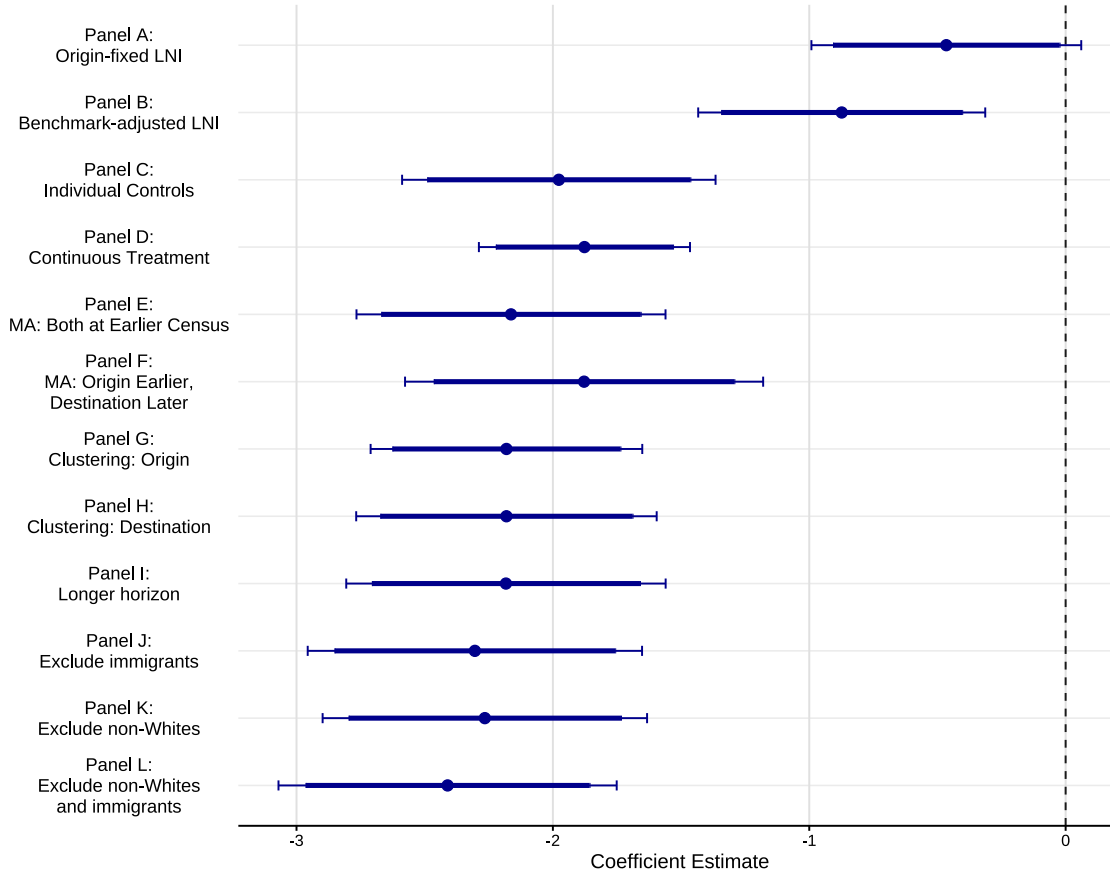


Figure C.1: Robustness of Cultural Adaptation Results

Note: This figure plots the estimates of $\bar{\beta}$ —the average post-migration treatment effect weighted by the number of treated children born in each relative year—and 90% and 95% confidence intervals from the difference-in-differences equation 1. Panel A uses origin-fixed LNI, scoring all names in the origin county with the baseline census-year rule. Panel B uses benchmark-adjusted LNI: after migration, actual LNI minus the sibling-name benchmark, with both names scored in the same destination county and census year. The benchmark draws uniformly from distinct same-sex sibling names given before migration; the adjusted outcome is zero before migration by construction. Panel C includes gender, birth order, and 5-year cohort fixed effects. Panel D uses a continuous treatment definition—the difference in log market access—and weights the post-migration treatment effects by the number of children born in each relative year with a positive log market access difference. Panels E–F vary the census dates used to define the binary market-access treatment: both counties at the earlier linked census (E), and the origin earlier and destination later (F). Panels G–H cluster at the county of origin and the county of destination, respectively. Panel I estimates effects over the maximal possible horizons—19 years before and after the year of migration—instead of aggregating effects 10 years before and after migration. Panels J–L exclude from the sample immigrants, non-Whites, and both.

Table C.1: Individual Controls in Within Origin-Destination-Decade Bins Analysis

Heterogeneity Variable:	Dependent variable: Less Parochial				
	Market Orientation	Higher paying job	Out of Agriculture	Further From Kin	Fewer Shared Ancestry
	(1)	(2)	(3)	(4)	(5)
Heterogeneity Variable	-0.0536** (0.0218)	0.0062 (0.0113)	0.0425 (0.0288)	0.0010 (0.0103)	0.0171 (0.0111)
Heterogeneity Variable $\times$ treat	0.0604** (0.0295)	-0.0051 (0.0173)	-0.0182 (0.0440)	-0.0077 (0.0193)	0.0102 (0.0146)
Observations	71,192	105,471	49,167	105,471	100,436
R ²	0.834	0.796	0.897	0.796	0.802
Origin $\times$ Destination $\times$ Year Fixed-Effects	Yes	Yes	Yes	Yes	Yes
Individual Ctrl. (demographics)	Yes	Yes	Yes	Yes	Yes
Individual Ctrl. (traits)	Yes	Yes	Yes	Yes	Yes

Note: This table reports estimates from the within origin-destination-decade bins framework (Eq. 3) when individual characteristics are additionally controlled for. Individual demographic controls include age, race, and birthplace fixed effects. Individual trait controls include fixed effects for ICM and urban origin. Each column focuses on a different margin of heterogeneity across domestic migrants, indicated by the column headers. Standard errors two-way clustered at the county of destination and the county of origin in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

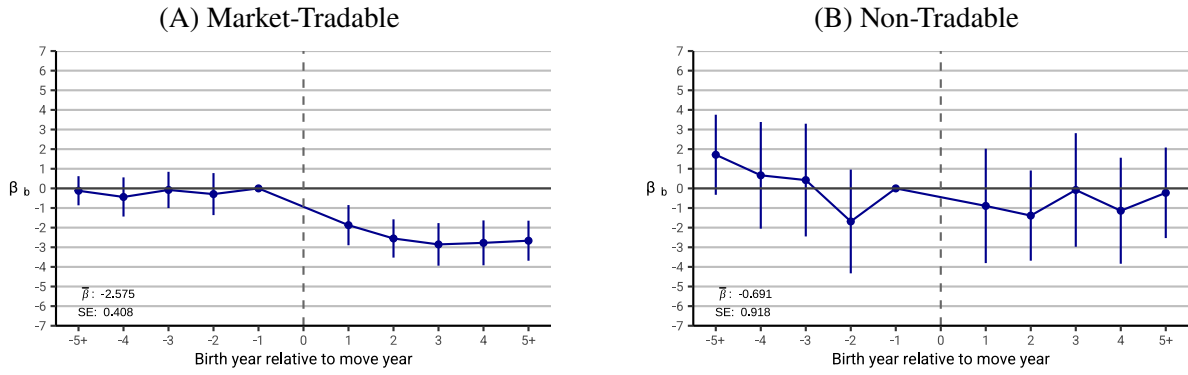


Figure C.2: Cultural Adaptation is Stronger in Tradable Industries

Note: This figure plots the estimates of β_b and 95% confidence intervals from the dynamic difference-in-differences equation (1). The dependent variable is children's LNI. In Panel A, the sample is restricted to households in which the father worked in a tradable industry both before and after the migration. In Panel B, the sample is restricted to households in which the father worked in a non-tradable industry both before and after the migration. $\bar{\beta}$ is the average treatment effect across post-migration periods weighted by the number of children born in each relative year.

Table C.2: Cultural Adaptation under Alternative Industry Classifications

	(1)	(2)
Heterogeneity Variable:	Market Orientation	Tradable
<i>Panel A:</i> Triple-Differences Designs Dependent variable: LNI		
Post Migration × Higher Market Access	-0.204 (0.905)	-0.692 (0.915)
Post Migration × Heterogeneity Variable	2.476*** (0.694)	2.603*** (0.732)
Post Migration × Higher Market Access × Heterogeneity Variable	-2.282** (0.953)	-1.884* (0.990)
Observations	298,474	263,859
R ²	0.346	0.345
<i>Panel B:</i> Within Origin-Destination-Decade Bins Dependent variable: Less Parochial		
Heterogeneity Variable	-0.0501** (0.0216)	-0.0381* (0.0219)
Higher Market Access × Heterogeneity Variable	0.0628** (0.0287)	0.0656** (0.0280)
Observations	71,192	62,161
R ²	0.831	0.849

Note: Panel A reports the average treatment effect on children's LNI score in post-migration periods, weighted by the number of children born in each relative year, from the triple-differences framework (Eq. 2). Panel B reports estimates from the within origin-destination-decade bins framework (Eq. 3). Each of columns 1-2 focuses on a different margin of heterogeneity across domestic migrants, indicated by the column headers. Standard errors two-way clustered at the county of destination and the county of origin in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

C.2 County-level analysis

Table C.3: County-level analysis: Exclusion of Immigrants and Non-Whites

Sample:	Dependent variable:			
	Baseline	Exclude foreign-born	Exclude non-whites	Exclude non-whites and foreign-born
	(1)	(2)	(3)	(4)
<i>Panel A: Composite Universalism Index</i>				
Log market access	0.5602*** (0.0642)	0.4954*** (0.0656)	0.6134*** (0.0643)	0.5523*** (0.0664)
Observations	16,574	16,422	16,560	16,404
R ²	0.805	0.794	0.818	0.812
<i>Panel B: Local Name Index</i>				
Log market access	-0.9250*** (0.1610)	-0.8056*** (0.1678)	-1.004*** (0.1650)	-0.8975*** (0.1740)
Observations	18,182	18,182	18,182	18,182
R ²	0.805	0.812	0.810	0.815
<i>Panel C: Share in Family Care</i>				
Log market access	-0.0123*** (0.0032)	-0.0100*** (0.0033)	-0.0126*** (0.0031)	-0.0102*** (0.0033)
Observations	18,173	18,157	18,163	18,142
R ²	0.722	0.717	0.721	0.724
<i>Panel D: Norms Tolerance Index</i>				
Log market access	0.1704*** (0.0280)	0.1349*** (0.0296)	0.1980*** (0.0278)	0.1592*** (0.0293)
Observations	17,112	16,897	17,091	16,868
R ²	0.702	0.696	0.678	0.675

Note: This table reports estimates of the baseline specification of equation (4) when the dependent variables are the composite universalism index (Panel A), the LNI (Panel B), the share of vulnerable individuals in family care (Panel C), and the NTI (Panel D). All columns control for county fixed effects, state-by-year fixed effects, and a cubic spatial polynomial interacted with year fixed effects. The column-1 outcomes use their respective baseline populations, as defined in Appendix A.3. In column 2 the sample excludes foreign-born, in column 3 it excludes non-whites, and in column 4 it excludes all non-whites and foreign-born. Standard errors clustered at arbitrary grid cells of 100 miles square in parentheses (Bester et al., 2011). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.4: County-level analysis: Different Market Access Measures

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	<i>Panel A: Baseline and Robustness</i>							
	Baseline	Robustness						
	$P = 35$	$P = 38.7$		$P = 35$				
	$\theta = 8.22$	$\theta = 3.05$	$\theta = 8.22$	$\theta = 1$	$\theta = 2$	$\theta = 3$	$\theta = 4$	$\theta = 5$
CUI	0.560*** (0.064)	1.60*** (0.187)	0.595*** (0.069)	4.61*** (0.534)	2.30*** (0.266)	1.53*** (0.177)	1.15*** (0.132)	0.916*** (0.106)
LNI	-0.925*** (0.161)	-2.43*** (0.450)	-0.963*** (0.171)	-7.08*** (1.30)	-3.54*** (0.647)	-2.37*** (0.430)	-1.78*** (0.322)	-1.44*** (0.258)
Family Care	-0.012*** (0.003)	-0.035*** (0.009)	-0.013*** (0.003)	-0.102*** (0.026)	-0.051*** (0.013)	-0.034*** (0.009)	-0.025*** (0.006)	-0.020*** (0.005)
NTI	0.170*** (0.028)	0.471*** (0.080)	0.181*** (0.030)	1.36*** (0.229)	0.678*** (0.114)	0.451*** (0.076)	0.339*** (0.057)	0.272*** (0.045)
RDI	0.268*** (0.035)	0.731*** (0.097)	0.282*** (0.037)	2.12*** (0.280)	1.06*** (0.140)	0.704*** (0.093)	0.528*** (0.070)	0.424*** (0.056)
Turnout	0.037*** (0.006)	0.107*** (0.018)	0.040*** (0.007)	0.309*** (0.051)	0.154*** (0.025)	0.103*** (0.017)	0.077*** (0.013)	0.061*** (0.010)
Trust	0.120** (0.048)	0.341** (0.139)	0.127** (0.052)	0.998** (0.400)	0.495** (0.198)	0.329** (0.132)	0.246** (0.099)	0.197** (0.079)

Note: This table reports estimates of β from the baseline specification of equation (4) when the dependent variables are: the composite universalism index, the LNI, the share of vulnerable individuals in family care, the NTI, the RDI, voter turnout, and social trust (rows), and market access is calculated using different average costs P and different trade elasticities θ (columns). All columns control for county fixed effects, state-by-year fixed effects, and a cubic spatial polynomial interacted with year fixed effects. Standard errors clustered at arbitrary grid cells of 100 miles square in parentheses (Bester et al., 2011). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The table continues to the next page.

Table C.4: County-level analysis: Different Market Access Measures (cont.)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	<i>Panel B: Robustness</i>							
	<i>P = 35</i>							
	$\theta = 6$	$\theta = 7$	$\theta = 8$	$\theta = 9$	$\theta = 10$	$\theta = 11$	$\theta = 12$	$\theta = 13$
CUI	0.764*** (0.088)	0.655*** (0.075)	0.575*** (0.066)	0.514*** (0.059)	0.466*** (0.053)	0.429*** (0.048)	0.399*** (0.044)	0.376*** (0.041)
LNI	-1.21*** (0.217)	-1.06*** (0.187)	-0.946*** (0.165)	-0.858*** (0.148)	-0.789*** (0.134)	-0.737*** (0.123)	-0.699*** (0.114)	-0.672*** (0.107)
Family Care	-0.017*** (0.004)	-0.015*** (0.004)	-0.013*** (0.003)	-0.011*** (0.003)	-0.010*** (0.003)	-0.009*** (0.002)	-0.009*** (0.002)	-0.008*** (0.002)
NTI	0.229*** (0.038)	0.198*** (0.033)	0.175*** (0.029)	0.156*** (0.026)	0.141*** (0.023)	0.129*** (0.021)	0.120*** (0.019)	0.112*** (0.018)
RDI	0.356*** (0.047)	0.309*** (0.040)	0.274*** (0.035)	0.248*** (0.032)	0.228*** (0.029)	0.212*** (0.026)	0.200*** (0.025)	0.190*** (0.023)
Turnout	0.051*** (0.008)	0.044*** (0.007)	0.038*** (0.006)	0.034*** (0.006)	0.031*** (0.005)	0.029*** (0.005)	0.027*** (0.004)	0.025*** (0.004)
Trust	0.164** (0.066)	0.141** (0.057)	0.123** (0.050)	0.110** (0.044)	0.100** (0.040)	0.092** (0.036)	0.087** (0.034)	0.082*** (0.031)

Note: This table reports estimates of β from the baseline specification of equation (4) when the dependent variables are: the composite universalism index, the LNI, the share of vulnerable individuals in family care, the NTI, the RDI, voter turnout, and social trust (rows), and market access is calculated using different average costs P and different trade elasticities θ (columns). All columns control for county fixed effects, state-by-year fixed effects, and a cubic spatial polynomial interacted with year fixed effects. Standard errors clustered at arbitrary grid cells of 100 miles square in parentheses (Bester et al., 2011). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

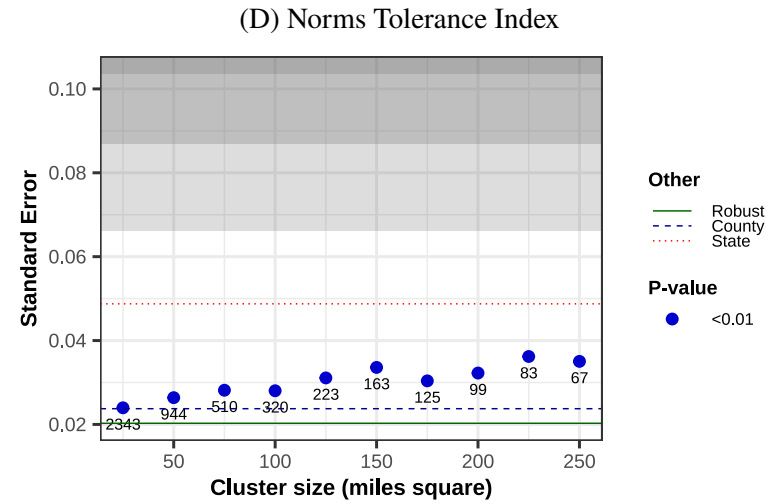
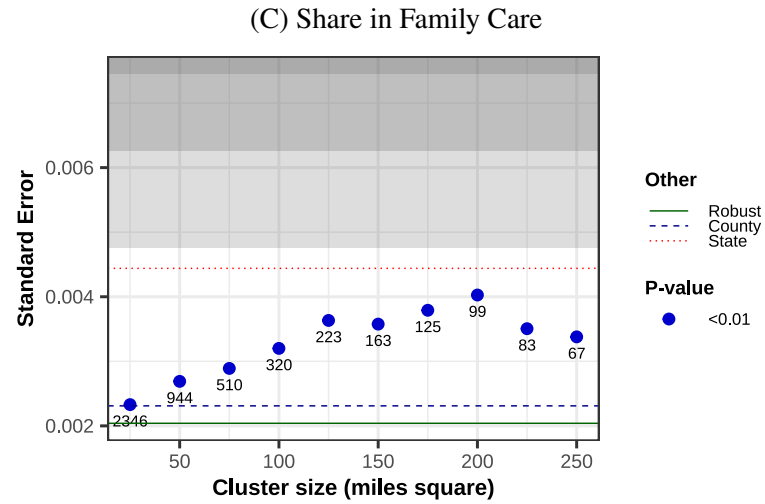
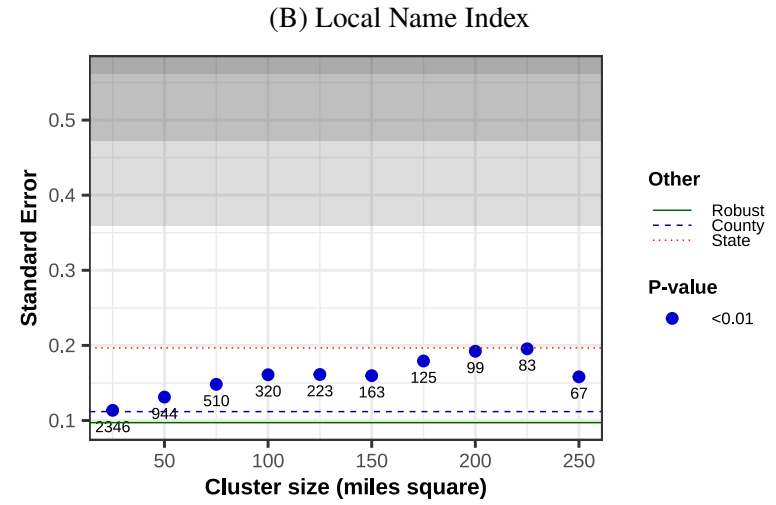
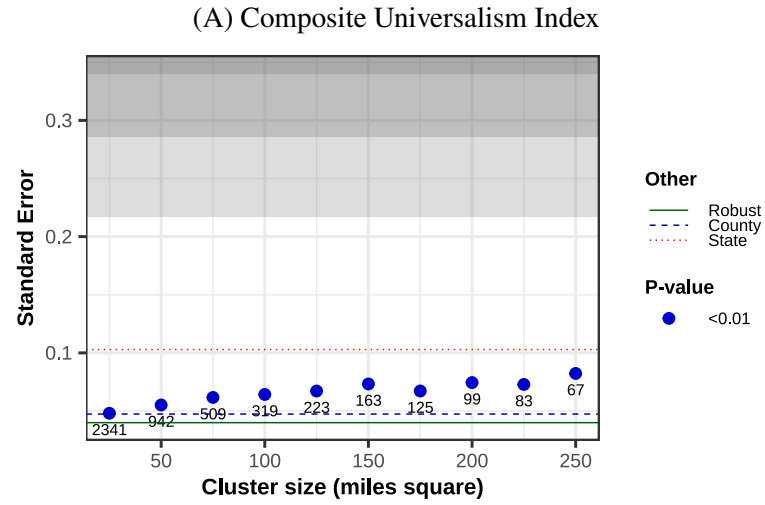


Figure C.3: Universalism, Impersonal Prosociality, and Tolerance: Different Standard Errors

Note: This figure plots the standard errors of β from the preferred specification of equation (4) using different approaches for inference. The blue dots represent the standard errors (on the y-axis) using arbitrary grid-cell of different sizes (on the x-axis), as proposed by [Bester et al. \(2011\)](#). The numeric label under each dot indicates the number of spatial clusters. The dotted dark green horizontal line plots the HC robust standard errors, the dashed dark blue horizontal line plots the standard errors when clustering at the county level, and the dash-dotted red horizontal line plots the standard errors when clustering at the state level. The background color is indicative of the level of statistical significance. The p-value is < 0.01 in the white area, and < 0.05, < 0.1 and > 0.1 in the light to dark shades of gray. The figure continues to the next page.

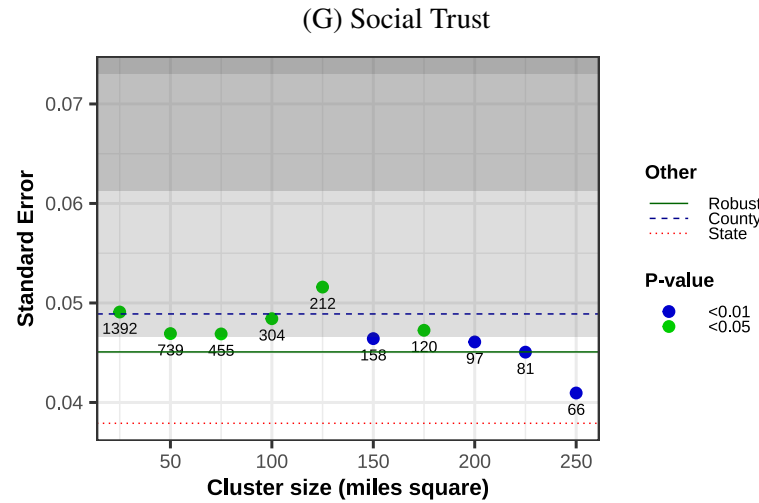
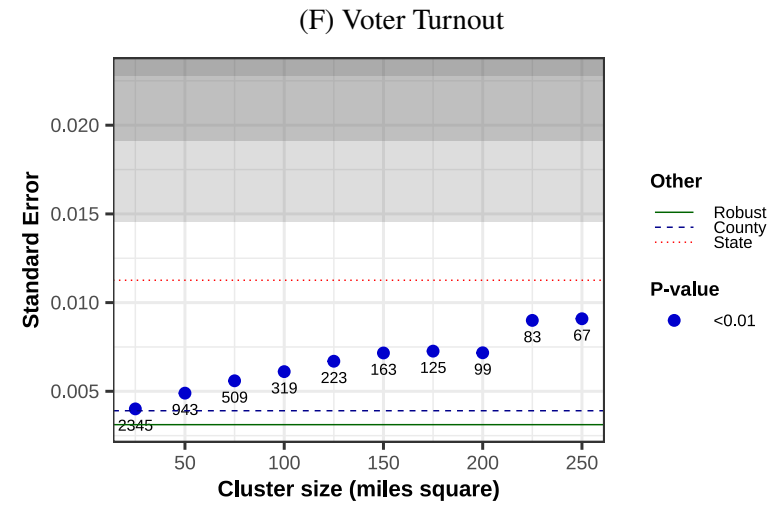
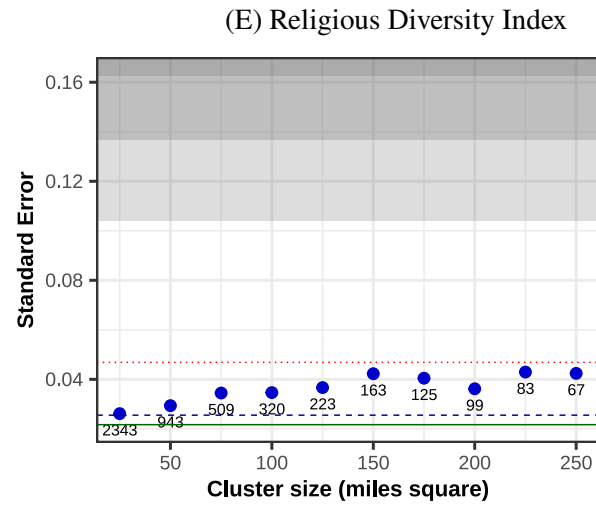


Figure C.3: Universalism, Impersonal Prosociality, and Tolerance: Different Standard Errors (cont.)

Note: This figure plots the standard errors of β from the preferred specification of equation (4) using different approaches for inference. The blue dots represent the standard errors (on the y-axis) using arbitrary grid-cell of different sizes (on the x-axis), as proposed by [Bester et al. \(2011\)](#). The numeric label under each dot indicates the number of spatial clusters. The dotted dark green horizontal line plots the HC robust standard errors, the dashed dark blue horizontal line plots the standard errors when clustering at the county level, and the dash-dotted red horizontal line plots the standard errors when clustering at the state level. The background color is indicative of the level of statistical significance. The p-value is < 0.01 in the white area, and < 0.05, < 0.1 and > 0.1 in the light to dark shades of gray.

Table C.5: County-level Analysis: Exclusion of Regions

Sample:	Dependent variable:				
	Baseline	Exclude Northeast	Exclude Midwest	Exclude South	Exclude West
	(1)	(2)	(3)	(4)	(5)
<i>Panel A: Composite Universalism Index</i>					
Log market access	0.5602*** (0.0642)	0.5733*** (0.0647)	0.5375*** (0.0802)	0.4584*** (0.0812)	0.5908*** (0.0675)
Observations	16,574	15,069	10,621	8,584	15,448
R ²	0.805	0.788	0.818	0.781	0.798
<i>Panel B: Local Name Index</i>					
Log market access	-0.9250*** (0.1610)	-0.9084*** (0.1614)	-0.9096*** (0.1931)	-0.8586*** (0.1944)	-1.016*** (0.1871)
Observations	18,182	16,677	11,573	9,882	16,414
R ²	0.805	0.789	0.813	0.830	0.799
<i>Panel C: Share in Family Care</i>					
Log market access	-0.0123*** (0.0032)	-0.0123*** (0.0032)	-0.0155*** (0.0033)	-0.0132*** (0.0046)	-0.0084** (0.0035)
Observations	18,173	16,668	11,572	9,873	16,406
R ²	0.722	0.716	0.761	0.715	0.701
<i>Panel D: Norms Tolerance Index</i>					
Log market access	0.1704*** (0.0280)	0.1697*** (0.0286)	0.1184*** (0.0339)	0.2181*** (0.0346)	0.1830*** (0.0305)
Observations	17,112	15,607	11,043	9,005	15,681
R ²	0.702	0.684	0.730	0.716	0.681

Note: This table reports estimates of the baseline specification of equation (4) when the dependent variables are the composite universalism index (Panel A), the LNI (Panel B), the share of vulnerable individuals in family care (Panel C), the NTI (Panel D), the RDI (Panel E), voters turnout (Panel F), and social trust (Panel G). Column 1 reports the baseline estimate. Columns 2-5 exclude different regions of the country: the Northeast, the Midwest, the South, and the West. All columns control for county fixed effects, state-by-year fixed effects, and a cubic spatial polynomial interacted with year fixed effects. Standard errors clustered at arbitrary grid cells of 100 miles square in parentheses (Bester et al., 2011). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The table continues to the next page.

Table C.5: County-level Analysis: Exclusion of Regions (cont.)

Sample:	Dependent variable:				
	Baseline (1)	Exclude Northeast (2)	Exclude Midwest (3)	Exclude South (4)	Exclude West (5)
<i>Panel E: Religious Diversity Index</i>					
Log market access	0.2678*** (0.0346)	0.2653*** (0.0350)	0.2254*** (0.0404)	0.2922*** (0.0386)	0.3022*** (0.0398)
Observations	17,303	15,798	11,203	9,191	15,717
R ²	0.681	0.676	0.704	0.739	0.616
<i>Panel F: Voter turnout</i>					
Log market access	0.0375*** (0.0061)	0.0392*** (0.0063)	0.0420*** (0.0077)	0.0187*** (0.0061)	0.0378*** (0.0065)
Observations	45,308	41,241	28,218	24,534	41,931
R ²	0.804	0.801	0.799	0.765	0.813
<i>Panel G: Social Trust</i>					
Log market access	0.1201** (0.0484)	0.1117** (0.0486)	0.0851 (0.0661)	0.1055* (0.0556)	0.1959*** (0.0586)
Observations	6,821	5,893	3,762	4,814	5,994
R ²	0.681	0.679	0.711	0.664	0.676

Note: This table reports estimates of the baseline specification of equation (4) when the dependent variables are the composite universalism index (Panel A), the LNI (Panel B), the share of vulnerable individuals in family care (Panel C), the NTI (Panel D), the RDI (Panel E), voters turnout (Panel F), and social trust (Panel G). Column 1 reports the baseline estimate. Columns 2-5 exclude different regions of the country: the Northeast, the Midwest, the South, and the West. All columns control for county fixed effects, state-by-year fixed effects, and a cubic spatial polynomial interacted with year fixed effects. Standard errors clustered at arbitrary grid cells of 100 miles square in parentheses (Bester et al., 2011). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.6: Family Care: Winsorize at Different Percentiles

Winsorize at:	Dependent variable: Share in Family Care					
	p(0)	p(1)	p(2)	p(3)	p(4)	p(5)
	(1)	(2)	(3)	(4)	(5)	(6)
Log market access	-0.0104** (0.0042)	-0.0120*** (0.0038)	-0.0123*** (0.0033)	-0.0121*** (0.0031)	-0.0119*** (0.0029)	-0.0118*** (0.0028)
Observations	18,173	18,173	18,173	18,173	18,173	18,173
R ²	0.674	0.701	0.717	0.726	0.730	0.732
DV mean	0.7740	0.7750	0.7770	0.7790	0.7800	0.7810
DV sd	0.1270	0.1200	0.1140	0.1090	0.1060	0.1050

Note: This table reports estimates of the baseline specification of equation (4) when the dependent variable is the share of vulnerable individuals in family care. In each column the outcome is winsorized at different percentiles at the bottom, between 0% (i.e., not winsorized, column 1) and 5% (column 6). All columns control for county fixed effects, state-by-year fixed effects, and a cubic spatial polynomial interacted with year fixed effects. Standard errors clustered at arbitrary grid cells of 100 miles square in parentheses (Bester et al., 2011). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

C.3 More Evidence on Mechanisms

C.3.1 Commerce, Exchange with strangers, and Economic Interdependence with distant others

Table C.7: Trade Share: Exclusion of Immigrants and Non-Whites

Sample:	Dependent variable: Wholesale and Retail Share			
	Baseline	Exclude foreign-born	Exclude non-whites	Exclude non-whites and foreign-born
	(1)	(2)	(3)	(4)
Log market access	0.0051*** (0.0008)	0.0048*** (0.0008)	0.0041*** (0.0009)	0.0038*** (0.0009)
DV mean	0.0550	0.0550	0.0620	0.0630
DV sd	0.0360	0.0370	0.0400	0.0400
Observations	18,266	18,212	18,257	18,197
R ²	0.780	0.790	0.781	0.784

Note: This table reports estimates of equation (4) when the dependent variable is the share of individuals working in the wholesale and retail trade industries. The base sample used to calculate the county-level measures in column 1 includes all of the population not residing in group quarters. In column 2, the sample excludes foreign-born, in column 3 it excludes non-whites, and in column 4 it excludes all non-whites and foreign-born. All columns control for county fixed effects, state-by-year fixed effects, and a cubic spatial polynomial interacted with year fixed effects. Standard errors clustered at arbitrary grid cells of 100 miles square in parentheses (Bester et al., 2011). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.8: Market Access and Commercial Relationships: Different Market Access Measures

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A: Baseline and Robustness</i>								
	Baseline	Robustness						
	$P = 35$	$P = 38.7$			$P = 35$			
	$\theta = 8.22$	$\theta = 3.05$	$\theta = 8.22$	$\theta = 1$	$\theta = 2$	$\theta = 3$	$\theta = 4$	$\theta = 5$
Share in Trade	0.005*** (0.0008)	0.015*** (0.002)	0.005*** (0.0009)	0.044*** (0.007)	0.022*** (0.003)	0.015*** (0.002)	0.011*** (0.002)	0.009*** (0.001)
Exchange with Strangers	0.220*** (0.053)	0.626*** (0.154)	0.233*** (0.057)	1.82*** (0.442)	0.906*** (0.220)	0.602*** (0.147)	0.450*** (0.109)	0.360*** (0.087)
Interdependence	0.206*** (0.044)	0.584*** (0.127)	0.219*** (0.047)	1.68*** (0.385)	0.837*** (0.182)	0.557*** (0.120)	0.417*** (0.091)	0.333*** (0.072)
<i>Panel B: Robustness</i>								
	$P = 35$							
	$\theta = 6$	$\theta = 7$	$\theta = 8$	$\theta = 9$	$\theta = 10$	$\theta = 11$	$\theta = 12$	$\theta = 13$
Share in Trade	0.007*** (0.001)	0.006*** (0.0009)	0.005*** (0.0008)	0.005*** (0.0007)	0.004*** (0.0007)	0.004*** (0.0006)	0.004*** (0.0006)	0.003*** (0.0005)
Exchange with Strangers	0.300*** (0.073)	0.258*** (0.062)	0.226*** (0.055)	0.202*** (0.049)	0.183*** (0.044)	0.167*** (0.040)	0.155*** (0.037)	0.144*** (0.034)
Interdependence	0.278*** (0.060)	0.240*** (0.052)	0.211*** (0.045)	0.189*** (0.040)	0.171*** (0.036)	0.157*** (0.033)	0.146*** (0.030)	0.137*** (0.028)

Note: This table reports estimates of β from the baseline specification of equation (4) when the dependent variables are three mechanism measures and market access is calculated using different average costs P and different trade elasticities θ . All columns control for county fixed effects, state-by-year fixed effects, and a cubic spatial polynomial interacted with year fixed effects. Standard errors clustered at arbitrary grid cells of 100 miles square in parentheses (Bester et al., 2011).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

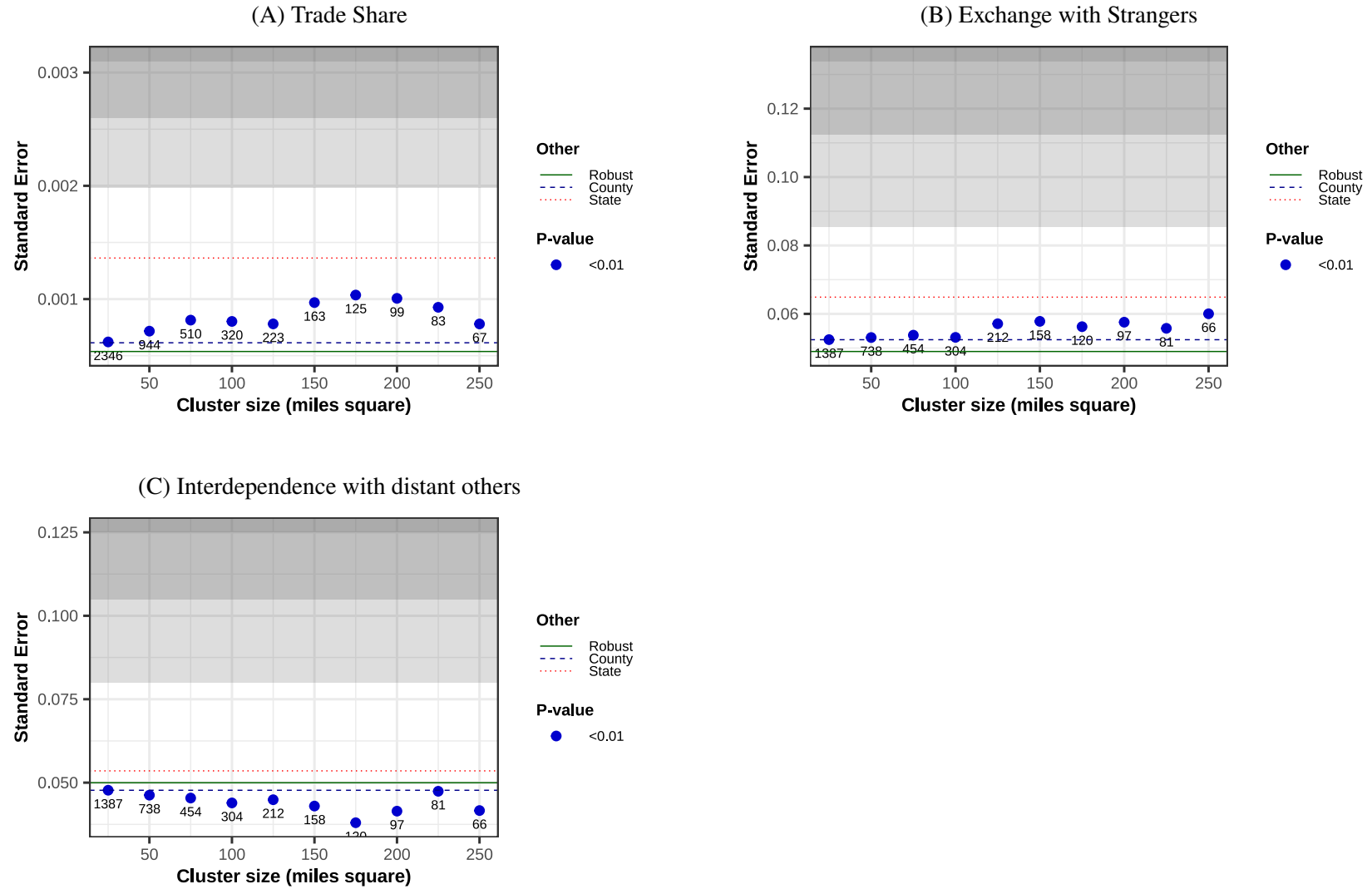


Figure C.4: Market Access and Commercial Relationships: Different Standard Errors

Note: This figure plots the standard errors of β from the baseline specification of equation (4) using different approaches for inference. The blue dots represent the standard errors (on the y-axis) using arbitrary grid-cells of different sizes (on the x-axis), as proposed by [Bester et al. \(2011\)](#). The numeric label under each dot indicates the number of spatial clusters. The dotted dark green horizontal line plots the HC robust standard errors, the dashed dark blue horizontal line plots the standard errors when clustering at the county level, and the dash-dotted red horizontal line plots the standard errors when clustering at the state level. The background color is indicative of the level of statistical significance. The p-value is < 0.01 in the white area, and < 0.05 , < 0.1 , and > 0.1 in the light to dark shades of gray.

Table C.9: Market Access and Commercial Relationships: Exclusion of Regions

Sample:	Dependent variable:				
	Baseline	Exclude Northeast	Exclude Midwest	Exclude South	Exclude West
	(1)	(2)	(3)	(4)	(5)
<i>Panel A: Wholesale and Retail Share</i>					
Log market access	0.0051*** (0.0008)	0.0050*** (0.0008)	0.0064*** (0.0010)	0.0040*** (0.0009)	0.0058*** (0.0009)
Observations	18,266	16,769	11,596	9,943	16,490
R ²	0.780	0.760	0.796	0.755	0.792
<i>Panel B: Exchange with Strangers</i>					
Log market access	0.2203*** (0.0531)	0.2104*** (0.0539)	0.1733** (0.0863)	0.1625*** (0.0468)	0.3372*** (0.0673)
Observations	6,809	5,884	3,752	4,809	5,982
R ²	0.640	0.644	0.673	0.484	0.653
<i>Panel C: Interdependence with Distant Others</i>					
Log market access	0.2057*** (0.0440)	0.2096*** (0.0452)	0.2786*** (0.0587)	0.1643*** (0.0511)	0.1727*** (0.0564)
Observations	6,809	5,884	3,752	4,809	5,982
R ²	0.647	0.636	0.666	0.614	0.650

Note: This table reports estimates of equation (4) when the dependent variables are three measures relating to channels: the share of residents working in the wholesale and retail trade industries (Panel A), exchange with strangers in local newspapers (Panel B), and economic interdependence with distant others in local newspapers (Panel C). Column 1 reports the baseline estimate. Columns 2-5 exclude different regions of the country: the Northeast, the Midwest, the South, and the West. All columns control for county fixed effects, state-by-year fixed effects, and a cubic spatial polynomial interacted with year fixed effects. Standard errors clustered at arbitrary grid cells of 100 miles square in parentheses (Bester et al., 2011). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.10: Wholesale and Retail Share: Winsorize at Different Percentiles

Dependent variable: Wholesale and Retail Share						
	p(96)	p(96.5)	p(97)	p(98)	p(98.5)	p(99)
	(1)	(2)	(3)	(4)	(5)	(6)
Log market access	0.0050*** (0.0007)	0.0051*** (0.0008)	0.0051*** (0.0008)	0.0051*** (0.0008)	0.0052*** (0.0009)	0.0052*** (0.0010)
Observations	18,266	18,266	18,266	18,266	18,266	18,266
R ²	0.780	0.781	0.781	0.778	0.777	0.773
DV mean	0.0540	0.0540	0.0550	0.0550	0.0550	0.0560
DV sd	0.0340	0.0340	0.0350	0.0370	0.0380	0.0390

Note: This table reports estimates of equation (4) when the dependent variable is the winsorized share of individuals working in the wholesale and retail trade industries. In each column, that outcome is winsorized at different percentiles at the top, between 4% (column 1) and 1% (column 6). All columns control for county fixed effects, state-by-year fixed effects, and a cubic spatial polynomial interacted with year fixed effects. Standard errors clustered at arbitrary grid cells of 100 miles square in parentheses (Bester et al., 2011). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.11: Wholesale and Retail Share: Alternative Ways to Address Skewed Distributions With Zeros

Dependent variable: Wholesale and Retail Share (mean = 0.056, sd = 0.041)				
	Poisson Regression	asinh(y)	log(1+y)	log(0.0001+y)
	(1)	(2)	(3)	(4)
Log market access	0.0843*** (0.0209)	0.0045*** (0.0013)	0.0044*** (0.0011)	0.2271*** (0.0325)
Observations	18,260	18,266	18,266	18,266
R ²		0.73832	0.74764	0.61196
Pseudo R ²	0.054			

Note: This table reports estimates of equation (4) when the dependent variable is the share of individuals working in the wholesale and retail trade industries. Each column deals with the skewness of the outcome that contains zeros differently: column 1 uses a Poisson regression, column 2 uses an inverse hyperbolic sine transformation of the dependent variable, and columns 3-4 use log transformations. All columns control for county fixed effects, state-by-year fixed effects, and a cubic spatial polynomial interacted with year fixed effects. Standard errors clustered at arbitrary grid cells of 100 miles square in parentheses (Bester et al., 2011). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

C.3.2 Returns to Adaptation

Table C.12: Cultural Adaptation and Real Property: Winsorize at Different Percentiles

	Dependent variable:						
	Real Property Value						
	p(96)	p(96.5)	p(97)	p(98)	p(98.5)	p(99)	p(100)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Less Parochial	-79.82 (110.9)	-79.82 (110.9)	-79.37 (119.1)	-73.43 (131.2)	-80.49 (139.1)	-81.30 (149.5)	-330.7 (290.2)
Higher Market Access × Less Parochial	531.7** (230.6)	531.7** (230.6)	585.2** (247.1)	655.3** (274.9)	703.2** (294.7)	742.1** (315.9)	1,254.7** (625.2)
Observations	23,769	23,769	23,769	23,769	23,769	23,769	23,769
R ²	0.898	0.898	0.896	0.894	0.893	0.892	0.910
DV mean	2,249.4	2,249.4	2,306.7	2,384.3	2,427.0	2,474.6	2,763.6
DV sd	2,422.0	2,422.0	2,617.6	2,924.9	3,119.8	3,366.0	7,353.8
Origin × Destination × Year Fixed-Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Individual Ctrl. (demographics)	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: This table reports estimates of equation (5) when the dependent variable is the value of real property. In each column, the outcome is winsorized at a different percentile at the top, between 4% (column 1) and 0% (i.e., not winsorized, column 7). Standard errors two-way clustered at the county of destination and the county of origin in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table C.13: Cultural Adaptation and Real Property: Different Transformations

Dependent variable: Real Property Value (mean = 2748.95, sd = 7248.46)			
	Poisson Regression (1)	asinh(y) (2)	log(y) (3)
Less Parochial	-0.0694 (0.0603)	-0.0774* (0.0464)	-0.0774* (0.0464)
Higher Market Access × Less Parochial	0.2188** (0.0974)	0.1795** (0.0834)	0.1795** (0.0834)
Observations	23,769	23,769	23,769
R ²		0.905	0.905
Pseudo R ²	0.979		
Origin × Destination × Year Fixed-Effects	Yes	Yes	Yes
Individual Ctrl. (demographics)	Yes	Yes	Yes

Note: This table reports estimates of variants of equation (5) when the dependent variable is the value of real property. Each columns deals with the skewness of the outcome differently: column 1 uses a Poisson regression, while columns 2-3 use inverse hyperbolic sine and log transformations of the dependent variable, respectively. Standard errors two-way clustered at the county of destination and the county of origin in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Supplementary Material References

- Bazzi, S., M. Fiszbein, and M. Gebresilasse**, “Frontier Culture: The Roots and Persistence of “Rugged Individualism” in the United States,” *Econometrica*, 2020, 88 (6), 2329–2368.
- Bester, A. C., T. G. Conley, and C. B. Hansen**, “Inference with dependent data using cluster covariance estimators,” *Journal of Econometrics*, 2011, 165 (2), 137–151.
- Raz, I. T.**, “Soil Heterogeneity, Social Learning, and the Formation of Close-Knit Communities,” *Journal of Political Economy*, 2025, 133 (8), 2643–2691.